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Chapter 1: Orienting Yourself - The Use of Coordinates

This chapter teaches you how to describe the exact position of any point on a flat surface using nothing but two numbers. That sounds small, but it is the foundation of coordinate geometry, which you will use in every higher class. The uploaded PDF covers the 2-D Cartesian system (axes, origin, ordered pairs, quadrants, plotting), distances parallel to the axes, the distance formula built on the Baudhayana-Pythagoras Theorem, reflections in the axes, midpoint and trisection ideas, the collinearity test, and points on a circle centred at the origin. In this practice book we push every one of those ideas one level higher, with harder numbers, missing-value reasoning, and proof-style thinking.

1.1 Complete Concept Cheat Sheet (read this before any question)

1.1.1 The coordinate system and its parts

What is it? A coordinate system is a grid-based framework that locates every point in a plane using an ordered pair of numbers. Two perpendicular number lines are drawn: the horizontal one is the **x-axis** and the vertical one is the **y-axis**. They meet at the **origin O = (0, 0)**. Values to the right of O on the x-axis and above O on the y-axis are positive; values to the left and below are negative.

When do I use it? Whenever a question gives you points, positions on a map or a room layout, distances between locations, or asks you to 'plot', 'locate', 'identify the quadrant', or 'find the distance between points'.

How do I recognise it? Any mention of (x, y) pairs, axes, quadrants, graph paper, or a floor plan with walls along two directions. **What does each part mean?** In the pair (x, y), the first number x is the **perpendicular distance from the y-axis** (measured along the x-axis direction), and the second number y is the **perpendicular distance from the x-axis**. The pair is called an *ordered pair* because the order is fixed: (3, 5) and (5, 3) are different points. Logically, you can understand it as 'walk x units horizontally, then y units vertically from O'.

Tiny example. The point (2, 3) is 2 units right of O and 3 units up. The point (-2, 3) is 2 units LEFT and 3 up. Same distances, different signs, different points entirely.

1.1.2 Points on the axes and the quadrant sign rules

- A point on the x-axis always looks like (x, 0). Its y-coordinate is 0 because it has zero height above the x-axis. Positive x means right of O, negative x means left of O.
- A point on the y-axis always looks like (0, y). Its x-coordinate is 0. Positive y means above O, negative y means below O.
- The origin (0, 0) lies on BOTH axes; it belongs to no quadrant.
- Quadrant I: (+, +). Quadrant II: (-, +). Quadrant III: (-, -). Quadrant IV: (+, -). Memorise the cycle: starting from I and going anticlockwise the signs are ++, -+, --, +-. A quick memory trick: signs of y follow the C-shape of Roman numerals I, II, III, IV written anticlockwise; signs of x are + in I, - in II, - in III, + in IV.
- Any point with x = 0 and y = 0 must be the origin. Any point with both coordinates non-zero cannot lie on either axis.

> **Identify-the-method trick**
> See a question that says 'on the x-axis' or 'on the y-axis'? Instantly write the fixed coordinate: (something, 0) or (0, something). This single habit unlocks a whole family of missing-coordinate questions.

1.1.3 Distances parallel to the axes

What is it? If two points share the same y-coordinate, the segment joining them is horizontal, and the distance is simply the absolute gap between the x-values: distance = $|x_2 - x_1|$. If two points share the same x-coordinate, the segment is vertical and the distance is $|y_2 - y_1|$.

When do I use it? Whenever two given points have one identical coordinate - check this FIRST before reaching for the distance formula. **How do I recognise it?** Same second number means a horizontal move (like walking along a corridor east-west); same first number means a vertical move. **Each term:** the absolute value bars guarantee a positive (non-negative) length; order of subtraction does not matter.

Tiny example. Distance between (2, 5) and (9, 5) is $|9 - 2| = 7$. Distance between (4, -3) and (4, 6) is $|6 - (-3)| = 9$. Note the classic sign trap: $6 - (-3) = 9$, not 3.

1.1.4 The distance formula (the star of the chapter)

What is it? For any two points $A(x_1, y_1)$ and $D(x_2, y_2)$, the distance is $AD = \sqrt{((x_2 - x_1)^2 + (y_2 - y_1)^2)}$. **Where does it come from?** Drop a right-angled helper triangle: the horizontal leg is $|x_2 - x_1|$ and the vertical leg is $|y_2 - y_1|$. The segment AD is the hypotenuse, so the Baudhayana-Pythagoras Theorem ($leg^2 + leg^2 = hypotenuse^2$) gives the formula. It is not a magic rule - it is Pythagoras wearing coordinates.

When do I use it? When the two points differ in BOTH coordinates. **How do I recognise it?** Words like 'find the length of AB', 'how far apart', 'show that the triangle is isosceles', 'find the perimeter' with slanted sides. **Each term:** subtract corresponding coordinates, square each difference (squares kill any minus sign, so the order of the points does not matter), add, then take the positive square root.

Can it be derived logically? Yes - see the helper-triangle picture in Figure 2. **Tiny example.** From (1, 2) to (4, 6): differences are 3 and 4, so distance = $\sqrt{(9 + 16)} = \sqrt{25} = 5$. Whenever the squared sum is a perfect square you get a clean integer - examiners love the triples 3-4-5, 5-12-13, 6-8-10, 8-15-17, 9-12-15, 7-24-25.

- **Method 1 (helper triangle):** sketch, mark legs |dx| and |dy|, apply Pythagoras. Best for learning and for verifying.
- **Method 2 (formula directly):** substitute into $\sqrt{(dx^2 + dy^2)}$. Best for speed once confident.
- **Method 3 (shift to origin):** subtract the first point from both - distance is unchanged - then compute from (0, 0). Great mental shortcut: distance of (x, y) from origin is $\sqrt{(x^2 + y^2)}$.

> **Common confusion**
> Students often compute $\sqrt{(dx)} + \sqrt{(dy)}$ or $dx + dy$ instead of $\sqrt{(dx^2 + dy^2)}$. Remember: square EACH difference first, THEN add, THEN take one single square root at the very end. Never take a root midway.

1.1.5 Reflections in the axes

What is it? Reflecting a point in the x-axis flips its height: (x, y) becomes (x, -y). Reflecting in the y-axis flips its horizontal position: (x, y) becomes (-x, y). Reflecting in BOTH axes (equivalent to a half-turn about O)

gives (-x, -y). **Why only one sign changes:** a mirror in the x-axis changes only how far up or down you are; your sideways position stays the same.

Key property: reflection preserves distances. If A reflects to A' and B reflects to B', then AB = A'B'. The PDF proves this by computing both distances - the squares of the coordinate differences come out identical. Use this to check your reflection answers. **Tiny example.** A(2, 3) reflects in the x-axis to A'(2, -3). Take another point B(6, 3); it reflects to B'(6, -3). Now AB = |6 - 2| = 4 and A'B' = |6 - 2| = 4: the distance is preserved because the reflection moves the WHOLE segment, not just one endpoint. (If you compare A' with the unreflected B you get √52 - but that was never a preserved pair; always compare each point with its own image.)

1.1.6 Midpoint and equal-distance ideas

What is it? The midpoint of a segment splits it into two equal halves. The PDF discovers the connection experimentally: if M(m_1, m_2) is the midpoint of S and T, then each coordinate of M is the average of the corresponding coordinates: $m_1 = (x_S + x_T)/2$ and $m_2 = (y_S + y_T)/2$. You can verify with the distance formula: distances from M to S and from M to T come out equal. **Reverse use:** if you know one endpoint A and the midpoint M, the other endpoint is B = ($2m_1 - x_A$, $2m_2 - y_A$) - just undo the averaging. **Trisection:** points P and Q that cut AB into three equal parts satisfy P = A + (B - A)/3 and Q = A + 2(B - A)/3; the midpoint of P and Q equals the midpoint of AB, which is a lovely self-check.

When do I use the average idea? Keywords: 'midpoint', 'bisects', 'equally spaced', 'middle point of the segment'. **Tiny example.** Midpoint of (-4, 1) and (6, 5) is ((-4 + 6)/2, (1 + 5)/2) = (1, 3).

1.1.7 Collinearity test without plotting

What is it? Three points are collinear (lie on one straight line) if one of the three distances equals the sum of the other two. Concretely, compute AB, BC, AC and check whether the largest equals the sum of the other two. **Why it works:** if B truly sits on the straight path from A to C, then walking A to B to C covers exactly the straight distance AC - no detour is possible on a straight line (triangle inequality with equality).

When do I use it? Questions like 'are these points on the same straight line?' or 'check without plotting'. **Trap:** the check must be done with EXACT values. If the distances involve roots, compare exact squared forms or keep radicals; do not round early. **Tiny example.** (0,0), (3,4), (6,8): AB = 5, BC = 5, AC = 10, and 5 + 5 = 10, so collinear.

1.1.8 Points on a circle centred at the origin

What is it? A circle with centre O(0, 0) and radius r is the set of points whose distance from O equals exactly r. So a point (x, y) lies ON the circle if $x^2 + y^2 = r^2$, INSIDE if $x^2 + y^2$ is less than r^2, and OUTSIDE if it is greater. This is just the distance-from-origin shortcut applied with an equality/inequality. **How do I recognise it?** 'Show that these points lie on a circle with centre the origin - find the radius' or 'is the point inside, on, or outside the circle?'. **Tiny example.** (1, -8), (-4, 7), (-7, -4): squares give 65, 65, 65 - all equal, so all three lie on the circle $x^2 + y^2 = 65$, radius √65.

1.1.9 Exam tips for this chapter

1. Before any distance computation, CHECK whether a coordinate is shared - the parallel-to-axes shortcut saves time and avoids errors.
2. Keep the ordered-pair order sacred. (x, y) and (y, x) coincide only when x = y.

3. Squaring removes minus signs, so $(x_2 - x_1)^2 = (x_1 - x_2)^2$. Do not waste time worrying about subtraction order.
4. Memorise Pythagorean triples; they turn ugly roots into integers.
5. For 'inside/on/outside circle' questions, compare SQUARES of distances - never square roots.
6. After finding a reflection, quickly verify one preserved distance.
7. In multi-part geometry questions, draw a rough sketch even if the question says 'without plotting' - your sketch guides the algebra; the algebra is what you present.

1.2 Question Types / Concept Map for Chapter 1

Almost every coordinate question in exams falls into one of the following types. Use this map to label what you are looking at before solving - half the battle is knowing WHICH type of fight it is.

Type	Name	How to recognise it in the question	Core method
T1	Direct identification	Given coordinates, find quadrant / axis / sign pattern	Sign rules of the four quadrants
T2	Plotting and reading	Mark points, read positions from a room/map figure	Ordered pair = walk x then y
T3	Axis-parallel distance	Two points share one coordinate	|dx| or |dy| shortcut
T4	Distance formula	Two points, both coordinates differ	$\sqrt{(dx^2 + dy^2)}$
T5	Reverse / missing value	Distance given, one coordinate unknown (find k)	Set up equation, solve, check BOTH roots
T6	Triangle / quadrilateral classification	Show isosceles / right / square / rhombus / parallelogram	Compute all sides (and diagonals), compare
T7	Reflection reasoning	Image of a point in x-axis / y-axis / both	Flip the relevant sign; verify preserved distance
T8	Midpoint / trisection	'M is the midpoint', 'points of trisection', 'bisects'	Average coordinates; reverse-average for missing endpoint
T9	Collinearity test	'On the same line?', 'without plotting'	Largest distance = sum of other two (exact values)
T10	Circle membership	'lie on circle with centre origin', inside/outside check	Compare $x^2 + y^2$ with r^2
T11	Applied grid problems	City streets, screen pixels, room maps	Model positions as coordinates, then use T3/T4
T12	Perimeter / area from coordinates	Vertices given, find perimeter or area	Distance formula on each side; 1/2 base x height for right triangles
T13	Conceptual trap questions	Order of pair, meaning of coordinates, statements to justify	Reason from definitions, not memory

Type	Name	How to recognise it in the question	Core method
T14	Multi-concept HOTS	Combines distance + midpoint + circle or reflection chains	Break into labelled mini-steps, verify at each junction

1.3 Solved Practice Set (20 Questions)

Har question apne BOX me hai, uske turant baad uska boxed step-by-step solution. Pehle question ko khud solve karne ki koshish karo - jab atak jao tabhi solution dekho. Har step ke saath "Why this step?" diya hai, aur end me Hinglish note jo concept ko dil me basa dega.

QUESTION 1 | Basic | Quadrant sign rules (T1)

Q1. Name the quadrant (or axis) in which each point lies, and state the sign pattern: A(7, -9), B(-6, 2), C(-8, -11), D(9, 4), E(0, -3), F(-12, 0).

SOLUTION — step by step

Step 1. Given: six points with their coordinates. Needed: location of each point.

Step 2. Recall the sign pattern: Quadrant I (+, +), Quadrant II (-, +), Quadrant III (-, -), Quadrant IV (+, -). Points with x = 0 lie on the y-axis; points with y = 0 lie on the x-axis.

Step 3. A(7, -9): x is positive, y is negative, so A is in Quadrant IV with signs (+, -).

Step 4. B(-6, 2): x negative, y positive, so Quadrant II with signs (-, +).

Step 5. C(-8, -11): both negative, so Quadrant III with signs (-, -).

Step 6. D(9, 4): both positive, so Quadrant I with signs (+, +).

Step 7. E(0, -3): x-coordinate is 0, so E lies on the y-axis, 3 units below the origin, in no quadrant.

Step 8. F(-12, 0): y-coordinate is 0, so F lies on the x-axis, 12 units left of the origin.

Answer: A in Q IV (+, -); B in Q II (-, +); C in Q III (-, -); D in Q I (+, +); E on the y-axis below O; F on the x-axis left of O.

Hinglish: *Beta, yahan sirf signs dekhne hain. Pehle number jo hai wo x ki duniya hai - positive matlab right side, negative matlab left. Doosra number y hai - positive matlab upar, negative matlab neeche. A mein x positive, y negative, matlab right aur neeche = Quadrant IV. Bas ye pattern yaad rakho: I mein dono plus, II mein x minus y plus, III mein dono minus, IV mein x plus y minus. Aur agar koi coordinate 0 hai toh point axis par baitha hai, quadrant mein nahi.*

QUESTION 2 | Basic | Axis-parallel distances (T3)

Q2. Find the distance between (i) A(2, 0) and B(9, 0), (ii) C(0, -4) and D(0, 6), (iii) E(-5, 3) and F(4, 3).

SOLUTION — step by step

Step 1. Given: three pairs of points. Needed: distances.

Step 2. Pair (i): both points lie on the x-axis (y = 0 for both), so use the horizontal shortcut |x2 - x1| = |9 - 2| = 7.

Step 3. Pair (ii): both on the y-axis, so distance = |y2 - y1| = |6 - (-4)| = |6 + 4| = 10.

Step 4. Pair (iii): both have y = 3, a horizontal segment: distance = |4 - (-5)| = |4 + 5| = 9.

Step 5. In every case the shared coordinate confirms the segment is parallel to one axis, so the single-difference shortcut is valid - no square roots needed.

» **Why this step?** Checking the shared coordinate FIRST is the habit that separates fast, accurate students from slow ones.

Answer: (i) 7 units (ii) 10 units (iii) 9 units

Hinglish: *Jab do points mein se koi ek coordinate same ho, toh seedha difference lo - formula ki zaroorat hi nahi. (ii) mein dhyan do: 6 - (-4) = 6 + 4 = 10. Minus ke saath minus mila toh plus ban gaya. Ye sign wala jhatka sabse common galti hai, isliye maine absolute value ke andar rakha hai. (iii) bhi same khel hai: 4 - (-5) = 9.*

QUESTION 3 | Basic | Perimeter from coordinates (T12/T3)

Q3. The corners of a rectangular room are O(0, 0), A(8, 0), B(8, 5) and C(0, 5). Find the perimeter and state which walls are parallel to which axes.

SOLUTION — step by step

Step 1. Given: four corners in order. Needed: perimeter and axis directions of walls.

Step 2. OA joins (0, 0) to (8, 0): both y-values are 0, so OA is horizontal with length |8 - 0| = 8.

Step 3. AB joins (8, 0) to (8, 5): both x-values are 8, so AB is vertical with length |5 - 0| = 5.

Step 4. BC joins (8, 5) to (0, 5): horizontal, length |8 - 0| = 8. CO joins (0, 5) to (0, 0): vertical, length 5.

Step 5. Perimeter = 8 + 5 + 8 + 5 = 26 units.

Step 6. OA and BC run along the x-direction (parallel to the x-axis); AB and CO run along the y-direction (parallel to the y-axis).

Answer: Perimeter = 26 units; OA and BC are parallel to the x-axis, AB and CO are parallel to the y-axis.

Hinglish: *Rectangle ke opposite sides barabar hoti hain, isliye do sides hi kaafi hain: ek horizontal 8, ek vertical 5. Perimeter = 2(8 + 5) = 26. Dekho, coordinates same hone se hi pata chal gaya ki kaun si deewar kis axis ke parallel hai - y same hai toh horizontal wall, x same hai toh vertical wall. Ye floor-plan wale questions mein bahut kaam aata hai.*

QUESTION 4 | Basic | Distance from origin (T4)

Q4. Find the distance of the point (6, 8) from the origin. Which Pythagorean triple is hiding inside this question, and how can the triple help you answer faster next time?

SOLUTION — step by step

Step 1. Given: point (6, 8). Needed: distance from O(0, 0).

Step 2. Distance from origin = $\sqrt{(x^2 + y^2)} = \sqrt{(6^2 + 8^2)}$.

Step 3. Compute the squares: 36 + 64 = 100.

Step 4. Take the root: $\sqrt{100} = 10$.

Step 5. The hidden triple is 3-4-5 scaled by 2: 6-8-10. If you spot (6, 8), you can jump straight to 10 without squaring.

Step 6. So the distance is 10 units.

» **Why this step?** Spotting the triple is a safe shortcut only when BOTH numbers of a known triple appear exactly; otherwise use the formula.

Answer: 10 units; the triple 6-8-10 (which is 2 x (3-4-5)).

Hinglish: *Origin se distance ka short formula: $\sqrt{(x^2 + y^2)}$. 6 aur 8 ko dekhte hi 3-4-5 wala triple yaad karo - dono numbers 2 se multiply ho rahe hain, isliye answer 5 ka 2 guna = 10. Triples ratt lo: 3-4-5, 5-12-13, 8-15-17, 7-24-25. Par dhyan rahe - triple tabhi use karo jab numbers EXACT match karein, warna seedha formula lagao.*

QUESTION 5 | Basic | Sign reasoning (T1/T13)

Q5. A point has coordinates (a, -a) where a is a non-zero real number. (i) In which quadrant does it lie when a is positive? (ii) In which quadrant when a is negative? (iii) What happens at a = 0? (iv) What is always true about the distances of this point from the two axes?

SOLUTION — step by step

Step 1. Given: the family of points (a, -a). Needed: quadrant analysis by sign of a.

Step 2. Case a positive: x = a is positive, y = -a is negative, so the point is in Quadrant IV.

Step 3. Case a negative: writing a = -m with m positive gives x = -m (negative) and y = m (positive), so the point is in Quadrant II.

Step 4. Case a = 0: the point is (0, 0), the origin itself - on both axes, in no quadrant.

Step 5. For every a, the distance from the y-axis is |a| and the distance from the x-axis is |-a| = |a|. These are always equal.

Step 6. So every point of this family stays exactly equally far from both axes - the family slides along the two lines y = -x.

Answer: (i) Quadrant IV when a > 0; (ii) Quadrant II when a < 0; (iii) the origin when a = 0; (iv) distances from both axes are always equal (each equals |a|).

Hinglish: *Ye question sign-logic ka gym hai. a positive toh x positive, y negative = Quadrant IV. a negative karo toh dono ke sign flip ho jaate hain: x negative, y positive = Quadrant II. a = 0 par origin. Aur sundar observation: is point ki dono axis se doori hamesha barabar hai |a|, kyunki x aur y ka size same hai, sirf sign ulta hai. Aage Ch 2 mein y = -x line padhoge, ye usi ka preview hai.*

QUESTION 6 | Moderate | Distance formula (T4)

Q6. Find the distance between A(1, 2) and B(4, 6) using (i) the helper-triangle method and (ii) the direct formula. Confirm both give the same value.

SOLUTION — step by step

Step 1. Given: A(1, 2), B(4, 6). Needed: AB by two methods.

Step 2. Helper triangle: the horizontal leg is $|4 - 1| = 3$ and the vertical leg is $|6 - 2| = 4$.

Step 3. By Pythagoras, $AB = \sqrt{(3^2 + 4^2)} = \sqrt{(9 + 16)} = \sqrt{25} = 5$.

Step 4. Direct formula: $AB = \sqrt{((4 - 1)^2 + (6 - 2)^2)} = \sqrt{(3^2 + 4^2)} = 5$.

Step 5. Both methods agree: AB = 5 units. The helper triangle simply explains where the formula comes from.

» **Why this step?** Showing both methods builds the habit of verifying with geometry - on hard questions this catches sign slips instantly.

Answer: AB = 5 units by both methods.

Hinglish: *Yahi chapter ka hero question hai. Helper triangle mein humne sirf differences liye - 3 aur 4 - aur wo 3-4-5 triple nikla, isliye answer 5. Formula bhi wahi karta hai, bas ek line mein. Dono ka jawab same aana hi chahiye kyunki formula Pythagoras ka hi shortcut hai. Aage se seedha formula use karo, par agar kabhi doubt ho, helper triangle bana lo.*

QUESTION 7 | Moderate | Collinearity test (T9)

Q7. Without plotting, check whether M(0, 0), A(3, 4) and G(6, 8) lie on the same straight line. State the reasoning principle you used.

SOLUTION — step by step

Step 1. Given: three points. Needed: collinearity verdict without plotting.

Step 2. Compute $OM = \sqrt{(3^2 + 4^2)} = \sqrt{25} = 5$.

Step 3. Compute $AG = \sqrt{((6 - 3)^2 + (8 - 4)^2)} = \sqrt{(9 + 16)} = \sqrt{25} = 5$.

Step 4. Compute $MG = \sqrt{(6^2 + 8^2)} = \sqrt{(36 + 64)} = \sqrt{100} = 10$.

Step 5. Test: the largest distance (10) equals the sum of the other two (5 + 5 = 10).

Step 6. Since MG = OM + AG exactly, the point A sits on the straight path from M to G; the three points are collinear.

Step 7. Principle used: on a straight line, the whole equals the sum of the parts (equality case of the triangle inequality). Any detour would make the total strictly larger.

Answer: Yes - OM + AG = MG (5 + 5 = 10), so M, A, G are collinear.

Hinglish: *Ye 'triangle inequality equality' ka test hai - bina graph banaye line check karne ka tareeka. Teen distances nikaalo: 5, 5, 10. Agar sabse bada = baaki dono ka sum, toh teeno ek hi line par hain. Agar sum bada aaye (strictly), toh beech ka point line se hat ke baitha hai. Yahan 5 + 5 = 10 exact match - collinear. Ek baat yaad rakho: values EXACT rakho, round mat karo, warna test galat ho sakta hai.*

QUESTION 8 | Moderate | Triangle classification (T6)

Q8. The vertices of a triangle are A(1, 2), B(5, 2) and C(3, 6). Find all three side lengths and classify the triangle. Is it equilateral? Is it right-angled?

SOLUTION — step by step

Step 1. Given: three vertices. Needed: side lengths and classification.

Step 2. AB: same y-coordinate (2), so AB = |5 - 1| = 4.

Step 3. AC = $\sqrt{((3 - 1)^2 + (6 - 2)^2)} = \sqrt{(4 + 16)} = \sqrt{20} = 2\sqrt{5}$.

Step 4. BC = $\sqrt{((3 - 5)^2 + (6 - 2)^2)} = \sqrt{(4 + 16)} = \sqrt{20} = 2\sqrt{5}$.

Step 5. Two sides equal (AC = BC), one different: the triangle is isosceles, not equilateral.

Step 6. Right-angle check by the converse of Pythagoras: is the square of the longest side equal to the sum of squares of the other two? Longest is $2\sqrt{5}$, its square 20. Others: $4^2 = 16$ and $(2\sqrt{5})^2 = 20$, and 16 + 20 = 36 which is not 20.

Step 7. Not right-angled. In fact $4 + 2\sqrt{5} + 2\sqrt{5}$ is the perimeter = $4 + 4\sqrt{5}$.

Answer: AB = 4, AC = BC = $2\sqrt{5}$; the triangle is isosceles; it is neither equilateral nor right-angled.

Hinglish: *Side nikaalne ke baad classification ka recipe: (1) do barabar? isosceles. (2) teen barabar? equilateral. (3) Pythagoras satisfy? right-angled. Yahan AC aur BC dono $2\sqrt{5}$ nikle - isosceles pakka. Right angle ka test: sabse lambi side ka square = baaki dono ke squares ka sum? 20 vs 36 - fail. Toh na equilateral, na right. Simple hai, bas teeno tests order mein lagao.*

QUESTION 9 | Moderate | Missing value, two roots (T5)

Q9. The distance between the points (k, 3) and (4, 3) is 7. Find all possible values of k. How many points satisfy the condition?

SOLUTION — step by step

Step 1. Given: two points sharing y = 3, distance 7. Needed: k.

Step 2. Both points have y-coordinate 3, so the segment is horizontal and distance = |x2 - x1|.

Step 3. So |4 - k| = 7. An absolute-value equation splits into two cases.

Step 4. Case 1: 4 - k = 7, giving k = -3. Case 2: 4 - k = -7, giving k = 11.

Step 5. Verification: from (4, 3) to (-3, 3): |4 - (-3)| = 7. From (4, 3) to (11, 3): |11 - 4| = 7. Both work.

Step 6. So there are TWO points, (-3, 3) and (11, 3), one on each side of (4, 3).

» Why this step? The absolute value MUST open into two cases. Dropping one case is the single most common error in 'find k' questions.

Answer: k = -3 or k = 11; there are two such points.

Hinglish: *Yahi wo question hai jahan aadhe marks ud jaate hain! |4 - k| = 7 ke DO solutions hote hain: point (4, 3) ke left mein bhi ek aur right mein bhi ek. k = -3 aur k = 11. Agar tumne sirf ek nikala, toh samajh lo absolute value ke do case khulane bhool gaye. Verification bhi kar li - dono se distance 7 hi aati hai.*

QUESTION 10 | Moderate | Missing y-value on a circle (T5/T10)

Q10. The point (3, y) lies at distance 5 from the origin. Find all possible values of y. Where do the two possible points sit relative to each other?

SOLUTION — step by step

Step 1. Given: distance from O equals 5. Needed: y.

Step 2. Distance from origin: $\sqrt{(3^2 + y^2)} = 5$.

Step 3. Square both sides (both sides are non-negative, so squaring is safe): $9 + y^2 = 25$.

Step 4. So $y^2 = 16$, and taking square roots gives y = 4 or y = -4.

Step 5. Verification: (3, 4) gives $\sqrt{(9 + 16)} = 5$, and (3, -4) also gives 5. Both are genuine solutions.

Step 6. The two points (3, 4) and (3, -4) are reflections of each other in the x-axis - same x, opposite y - and both lie on the circle of radius 5.

Answer: y = 4 or y = -4; the two points are mirror images in the x-axis, both on the circle of radius 5.

Hinglish: *Yahan bhi do answers aayenge, aur wajah samjho: circle ke ek vertical line par do points hote hain - ek upar, ek neeche. y = 4 aur y = -4. Dono verify kiye, dono sahi. Aur ye dono points x-axis ke reflection partners hain. Ye pattern (quadratic se do roots) Chapter 1 ka signature hai - jab bhi circle ya distance se equation bane, do answers ki ummeed rakho.*

QUESTION 11 | Difficult | Perimeter + classification (T6/T12)

Q11. Find the perimeter of the triangle with vertices A(2, -2), B(-2, 1) and C(5, 2). Along the way, show that the triangle is isosceles.

SOLUTION — step by step

Step 1. Given: three vertices. Needed: all sides, isosceles check, perimeter.

Step 2. AB: differences are (-2 - 2) = -4 and (1 - (-2)) = 3. AB = √(16 + 9) = √25 = 5.

Step 3. BC: differences are (5 - (-2)) = 7 and (2 - 1) = 1. BC = √(49 + 1) = √50 = 5√2.

Step 4. CA: differences are (2 - 5) = -3 and (-2 - 2) = -4. CA = √(9 + 16) = √25 = 5.

Step 5. AB = CA = 5, so the triangle is isosceles with apex at A.

Step 6. Perimeter = AB + BC + CA = 5 + 5√2 + 5 = 10 + 5√2 units (approximately 17.07).

Step 7. Keep the exact form 10 + 5√2 in the answer; convert to decimals only if the question asks.

Answer: Perimeter = 10 + 5√2 units; the triangle is isosceles (AB = CA = 5).

Hinglish: *Do sides 5 nikli aur ek 5√2 - isosceles confirm. Perimeter mein exact value rakho: 10 + 5√2. Exam mein log √50 ko 7.07 leke ghumaa dete hain aur final answer gandaa ho jata hai. Rule yaad rakho: jab tak decimal manga nahi, root form hi rakho. Aur ek observation: (4, 3) wala 3-4-5 triple do baar mila - coordinate questions mein triples baar-baar ghuumte hain, unhe pakadna speed deta hai.*

QUESTION 12 | Difficult | Reverse midpoint (T8)

Q12. M(-1, 3) is the midpoint of the segment joining A(2, -4) and B(x, y). Find B, and verify your answer by checking that MA = MB.

SOLUTION — step by step

Step 1. Given: midpoint and one endpoint. Needed: the other endpoint.

Step 2. Midpoint equations: (-1) = (2 + x)/2 and 3 = (-4 + y)/2.

Step 3. Solve the first: multiply by 2: -2 = 2 + x, so x = -4.

Step 4. Solve the second: 6 = -4 + y, so y = 10.

Step 5. So B = (-4, 10).

Step 6. Verify MA: differences (2 - (-1)) = 3 and (-4 - 3) = -7, MA = √(9 + 49) = √58.

Step 7. Verify MB: differences (-4 - (-1)) = -3 and (10 - 3) = 7, MB = √(9 + 49) = √58. Equal, so B is correct.

» Why this step? Reverse midpoint = undo the average: multiply by 2, then subtract the known endpoint. Verification with equal distances is cheap and catches slips.

Answer: B = (-4, 10), verified since MA = MB = √58.

Hinglish: *Midpoint wala equation ulta chalana hai: average diya hai, original nikalna hai. 2 se multiply karo, phir known endpoint minus karo. x ke liye: (-1)(2) - 2 = -4. y ke liye: (3)(2) - (-4) = 10. Verification zaroor karo - MA aur MB dono √58 aaye, matlab M sach mein beech mein baitha hai. Bina verification ke answer likhna adhoora hai.*

QUESTION 13 | Difficult | Equidistant point on an axis (T5/T8)

Q13. Find the point on the y-axis that is equidistant from P(-2, 0) and Q(6, 4). Verify your answer.

SOLUTION — step by step

Step 1. Given: unknown point R on the y-axis. Needed: R equidistant from P and Q.

Step 2. On the y-axis the point has the form R(0, y) - fix the zero first.

Step 3. $RP^2 = (0 - (-2))^2 + (y - 0)^2 = 4 + y^2$.

Step 4. $RQ^2 = (0 - 6)^2 + (y - 4)^2 = 36 + (y - 4)^2$.

Step 5. Set them equal: $4 + y^2 = 36 + y^2 - 8y + 16$.

Step 6. Cancel y^2 from both sides (this is why the equation stays linear and easy): $4 = 52 - 8y$.

Step 7. So $8y = 48$, giving $y = 6$. Hence R = (0, 6).

Step 8. Verify: $RP = \sqrt{(4 + 36)} = \sqrt{40} = 2\sqrt{10}$. $RQ = \sqrt{(36 + (6 - 4)^2)} = \sqrt{(36 + 4)} = \sqrt{40} = 2\sqrt{10}$. Equal, so R is correct.

» **Why this step?** Comparing squared distances avoids square roots entirely; and y^2 cancels, leaving a linear equation.

Answer: R = (0, 6), verified since RP = RQ = 2√10.

Hinglish: *Y-axis par point = (0, y) - pehle yeh lock karo. Phir dono distances ke SQUARES barabar rakho, root ka jhanjhat khatam. Sabse sundar cheez: y^2 dono taraf cancel ho gaya aur seedha linear equation mil gaya, y = 6. Verify karo: dono distances 2√10. Aisa equidistant point technically perpendicular bisector par baithta hai - yahan humne uska sirf y-axis wala intersection nikala. 2√10 ko 6.32 likhne ki zaroorat nahi.*

QUESTION 14 | High | Missing rectangle vertex (T6/T8)

Q14. Three vertices of a rectangle are A(1, 2), B(5, 2) and C(5, 7). Find the fourth vertex D, and verify the rectangle property that its diagonals are equal.

SOLUTION — step by step

Step 1. Given: three consecutive vertices. Needed: D and the diagonal check.

Step 2. AB is horizontal (y = 2 for both) with length 4; BC is vertical (x = 5 for both) with length 5.

Step 3. In a rectangle the fourth side DA must be vertical (above A) and CD horizontal - so D must sit directly above A at the same height as C.

Step 4. Directly above A(1, 2) means x = 1; at the same height as C means y = 7. So D = (1, 7).

Step 5. Verify with side lengths: DA from (1, 7) to (1, 2) has length 5, CD from (5, 7) to (1, 7) has length 4 - matching the opposite sides.

Step 6. Diagonal AC = $\sqrt{((5 - 1)^2 + (7 - 2)^2)} = \sqrt{(16 + 25)} = \sqrt{41}$.

Step 7. Diagonal BD = $\sqrt{((1 - 5)^2 + (7 - 2)^2)} = \sqrt{(16 + 25)} = \sqrt{41}$. Equal, so the rectangle property holds.

Answer: D = (1, 7); diagonals AC = BD = √41, confirming the rectangle.

Hinglish: *Rectangle ka chautha vertex pattern se milta hai: A ke seedha upar, C ke level par. x = 1 (A jaisa), y = 7 (C jaisa). Phir property check - diagonals barabar? Dono √41. Yahan √41 ko simplify nahi kar sakte (41 prime hai), toh root form hi final hai. Notice karo: har rectangle mein diagonals barabar hote hain - ye test humne parallelogram question (Q23) mein bhi use kiya tha, wahan fail hua tha, yahan pass.*

QUESTION 15 | High | Area of right triangle from coordinates (T12)

Q15. A triangle has vertices O(0, 0), A(8, 0) and B(0, 6). Show that it is right-angled, find its hypotenuse, and compute its area and perimeter.

SOLUTION — step by step

Step 1. Given: three vertices, two of them on the axes. Needed: right angle, hypotenuse, area, perimeter.

Step 2. OA lies along the x-axis: length 8. OB lies along the y-axis: length 6.

Step 3. The x-axis and y-axis are perpendicular, and OA, OB lie along them, so the angle at O is 90 degrees.

Step 4. Hypotenuse AB = $\sqrt{(8^2 + 6^2)} = \sqrt{(64 + 36)} = \sqrt{100} = 10$ (the 6-8-10 triple).

Step 5. Area = (1/2) x OA x OB = (1/2) x 8 x 6 = 24 square units.

Step 6. Perimeter = 8 + 6 + 10 = 24 units. A charming coincidence: area 24 and perimeter 24.

Answer: Right angle at O; hypotenuse AB = 10; area = 24 sq units; perimeter = 24 units.

Hinglish: *Jab triangle ke do sides axes par baithe hon, right angle free mein milta hai - axis parallel hi perpendicular hote hain. Phir 8-6-10 triple se hypotenuse 10. Area = 24, perimeter bhi 24 - dono number same! Aisi coincidences questions mein jaan-bujh ke rakhi jaati hain, toh apne practice questions mein bhi aisi numbers chuno toh checking easy hoti hai.*

QUESTION 16 | High | Square from its diagonal (T14)

Q16. Two opposite corners of a square are P(-3, 2) and R(5, -4). Find the length of the diagonal, the side of the square, its area, and the coordinates of the other two corners.

SOLUTION — step by step

Step 1. Given: diagonal endpoints of a square. Needed: diagonal, side, area, remaining vertices.

Step 2. Diagonal PR = $\sqrt{((5 - (-3))^2 + (-4 - 2)^2)} = \sqrt{(8^2 + (-6)^2)} = \sqrt{(64 + 36)} = \sqrt{100} = 10$.

Step 3. For a square of side s, the diagonal is $s\sqrt{2}$, so $s = 10/\sqrt{2} = 5\sqrt{2}$.

Step 4. Area = $s^2 = (5\sqrt{2})^2 = 50$ square units. (Shortcut: area = diagonal^2/2 = 100/2 = 50.)

Step 5. Centre of the square = midpoint of PR = ((-3 + 5)/2, (2 - 4)/2) = (1, -1).

Step 6. Half-diagonal vector from centre to R is (5 - 1, -4 - (-1)) = (4, -3). To build the other diagonal, swap the components and flip one sign: (3, 4).

Step 7. Other corners: centre + (3, 4) = (4, 3) and centre - (3, 4) = (-2, -5).

Step 8. Verify: Q(4, 3) to P(-3, 2): $\sqrt{(7^2 + 1)} = \sqrt{50} = 5\sqrt{2}$ (a side). Q to R(5, -4): $\sqrt{(1 + 49)} = \sqrt{50}$ (a side). Q to S(-2, -5): $\sqrt{(36 + 64)} = 10$ (the other diagonal). All consistent.

» **Why this step?** The swap-and-flip move rotates the half-diagonal by 90 degrees - it is just Pythagoras thinking in vector form, and it is fully verifiable with the distance formula.

Answer: Diagonal = 10; side = $5\sqrt{2}$; area = 50 sq units; other corners Q(4, 3) and S(-2, -5).

Hinglish: *Square ka diagonal diya hai toh side = diagonal/$\sqrt{2}$ = $5\sqrt{2}$, area = 50. Baaki do corners nikalne ka tareeka: pehle centre (midpoint) lo - (1, -1). Half-diagonal (4, -3) hai. Uske components SWAP karo aur ek sign flip karo: (3, 4) ban jayega - ye 90 degree ghuma hua half-diagonal hai. Centre mein plus-minus lagao: (4, 3) aur (-2, -5). Phir verification: sab sides $5\sqrt{2}$, doosra diagonal 10. Ye wala question HOTS-zone ka gateway hai - rotation ka idea samajh gaye toh aage Olympiad-level coordinate proofs easy lagenge.*

QUESTION 17 | High | Applied screen/circle problem (T11/T10)

Q17. A computer screen is a rectangle 800 pixels wide and 600 pixels high, with the origin at the bottom-left corner. A circular icon of radius 80 has centre A(100, 150); another of radius 100 has centre B(250, 230). (i) Does any part of either icon stick out of the screen? (ii) Do the two icons overlap each other?

SOLUTION — step by step

Step 1. Given: screen 800 x 600, two circles with centres and radii. Needed: boundary check and overlap check.

Step 2. Icon A: leftmost x = 100 - 80 = 20 (inside, positive), rightmost x = 100 + 80 = 180 (inside, below 800), lowest y = 150 - 80 = 70 (inside), highest y = 150 + 80 = 230 (inside, below 600). Icon A is fully on the screen.

Step 3. Icon B: leftmost x = 250 - 100 = 150, rightmost = 250 + 100 = 350, lowest y = 230 - 100 = 130, highest y = 330. All within 0 to 800 and 0 to 600. Fully on screen too.

Step 4. Overlap test for two circles: compare the centre distance d with r1 + r2 and |r1 - r2|.

Step 5. Centre distance AB = $\sqrt{((250 - 100)^2 + (230 - 150)^2)} = \sqrt{(150^2 + 80^2)} = \sqrt{(22500 + 6400)} = \sqrt{28900} = 170$ (the 8-15-17 triple scaled by 10).

Step 6. Now r1 + r2 = 180 and |r1 - r2| = 20. Since 20 is less than 170 and 170 is less than 180, the circles intersect in two points - they overlap partially.

Step 7. So: no icon leaves the screen, and the icons DO overlap.

Answer: (i) Both icons lie fully inside the screen. (ii) Yes, the icons overlap: AB = 170 lies strictly between |100 - 80| = 20 and 80 + 100 = 180.

Hinglish: *Screen check: har icon ka extreme left/right/top/bottom compute karo aur 0-800, 0-600 range se compare karo. Dono andar hain. Overlap ka test yaad rakho: centres ki doori vs radii ka sum. Distance 170 nikla (8-15-17 triple x 10!). 170 < 180 isliye overlap hota hai, aur 170 > 20 isliye poora ek doosre ke andar nahi hai - partial overlap. Ye 2-line test high-school geometry ka swiss knife hai.*

QUESTION 18 | HOTS | Circumcentre discovery (T10/T14)

Q18. For the right triangle with vertices A(0, 0), B(8, 0) and C(8, 6), find the point that is equidistant from all three vertices. Where does it sit, and what pattern do you notice?

SOLUTION — step by step

Step 1. Given: right triangle with legs on the axes... (adapted: right angle at B since AB is horizontal and BC is vertical). Needed: the common-equidistance point.

Step 2. AB = 8 (horizontal), BC = 6 (vertical), so the right angle is at B(8, 0), and AC is the hypotenuse.

Step 3. AC = $\sqrt{((8 - 0)^2 + (6 - 0)^2)} = \sqrt{(64 + 36)} = \sqrt{100} = 10$.

Step 4. Claim: the midpoint of the hypotenuse is equidistant from all three vertices. Midpoint M of AC = ((0 + 8)/2, (0 + 6)/2) = (4, 3).

Step 5. Check MA = $\sqrt{(4^2 + 3^2)} = 5$. MC = $\sqrt{((8 - 4)^2 + (6 - 3)^2)} = \sqrt{(16 + 9)} = 5$.

Step 6. MB = $\sqrt{((8 - 4)^2 + (0 - 3)^2)} = \sqrt{(16 + 9)} = 5$. All three distances are 5 - verified.

Step 7. Pattern: in a right triangle, the midpoint of the hypotenuse is equidistant from all three vertices (it is the centre of the circle passing through the three corners). This is the coordinate-proof version of a famous theorem.

Answer: M(4, 3), the midpoint of the hypotenuse, is equidistant (distance 5) from all three vertices.

Hinglish: *Ye question ek theorem khud discover karwa raha hai! Right angle B par hai (horizontal + vertical legs), hypotenuse AC = 10. Uska midpoint (4, 3) lo - teeno vertices se distance 5, 5, 5. Matlab: right triangle mein hypotenuse ka midpoint hi wo magical point hai jo teeno corners se barabar door baithta hai (circumcentre). 3-4-5 triple yahan bhi kaam kar gaya. Ye result yaad rakho - contest maths mein baar-baar kaam aayega.*

QUESTION 19 | HOTS | Box method for area (T12/T14)

Q19. A triangle has vertices A(0, 0), B(1, 2) and C(3, 1). Show that it is isosceles, and find its area using the box (enclosing rectangle) method.

SOLUTION — step by step

Step 1. Given: three lattice points. Needed: isosceles proof and area.

Step 2. AB = $\sqrt{(1^2 + 2^2)} = \sqrt{5}$. BC = $\sqrt{((3 - 1)^2 + (1 - 2)^2)} = \sqrt{(4 + 1)} = \sqrt{5}$. AC = $\sqrt{(3^2 + 1^2)} = \sqrt{10}$.

Step 3. AB = BC = $\sqrt{5}$, so the triangle is isosceles (apex at B).

Step 4. Box method: enclose the triangle in the smallest axis-aligned rectangle: corners (0, 0), (3, 0), (3, 2), (0, 2). Box area = 3 x 2 = 6.

Step 5. The box minus the triangle leaves three right triangles: T1 below AB with legs 1 and 2 (area 1), T2 right of BC with legs 2 and 1 (area 1), T3 above AC with legs 3 and 1 (area 1.5).

Step 6. Area of ABC = 6 - 1 - 1 - 1.5 = 2.5 square units.

Step 7. Cross-check by another triangle with same side data is not needed; instead sanity-check magnitude: the triangle fits inside a 3 x 2 box, so 2.5 is plausible.

» Why this step? When the triangle is not right-angled and the base-height pair is not obvious, the box method computes area using only right triangles - fully within Chapter 1 tools.

Answer: Isosceles (AB = BC = $\sqrt{5}$); area = 2.5 square units via the box method.

Hinglish: *Jab triangle tedha ho aur base-height na dikhe, box method use karo: triangle ko chhote rectangle mein band karo, rectangle ka area lo, aur charo taraf ke right triangles ghata do. 6 - 1 - 1 - 1.5 = 2.5. Ye technique squares ke root wale triangles ka area nikalne ka desi jugaad hai - bina kisi naye formula ke. Aur dekho, AB = BC = $\sqrt{5}$ se isosceles bhi proof ho gaya. Do kaam ek question mein.*

QUESTION 20 | HOTS | Grid-walk vs direct distance (T11/T4)

Q20. A delivery robot moves on a grid of unit squares. It must travel from station S(1, 1) to station T(7, 4). (i) What is the length of the shortest path using only horizontal and vertical moves? (ii) What is the straight-line distance? (iii) A second robot flies directly. How much distance does it save compared to the ground robot?

SOLUTION — step by step

Step 1. Given: grid unit 1, S(1, 1) to T(7, 4). Needed: street length, direct length, saving.

Step 2. Horizontal gap = 7 - 1 = 6 units; vertical gap = 4 - 1 = 3 units.

Step 3. Ground robot (only axis moves): 6 + 3 = 9 units. Every such path has exactly this length - no order of moves can beat it.

Step 4. Flying robot (straight line): $\sqrt{(6^2 + 3^2)} = \sqrt{(36 + 9)} = \sqrt{45} = 3\sqrt{5}$ (approximately 6.71 units).

Step 5. Saving = $9 - 3\sqrt{5}$ (exactly), approximately 9 - 6.71 = 2.29 units.

Step 6. Keep the exact answer 9 - 3√5 in the final statement; the decimal is only for intuition.

Answer: (i) 9 units (ii) 3√5 units (approx 6.71) (iii) saving 9 - 3√5 approx 2.29 units.

Hinglish: *Grid-walk ka funda: horizontal steps + vertical steps hamesha fix hote hain (yahan 6 + 3 = 9), chahe kisi bhi order mein chalo. Direct distance Pythagoras se 3√5. Saving 9 - 3√5 approx 2.3 units. Ye question Q26 (city auto) ka bhai hai - wahan 200 m blocks the, yahan 1 unit. Same skeleton, naye numbers - yahi 'question structure' samajhna original practice hai, ratna nahi.*

Chapter 2: Introduction to Linear Polynomials

From locating points we now move to describing relationships with expressions. The uploaded PDF builds the idea of a polynomial from daily life (boxes of pens, garden fencing, chess-club fees), defines degree, focuses on linear polynomials, shows them as input-output machines (functions), develops linear patterns and growth/decay, teaches how to fit the relationship y = ax + b from data, and finishes by plotting straight lines and reading the meaning of a (slope) and b (y-intercept). This book sharpens every one of those ideas with reverse questions, two-equation fittings, plan comparisons and pattern proofs.

2.1 Complete Concept Cheat Sheet (read this before any question)

2.1.1 Algebraic expressions: terms, coefficients, constants, variables

What is it? An algebraic expression combines numbers, variables (letter-numbers) and operation signs. In 4x + 5y + 3, the parts 4x, 5y and 3 are the **terms**; x and y are **variables** (also called letter-numbers); 4 and 5 are the **coefficients**; 3 is a **constant**. **Why coefficients matter:** the coefficient tells you how many copies of the variable you have, and it is the number that multiplies the variable - including its sign. In -7x the coefficient is -7, not 7.

When do I use this? Every 'identify the terms / coefficient / constant' question, and every modelling question where a real situation must become an expression (cost of fencing = rate x length, and so on). **Tiny example.** For the garden in the PDF: wire fence cost 100 x 2l = 200l, wooden fence 80 x 2w = 160w, seeds 50 x area = 50lw; total 200l + 160w + 50lw - a three-term expression in two variables.

> **Hidden coefficient trap**
> A term like x has coefficient 1 (not 0!), and -x has coefficient -1. A term like $5x^2y$ has coefficient 5. Missing numbers never mean missing coefficients.

2.1.2 Degree and the polynomial family

What is it? A polynomial in ONE variable is an expression made of that variable and its powers (whole-number powers). The **degree** is the highest power of the variable. Degree names: 1 = linear, 2 = quadratic, 3 = cubic, 0 = constant. Examples: 3z + 7 is linear (degree 1); $x^2 + 5x + 1$ is quadratic; $5y^3 + y^2 - 8$ is cubic; the number 8 alone is a constant polynomial (it can be written $8x^0$).

How do I find the degree quickly? Scan the powers of the variable and take the maximum. In $2x^4 - 3x^3 + 6x^2 - 2x + 7$ the powers are 4, 3, 2, 1, 0, so the degree is 4. **Sign-awareness:** the degree ignores the sign and size of coefficients - $-9x^5$ still has degree 5. **Tiny example.** -13 is a constant polynomial of degree 0; 6z + 1 is linear.

Common confusion: 'linear polynomial' (degree exactly 1, like 2x + 3) versus 'linear equation' (a statement like 2x + 3 = 64 that equates a linear polynomial to something). The polynomial is the expression; the equation is the claim that two expressions are equal.

2.1.3 Evaluation: the input-output machine

What is it? Substituting a value of the variable into a polynomial gives its value at that input. The PDF pictures this as a machine: feed x = 4 into the machine 2x + 3, out comes 11; feed x = -6, out comes -9. This

machine-view is the seed of the 'function' idea. **Golden rule of substitution:** write the substitution with brackets: 2(-6) + 3, never 2 -6 + 3. Brackets prevent the sign disaster.

Tiny example. For p(x) = 7x - 4: p(0) = -4, p(2) = 10, p(-3) = -25. **Reverse use:** given the output, find the input - that is solving a linear equation: 7x - 4 = 24 gives x = 4. Evaluation forward, equation backward - the same machine run both ways.

2.1.4 Linear equations from words

What is it? Translate a sentence into an equation with ONE unknown, solve, and answer the original question. Standard moves: 'one number is 10 more than the other' means x and x + 10; 'three times as many' means 3x; 'their sum is 64' means x + (x + 10) = 64.

Method (always the same skeleton). Step 1: define the unknown precisely ('let the smaller number be x'). Step 2: express every other quantity in terms of x. Step 3: write the balancing condition (sum, difference, total money). Step 4: solve. Step 5: ANSWER THE QUESTION ASKED - if the question asked for both numbers, report both; a bare x earns no full marks. **Sanity check:** substitute back into the words, not just the equation.

2.1.5 Linear patterns and the nth term

What is it? A linear pattern is a sequence whose consecutive terms differ by a CONSTANT. The tile pattern 1, 3, 5, 7, ... adds 2 tiles each stage, and its nth-stage count is 2n - 1. The constant difference becomes the coefficient of n; the constant term adjusts the start. **Finding the rule from the table:** if the difference is d, the rule looks like an = dn + c; find c by matching one known term (e.g. for 1, 3, 5: d = 2 and term 1 = 1 gives 1 = 2(1) + c, so c = -1).

Why the difference is the slope: on the graph of the rule, moving one stage right moves the value up by exactly d - the line rises by d per step. **Tiny example.** Matchstick squares in a row: 4, 7, 10, ... difference 3, rule 3n + 1. Stage 10 needs 31 matches. **Reverse use:** 'can the pattern ever equal 100?' means solving 3n + 1 = 100; here n = 33 - a whole number, so yes, stage 33 uses exactly 100 matches. If n had come out fractional, the answer would be NO - patterns only exist at whole stages.

2.1.6 Linear growth and linear decay

What is it? Growth: a quantity increases by a fixed amount over equal intervals - C(d) = 100 + 60d (journey cost). Decay: decreases by a fixed amount - h(t) = 3 - 0.5t (water level). The fixed change is the slope: positive for growth, negative for decay. **Structure:** value = starting value + (change per step) x (number of steps). On a graph, growth tilts up, decay tilts down.

How do I recognise it? Words like 'every month adds', 'decreases by 800 each year', 'reduces by 15 per day'. **Two standard questions:** forward (value after n steps - substitute) and backward (when does the value reach V? - solve). Decay also has the 'run out' question: set the expression to 0 (or the floor value) and solve. **Tiny example.** Phone bought at 10000, losing 800 per year: v = 10000 - 800t; after 3 years 7600; it touches 3600 when t = 8.

2.1.7 Fitting y = ax + b from two observations

What is it? Many real bills have the shape y = ax + b: a fixed part b plus a per-unit part a. Given two observations (x_1, y_1) and (x_2, y_2), substitute both into y = ax + b to get TWO equations, then eliminate b by subtraction: $a = (y_2 - y_1)/(x_2 - x_1)$. Then recover b from either observation: $b = y_1 - a\,x_1$.

Worked skeleton (the telecom example). 10 GB cost 350 and 20 GB cost 550. Equations: 350 = 10a + b and 550 = 20a + b. Subtract: 200 = 10a, so a = 20 (each extra GB costs Rs 20). Back-substitute: b = 350 - 200 = 150 (the fixed monthly fee). Answer: y = 20x + 150. **Meaning check:** a is the RATE per unit, b is the ENTRY fee - interpreting them is as important as computing them.

> **Subtraction is the whole trick**
> Subtracting the two equations kills b because b is the SAME in both. What remains is a clean one-step equation for a. If your two equations accidentally have different b values, you have copied the data wrongly - stop and re-read.

2.1.8 Graphs of lines: slope a and y-intercept b

What is it? Every equation y = ax + b draws a straight line. Plotting needs just two points: pick any two x-values, compute the y-values, join with a ruler and extend. Three facts carry all the exam weight: (1) y = ax lines all pass through the ORIGIN (0, 0); (2) larger positive a makes a steeper line, a between 0 and 1 makes a flatter line, negative a tilts the line down-left to up-right reversed (decay); (3) b is where the line CUTS THE Y-AXIS, at the point (0, b) - the y-intercept.

Effect of changing a vs changing b. Fix b and vary a: the line pivots about (0, b), changing steepness. Fix a and vary b: the line SLIDES up or down, staying PARALLEL - lines with the same slope are parallel and never meet. Growth means positive slope, decay means negative slope; the slope numerically equals the constant difference of the underlying pattern. **Tiny example.** y = 2x - 1, y = 2x + 1, y = 2x + 5 are three parallel lines cutting the y-axis at (0, -1), (0, 1) and (0, 5).

Converting to standard shape. Equations often arrive disguised: 2y = 4x + 7 means y = 2x + 3.5 after dividing by 2; 5y = 6x - 10 means y = 1.2x - 2. ALWAYS isolate y first; only then read a and b. Reading the coefficient of x before this conversion is a classic error.

2.1.9 Exam tips for this chapter

1. Before reading slope and intercept, rewrite the equation as y = ...
2. Substitute with brackets - especially for negative inputs.
3. In word problems, define the unknown in words first; the equation follows automatically.
4. For pattern rules, lock the constant difference first, then fit the constant term from ONE known term.
5. When fitting y = ax + b, subtract the two data equations; never solve them separately for b first (double work, double error chances).
6. Reverse questions (which stage gives 100 matches?) need INTEGER answers - a fractional n means 'no such stage', and saying so IS the answer.
7. On graphs, label the y-intercept point (0, b) explicitly - it is usually the mark-earning feature.

2.2 Question Types / Concept Map for Chapter 2

Type	Name	How to recognise it	Core method
T1	Anatomy of expressions	Identify terms, coefficients, variables, constants	Term-wise reading, sign-aware coefficients

Type	Name	How to recognise it	Core method
T2	Degree and classification	Find degree; label linear/quadratic/cubic/constant	Highest power rule
T3	Evaluation	Find value of polynomial at given input	Bracket substitution
T4	Reverse evaluation	Given output, find input	Solve the linear equation
T5	Word-problem modelling	Numbers/ages/coins/fences in words	Define x, translate, solve, answer in words
T6	Linear patterns	Table of stages; find nth term or a target stage	Constant difference = coefficient of n
T7	Reachability question	Can the pattern equal a value?	Solve dn + c = target; n must be a whole number
T8	Growth and decay	Increases/decreases by fixed amount per interval	value = start + rate x steps
T9	Fit y = ax + b	Two observations given	Subtract equations to get a, back-substitute for b
T10	Interpret a and b	What do the numbers mean?	a = rate per unit, b = fixed part
T11	Slope and intercept reading	Find slope/y-intercept of disguised equations	Isolate y first
T12	Parallel lines	Same slope, different intercepts	Compare a values after isolating y
T13	Graph plotting / reading	Draw the line; identify where it cuts axes	Two points; (0, b) on y-axis; y = 0 for x-axis
T14	Multi-concept HOTS	Composition of evaluations, plan comparisons, concurrence	Break into labelled steps, verify each

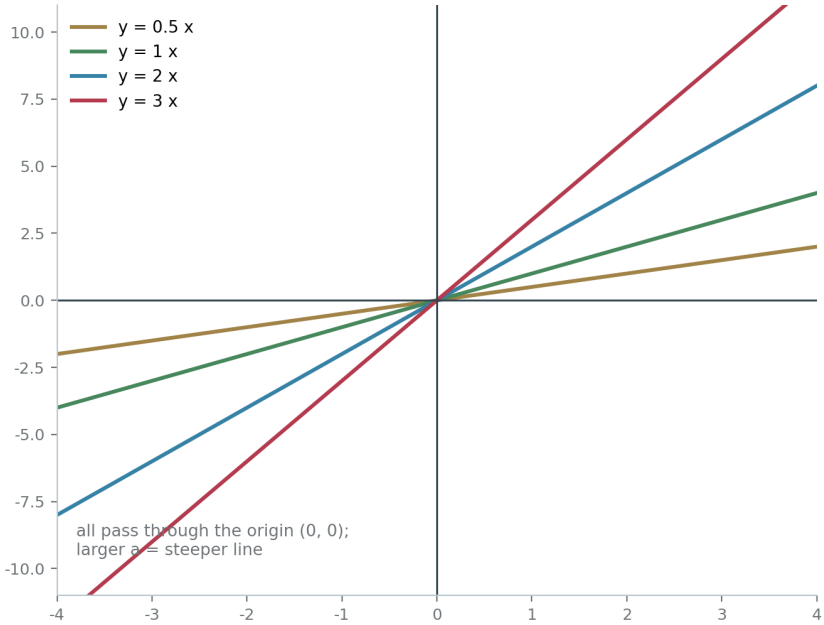


Figure 2.1: The family y = ax - all lines pass through the origin; larger a means steeper.

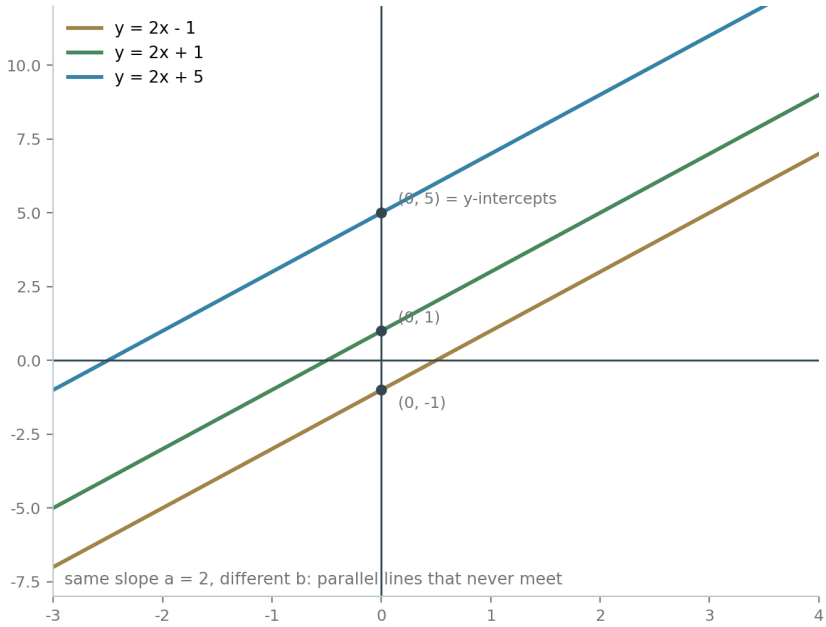


Figure 2.2: Same slope (a = 2), different intercepts b - parallel lines that never meet.

2.3 Solved Practice Set (20 Questions)

Har question apne BOX me hai, uske turant baad uska boxed step-by-step solution. Pehle question ko khud solve karne ki koshish karo - jab atak jao tabhi solution dekho. Har step ke saath "Why this step?" diya hai, aur end me Hinglish note jo concept ko dil me basa dega.

QUESTION 1 | Basic | Degree (T2)

Q1. Find the degree of each polynomial: (i) $3x^2 + 7x - 2$ (ii) $y^3 - y + 5$ (iii) -13 (iv) $6z + 1$ (v) $8 - t^4 + t^2$.

SOLUTION — step by step

Step 1. Degree = highest power of the variable present.

Step 2. (i) powers are 2, 1, 0, so degree 2 (quadratic).

Step 3. (ii) powers 3, 1, 0: degree 3 (cubic).

Step 4. (iii) the constant -13 can be written $-13x^0$, so degree 0 (constant polynomial).

Step 5. (iv) powers 1, 0: degree 1 (linear).

Step 6. (v) powers 0, 4, 2: the highest is 4, so degree 4 (a quartic - just report the number 4).

Answer: (i) 2 (ii) 3 (iii) 0 (iv) 1 (v) 4.

Hinglish: *Degree nikalne ka ek hi kaam: powers scan karo, sabse badi utha lo. Constant ka degree 0 hota hai - ye MCQ favourite hai. (v) mein dhyan do, powers beech mein ghumte hain (8, t^4, t^2) - order koi baat nahi, max chahiye.*

QUESTION 2 | Basic | Evaluation (T3)

Q2. Find the value of the linear polynomial 7x - 4 at (i) x = 0, (ii) x = 2, (iii) x = -3.

SOLUTION — step by step

Step 1. Substitute with brackets every time.

Step 2. (i) 7(0) - 4 = 0 - 4 = -4.

Step 3. (ii) 7(2) - 4 = 14 - 4 = 10.

Step 4. (iii) 7(-3) - 4 = -21 - 4 = -25.

Step 5. Note how the outputs drop by exactly 7 each time x drops by 1 going 2, 0, -3: (10, -4, -25) - the constant difference at work.

Answer: -4, 10, -25.

Hinglish: *Bracket lagana mandatory hai: 7(-3) = -21. Bina bracket ke log 7 - 3 - 4 = 0 likh dete hain - jabardast galti. Aur observation: x ek kadam gira toh output 7 gira - linear ka dhadakta hua constant difference.*

QUESTION 3 | Basic | Function machine (T3)

Q3. A machine computes y = 3x - 1 for every input x. Find the outputs for x = 5 and x = -2. Then find the input that produces the output 20.

SOLUTION — step by step

Step 1. Forward: x = 5 gives y = 3(5) - 1 = 15 - 1 = 14.

Step 2. Forward: x = -2 gives y = 3(-2) - 1 = -6 - 1 = -7.

Step 3. Reverse: we need x such that 3x - 1 = 20.

Step 4. Add 1 to both sides: 3x = 21. Divide by 3: x = 7.

Step 5. Verify: 3(7) - 1 = 20. Correct.

Answer: Outputs 14 and -7; the input producing 20 is x = 7.

Hinglish: *Machine aage bhi chalti hai peeche bhi! Aage = substitute karo (14, -7). Peeche = equation solve karo: 3x - 1 = 20 se x = 7. Function ka yahi asli matlab hai - input-output ka judgementa rishta.*

QUESTION 4 | Basic | Word problem: sum and difference (T5)

Q4. The sum of two numbers is 50, and one of them is 8 more than the other. Find the numbers.

SOLUTION — step by step

Step 1. Let the smaller number be x. Then the larger is x + 8.

Step 2. Sum condition: x + (x + 8) = 50.

Step 3. Simplify: 2x + 8 = 50, so 2x = 42, giving x = 21.

Step 4. The numbers are 21 and 29.

Step 5. Check in words: 21 + 29 = 50 and 29 - 21 = 8. Both conditions hold.

Answer: The numbers are 21 and 29.

Hinglish: *Chota number x maan lo, bada apne aap x + 8 ban jaata hai. Sum equation se x = 21. Aur LAST STEP yaad rakho: dono numbers report karo - sirf 'x = 21' likha toh aadha answer hai. Words mein wapas check karo, equation mein hi nahi.*

QUESTION 5 | Basic | Fee structure modelling (T5/T8)

Q5. A badminton academy charges a joining fee of Rs 300 plus Rs 60 for every session attended. (i) Write the total cost for n sessions. (ii) Find the cost for 12 sessions. (iii) A player paid Rs 780 in total. How many sessions did she attend?

SOLUTION — step by step

Step 1. Cost = fixed part + variable part = 300 + 60n.

Step 2. For 12 sessions: 300 + 60(12) = 300 + 720 = Rs 1020.

Step 3. Reverse: 300 + 60n = 780.

Step 4. Subtract 300: 60n = 480, so n = 8.

Step 5. Check: 300 + 60(8) = 300 + 480 = 780. Correct - she attended 8 sessions.

Answer: (i) 300 + 60n (ii) Rs 1020 (iii) 8 sessions.

Hinglish: *Fixed + rate x count - yahi har bill ka formula hai. Aage jaane ke liye substitute, peeche aane ke liye solve. 780 mein se 300 hataya toh sessions ka khel bacha: 480/60 = 8.*

QUESTION 6 | Moderate | Pattern rule (T6/T7)

Q6. A pattern of square tiles starts with 5 tiles in stage 1, and each new stage adds 3 tiles. (i) Write the number of tiles in stage n. (ii) How many tiles in stage 12? (iii) Which stage has exactly 65 tiles?

SOLUTION — step by step

Step 1. The sequence is 5, 8, 11, 14, ... with constant difference 3.

Step 2. Rule: tiles(n) = 3n + c. Match stage 1: 3(1) + c = 5 gives c = 2. So tiles(n) = 3n + 2.

Step 3. Stage 12: 3(12) + 2 = 38 tiles.

Step 4. Reverse: 3n + 2 = 65 gives 3n = 63, n = 21.

Step 5. n = 21 is a whole number, so stage 21 has exactly 65 tiles.

Answer: (i) 3n + 2 (ii) 38 tiles (iii) stage 21.

Hinglish: *Constant difference pehle pakdo (3), phir ek known term se c nikaalo (yahan 2). Aage ke liye substitute, peeche ke liye solve - aur n WHOLE NUMBER aana chahiye, warna stage exist nahi karti. Yahan 21 nikla, perfect.*

QUESTION 7 | Moderate | Linear decay money (T8)

Q7. Bela has Rs 500 and spends Rs 30 every day. (i) Write the amount left after n days. (ii) After how many days will she be left with Rs 200? (iii) What is the amount left after 15 days, and why does the model stay valid only up to 10 days of reaching zero? Actually find the exact day her money runs out.

SOLUTION — step by step

Step 1. Amount(n) = 500 - 30n.

Step 2. Set 500 - 30n = 200: 30n = 300, n = 10 days.

Step 3. After 15 days: 500 - 30(15) = 500 - 450 = Rs 50.

Step 4. Running out: 500 - 30n = 0 gives n = 500/30 = 16.67 days. Money reaches zero partway through the 17th day.

Step 5. So the linear model describes her cash only until about 16.67 days; beyond that the amount cannot go below 0.

» **Why this step?** Real decay models hit a floor. Reporting WHEN the floor is reached (and that the model stops there) is the mark of a careful modeller.

Answer: (i) 500 - 30n (ii) after 10 days (iii) Rs 50 after 15 days; money hits zero during the 17th day (n = 500/30 = 16.67), where the model ends.

Hinglish: *Decay question mein teen sawaal hote hain: kitna bacha (substitute), kab itna bacha (solve), kab khatam (zero set karo). Yahan 16.67 din - matlab 17ve din ke beech mein khatam. Model ki limit batana bhi answer ka hissa hai - paisa negative nahi ho sakta!*

QUESTION 8 | Moderate | Piecewise-style fare (T8)

Q8. A taxi charges Rs 40 for the first 3 km (and any part of it), then Rs 18 for every additional km. (i) Find the fare for a 12 km trip. (ii) Write the fare for a trip of n km where n is at least 3. (iii) For what distance is the fare Rs 148?

SOLUTION — step by step

Step 1. Beyond 3 km, chargeable extra distance = (n - 3) km.

Step 2. Fare for 12 km: 40 + 18 x (12 - 3) = 40 + 162 = Rs 202.

Step 3. General: F(n) = 40 + 18(n - 3) = 18n - 54 + 40 = 18n - 14, valid for n at least 3.

Step 4. Set 18n - 14 = 148: 18n = 162, n = 9 km.

Step 5. Check with the original description: 40 for the first 3 km plus 18 x 6 = 108 for the next 6 km: total 148. Correct.

Answer: (i) Rs 202 (ii) F(n) = 18n - 14 for n at least 3 (iii) 9 km.

Hinglish: *PDF ke auto-fare jaisa hi question, bas numbers naye. Formula banane ke liye (n - 3) ka idea zaroori hai - pehle 3 km ka paisa already fixed 40 mein hai. Simplify karke 18n - 14 milta hai aur reverse solve se 9 km. Hamesha original description se verify karo.*

QUESTION 9 | Moderate | Age problem (T5)

Q9. A mother is 4 times as old as her daughter. After 6 years, the sum of their ages will be 82 years. Find their present ages.

SOLUTION — step by step

Step 1. Let the daughter be x now; the mother is 4x.

Step 2. After 6 years: daughter x + 6, mother 4x + 6.

Step 3. Sum: (x + 6) + (4x + 6) = 82.

Step 4. Simplify: 5x + 12 = 82, so 5x = 70, x = 14.

Step 5. Present ages: daughter 14, mother 56.

Step 6. Check: after 6 years they are 20 and 62, sum 82. Correct.

Answer: Daughter 14 years, mother 56 years.

Hinglish: *Age questions mein 'after' ka matlab dono ke age mein addition. 5x + 12 = 82 se x = 14. Check zaroori: 20 + 62 = 82. Mother 4x isliye 56 - equation se nahi, definition se.*

QUESTION 10 | Moderate | Rectangle dimensions (T5)

Q10. The length of a rectangle is 3 more than twice its width, and its perimeter is 36 cm. Find the dimensions.

SOLUTION — step by step

Step 1. Let the width be w; then the length is 2w + 3.

Step 2. Perimeter: 2(w + 2w + 3) = 36.

Step 3. Inside out: 3w + 3 = 18, so 3w = 15, w = 5.

Step 4. Width 5 cm, length 2(5) + 3 = 13 cm.

Step 5. Check: 2(5 + 13) = 36 and 13 is indeed 3 more than twice 5. Correct.

Answer: Width 5 cm, length 13 cm.

Hinglish: *Perimeter formula ke andar hi expression daal do, phir dheere-dheere simplify: pehle 2 se divide, phir 3 se. '3 more than twice' ka translation (2w + 3) hi question ka dil hai.*

QUESTION 11 | Difficult | Fit y = ax + b (T9)

Q11. A learning platform charges a fixed monthly fee plus a per-module charge. A student observed that accessing 10 modules cost Rs 400 and accessing 14 modules cost Rs 500. Assuming the bill is y = ax + b, find a and b, and interpret both numbers.

SOLUTION — step by step

Step 1. Substitute both observations into y = ax + b: 400 = 10a + b and 500 = 14a + b.

Step 2. Subtract the first from the second: 100 = 4a, so a = 25.

Step 3. Back-substitute into the first: 400 = 10(25) + b = 250 + b, so b = 150.

Step 4. The relationship is y = 25x + 150.

Step 5. Interpretation: a = 25 is the charge per module (Rs 25 per module); b = 150 is the fixed monthly fee (Rs 150 even with zero modules).

Step 6. Verify with the second point: 25(14) + 150 = 350 + 150 = 500. Correct.

» **Why this step?** Subtracting the equations kills b instantly - the entire technique rests on b being identical in both lines.

Answer: a = 25 (per-module rate), b = 150 (fixed fee); y = 25x + 150.

Hinglish: *Do observation, do equation, ek SUBTRACTION - b khatam, a seedha mil gaya (25). Phir wapas jao, b nikaalo (150). Interpretation zaroor likho: 25 = per-module charge, 150 = fixed fee. Ye interpretation wali line hi examiner ko dikhati hai ki tumne samjha hai, sirf calculation nahi kiya.*

QUESTION 12 | Difficult | Temperature scales (T9)

Q12. The relation between Celsius C and Fahrenheit F has the linear form F = aC + b. Water melts at 0 degrees Celsius and 32 degrees Fahrenheit, and boils at 100 degrees Celsius and 212 degrees Fahrenheit. Find a and b.

SOLUTION — step by step

Step 1. First data point: C = 0, F = 32 gives 32 = a(0) + b, so b = 32 directly - a gift.

Step 2. Second data point: C = 100, F = 212 gives 212 = 100a + 32.

Step 3. So 100a = 180, giving a = 1.8 (which is 9/5).

Step 4. The relation is F = 1.8C + 32, the famous conversion formula.

Step 5. Verification: C = 100 gives 180 + 32 = 212. Correct.

Answer: a = 1.8 (= 9/5), b = 32; F = 1.8C + 32.

Hinglish: *Yahan data ki kripa se b PEHLE mil gaya (zero input pe output hi b hai!). Phir sirf ek equation bachi, a = 1.8. Ye formula tum pehle se jaante ho - ab tumne use DERIVE kiya hai. Fitting ka asli maza yahi hai.*

QUESTION 13 | Difficult | Matchstick pattern (T6/T7)

Q13. Squares are made with matchsticks placed in a row, each new square sharing one side with the previous one. The counts are 4, 7, 10, ... (i) How many matchsticks for n squares? (ii) For 22 squares? (iii) Can exactly 100 matchsticks form such a row of squares? Justify.

SOLUTION — step by step

Step 1. Sequence 4, 7, 10 has constant difference 3: rule M(n) = 3n + c. From stage 1: 3 + c = 4, so c = 1. Rule: M(n) = 3n + 1.

Step 2. For 22 squares: M(22) = 66 + 1 = 67 matchsticks.

Step 3. Reachability: set 3n + 1 = 100, giving 3n = 99, n = 33.

Step 4. n = 33 is a whole number, so YES - a row of 33 squares uses exactly 100 matchsticks.

Step 5. The justification is the integrality of n: patterns exist only at whole-number stages.

Answer: (i) 3n + 1 (ii) 67 (iii) Yes - stage n = 33 uses exactly 100 matchsticks.

Hinglish: *3n + 1 wala rule - ek square 4, har naya square sirf 3 extra le raha hai kyunki ek side share hoti hai. 'Can 100 form a stage?' ka matlab: solve karo aur INTEGRITY check karo. Yahan n = 33 whole number - YES. Agar 33.33 aata toh answer 'No, kabhi nahi' hota - aur wahi asli answer hota.*

QUESTION 14 | Difficult | Rally decay (T8)

Q14. A rally starts with 120 members. Every hour, exactly 9 members drop out. (i) Members after n hours? (ii) After how many hours will 39 members remain? (iii) Will the rally reach exactly 12 members at the end of some hour? Justify.

SOLUTION — step by step

Step 1. Members(n) = 120 - 9n.

Step 2. Set 120 - 9n = 39: 9n = 81, n = 9 hours.

Step 3. Reach 12? 120 - 9n = 12 gives 9n = 108, n = 12 - a whole number, so YES, exactly at the end of the 12th hour.

Step 4. Check: 120 - 9(12) = 120 - 108 = 12. Correct.

Step 5. For completeness: the count would hit zero when 120 - 9n = 0, n = 13.33, so the 12-member moment is real and comfortably inside the model's range.

Answer: (i) 120 - 9n (ii) 9 hours (iii) Yes - exactly at n = 12 hours.

Hinglish: *Do reverse questions ek saath - 39 aur 12. Dono ke n whole numbers nikle, dono possible. Yahan n = 12 bilkul clean aaya. Kabhi-kabhi dono mein se ek fraction dega - tab wala 'No, kyunki n whole nahi' answer bhi likhna aana chahiye.*

QUESTION 15 | Difficult | Tank decay (T8)

Q15. A tank holds 900 litres and drains at 45 litres per hour. (i) Volume after t hours? (ii) When will 495 litres remain? (iii) When does the tank empty exactly?

SOLUTION — step by step

Step 1. V(t) = 900 - 45t.

Step 2. Set 900 - 45t = 495: 45t = 405, t = 9 hours.

Step 3. Empty: 900 - 45t = 0 gives t = 900/45 = 20 hours exactly.

Step 4. Both answers are whole numbers, so both events happen cleanly at hour marks.

Step 5. Check: 900 - 45(20) = 0. Correct.

Answer: (i) V(t) = 900 - 45t (ii) 9 hours (iii) exactly 20 hours.

Hinglish: *Draining = decay with a definite end. 495 pe 9 hours, empty pe 20 hours - dono clean. 900/45 mental math trick: dono ko 5 se divide karo, 180/9 = 20.*

QUESTION 16 | High | Line through two points (T9/T13)

Q16. The graph of a linear polynomial p(x) passes through (2, 3) and (6, 11). (i) Find p(x). (ii) Find the coordinates where the graph cuts the x-axis and the y-axis. (iii) State the slope and interpret it.

SOLUTION — step by step

Step 1. p(x) = ax + b. Point (2, 3): 2a + b = 3. Point (6, 11): 6a + b = 11.

Step 2. Subtract: 4a = 8, so a = 2. Back-substitute: 2(2) + b = 3, b = -1.

Step 3. p(x) = 2x - 1.

Step 4. x-axis cut: set y = 0: 2x - 1 = 0, x = 1/2. Point (1/2, 0).

Step 5. y-axis cut: set x = 0: y = -1. Point (0, -1).

Step 6. Slope 2 means every extra unit of x raises the output by 2 - the line climbs two steps for one step across.

Answer: (i) p(x) = 2x - 1 (ii) cuts x-axis at (1/2, 0), y-axis at (0, -1) (iii) slope 2 - rises 2 units per 1 unit of x.

Hinglish: *Do points se line - wahi subtract-trick (4a = 8). Intercept nikaalne ka rule: x-axis cut = y ko 0 rakho; y-axis cut = x ko 0 rakho. (1/2, 0) fraction mein aata hai toh fraction mein chhod do. Slope ka meaning bolo: '1 right, 2 up'.*

QUESTION 17 | High | Disguised slope/intercept (T11)

Q17. Find the slope and y-intercept of each line, and say which is steeper: (i) 2y = 4x + 7, (ii) 5y = 6x - 10.

SOLUTION — step by step

Step 1. Isolate y first - never read the coefficient of x before this step.

Step 2. (i) Divide by 2: y = 2x + 3.5. Slope 2, y-intercept 3.5 (cuts at (0, 3.5)).

Step 3. (ii) Divide by 5: y = 1.2x - 2. Slope 1.2, y-intercept -2 (cuts at (0, -2)).

Step 4. Steepness compares slopes: 2 > 1.2, so line (i) is steeper.

Step 5. Interpretation: line (i) rises 2 units per unit of x; line (ii) rises 1.2 units per unit.

Answer: (i) slope 2, intercept 3.5 (ii) slope 1.2, intercept -2 (i) is steeper.

Hinglish: *Golden rule: pehle y = ... banao. 2y = 4x + 7 ko dekh ke log slope 4 likh dete hain - GALAT, 4 toh 2y ka hai! Divide karne ke baad hi a = 2 aur b = 3.5. Steepness = slope comparison: 2 vs 1.2.*

QUESTION 18 | High | Reverse data bill (T9/T10)

Q18. A telecom bill follows y = 20x + 150 (Rs 20 per GB plus Rs 150 fixed). (i) What will the bill be for 30 GB? (ii) A student received a bill of Rs 750. How much data did she use? (iii) What does the 150 mean for a month in which she used no data at all?

SOLUTION — step by step

Step 1. (i) y = 20(30) + 150 = 600 + 150 = Rs 750.

Step 2. (ii) Set 20x + 150 = 750: 20x = 600, x = 30 GB.

Step 3. (iii) With x = 0, y = 150: even with zero data the fixed monthly fee of Rs 150 applies.

Step 4. Notice (i) and (ii) are inverses of each other - forward and backward through the same machine.

Answer: (i) Rs 750 (ii) 30 GB (iii) the Rs 150 fixed fee is charged even for zero usage.

Hinglish: *Yehi bill equation aage-peeche do baar: substitute aur solve. Mazedaar baat: (i) aur (ii) ka jawab same pair (30, 750) hai - kyunki ek hi line ke do raaste hain. (iii) zero usage par bhi 150 - fixed fee ka matlab hi yahi hai.*

QUESTION 19 | HOTS | Wire rectangle area function (T14)

Q19. A wire of length 36 cm is bent into a rectangle. Let the length be x cm. (i) Write the area as a function of x. (ii) Find ALL lengths x that give an area of 77 sq cm. (iii) Which single length gives the largest area, and what is that area?

SOLUTION — step by step

Step 1. Perimeter 36 means half-perimeter 18: length + width = 18. Width = 18 - x.

Step 2. Area A(x) = x(18 - x) = 18x - x^2 - a quadratic function (the PDF builds exactly this in its Example 3 with a 20 cm wire).

Step 3. Set x(18 - x) = 77: x^2 - 18x + 77 = 0. Look for two numbers with product 77 and sum 18: 7 and 11.

Step 4. So x = 7 or x = 11. Both are genuine: a 7 x 11 rectangle and an 11 x 7 rectangle are the SAME rectangle turned around - one answer in two descriptions.

Step 5. Largest area: test symmetric candidates. x = 9 gives 9 x 9 = 81; x = 8 gives 8 x 10 = 80; x = 10 gives 10 x 8 = 80. The square (x = 9) wins with area 81.

Step 6. Structure noticed: areas rise toward x = 9 and fall after it - symmetric about the square.

Answer: (i) A(x) = 18x - x^2 (ii) x = 7 or 11 (the same rectangle, two descriptions) (iii) x = 9 gives the maximum area 81 sq cm.

Hinglish: *Ye PDF ke wire example ka bada bhai. Area quadratic nikla (18x - x^2) - chapter linear hai par evaluation quadratic ki bhi practice hai (Ex 2.2 mein tha). 77 ke liye do answers (7 aur 11) ek hi rectangle ke do naam hain - symmetry! Aur square (9 x 9 = 81) max deta hai - ye observation Grade 10 mein formally aayega, aaj ke liye table-test kaafi hai.*

QUESTION 20 | HOTS | Plan comparison (T14)

Q20. Mobile Plan A costs Rs 200 per month plus Rs 2 per GB. Plan B costs Rs 350 per month plus Rs 0.50 per GB. (i) Write both cost formulas. (ii) For what data usage do the plans cost the same? (iii) Which plan is cheaper for a user consuming 60 GB? (iv) For 150 GB?

SOLUTION — step by step

Step 1. Plan A: C_A = 200 + 2x. Plan B: C_B = 350 + 0.5x (x in GB).

Step 2. Equal cost: 200 + 2x = 350 + 0.5x. Move terms: 2x - 0.5x = 350 - 200, so 1.5x = 150, x = 100 GB.

Step 3. At 100 GB both cost 200 + 200 = Rs 400 (and 350 + 50 = 400 - consistent).

Step 4. At x = 60: A costs 200 + 120 = Rs 320; B costs 350 + 30 = Rs 380. A is cheaper (by Rs 60).

Step 5. At x = 150: A costs 200 + 300 = Rs 500; B costs 350 + 75 = Rs 425. B is cheaper (by Rs 75).

Step 6. Structure: below 100 GB choose the small-fixed-high-rate plan (A); above 100 GB the high-fixed-low-rate plan (B) wins. The crossing point 100 GB is the decision boundary.

Answer: (i) A: 200 + 2x; B: 350 + 0.5x (ii) equal at 100 GB (both Rs 400) (iii) A cheaper at 60 GB (320 vs 380) (iv) B cheaper at 150 GB (425 vs 500).

Hinglish: *Do plans ka muqabla = do lines ka intersection. 100 GB par dono 400 - crossing point. Usse kam data toh A sasta (fixed fee kam), zyada data toh B (rate kam). Ye 'decision boundary' thinking real life mein har jagah hai - OTT plans, taxi vs auto, sab yahi maths hai.*

Chapter 3: The World of Numbers (Real Numbers)

Numbers are not one big happy family - they live in clearly separated neighbourhoods, and exams love to test whether you know who lives where. The uploaded PDF builds the full number map: natural numbers N, integers Z, rationals Q, irrationals, and the master set R of real numbers. It studies decimal expansions (terminating, recurring, never-ending and never-repeating), shows how to convert recurring decimals back into fractions, constructs roots like $\sqrt{2}$ and $\sqrt{3}$ on the number line using right-angled triangles, proves that $\sqrt{2}$ is irrational, simplifies surds, rationalises denominators, and closes with the laws of exponents. In this book every one of those ideas is pushed to competition level: reverse questions, contradiction proofs, nested radicals, and exponent equations that look impossible until you spot the base.

3.1 Complete Concept Cheat Sheet (read this before any question)

3.1.1 The number family map: N, Z, Q, I, R

What is it? A classification of all numbers you will meet in Grade 9. **N** (natural numbers) = 1, 2, 3, ... **Z** (integers) = ..., -2, -1, 0, 1, 2, ... - N plus zero plus the negatives. **Q** (rationals) = every number writable as p/q with p, q integers and $q \neq 0$ - this includes all terminating decimals and all recurring decimals. **I** (irrationals) = real numbers that can NOT be written as p/q, like $\sqrt{2}$, $\sqrt{3}$, π, and decimals that go on forever without any repeating block. **R** (reals) = $Q \cup I$, every point on the number line. The nesting is strict: $N \subset Z \subset Q \subset R$, while I sits beside Q inside R with $I \cap Q = \emptyset$.

When do I use it? Whenever a question says 'classify', 'true or false', 'which of these is irrational', or asks about closure of operations. **How do I recognise the type?** Terminating decimal → rational. Recurring decimal → rational. Square root of a non-perfect-square → irrational. π and e → irrational. A fraction of integers → rational (even -7/3). **Tiny example.** -4 is Z and Q and R but not N; 0.333... = 1/3 is Q; $\sqrt{7}$ is I; 22/7 is Q (it is only an approximation of π, never equal to it).

Figure 3.1: The number family - each set sits inside the next bigger one; irrationals live beside Q inside R.

3.1.2 Decimal expansions and the denominator test

What is it? Every rational number has a decimal expansion that either STOPS (terminates) or STOPS NEVER BUT REPEATS a block (non-terminating recurring). Every irrational number has a decimal expansion that goes on forever WITHOUT any repeating block (non-terminating non-recurring). **The denominator test:** write the rational in lowest terms p/q. If $q = 2^m \times 5^n$ (only 2s and 5s, which are exactly the prime factors of 10), the decimal terminates; otherwise it recurs. Why 2 and 5? Because $10 = 2 \times 5$, so only these primes can be 'absorbed' into a power of ten.

When do I use it? Questions like 'without long division, state whether the decimal will terminate'. **Tiny example.** 17/8: $8 = 2^3$, so terminating (2.125). 13/3125: $3125 = 5^5$, terminating. 7/12: $12 = 2^2 \times 3$ - the stray 3 forces a recurring decimal. 6/15 = 2/5 in lowest terms - always REDUCE FIRST, the reduced denominator decides.

> **The reduce-first rule**
> The $2^l \times 5^n$ test works only in LOWEST TERMS. 6/15 looks non-terminating until you cancel the 3 and get 2/5 = 0.4. Reducing first is the difference between a right and a wrong answer.

3.1.3 Recurring decimals back to fractions (p/q form)

What is it? A reversing technique. **Pure recurring (repeat starts immediately):** count the repeating digits k, then the number = (repeating block)/(same number of 9s). Example: 0.555... = 5/9, 0.373737... = 37/99, 0.090909... = 9/99 = 1/11. **Mixed recurring (some non-repeating digits first):** (all digits once, minus the non-repeating part) / (as many 9s as repeating digits, followed by as many 0s as non-repeating digits). Example: 0.4777... = (47 - 4)/90 = 43/90.

Where does the trick come from? Multiply by the right power of 10 so two copies of the repeating block line up, then subtract - the infinite tails cancel exactly and you are left with a finite equation. **Full derivation for 0.777...:** let x = 0.777...; then 10x = 7.777...; subtract: 10x - x = 7.777... - 0.777... = 7; so 9x = 7 and x = 7/9. **Tiny example (mixed).** x = 0.2333...; 10x = 2.333..., 100x = 23.33...; subtract: 90x = 21, x = 21/90 = 7/30. Always finish by reducing the fraction.

3.1.4 Locating roots on the number line

What is it? A compass-ruler construction. To place $\sqrt{n}$: build a right triangle where the hypotenuse has length $\sqrt{n}$ using the Baudhayana-Pythagoras theorem. For $\sqrt{2}$: take OA = 1 on the line, raise AB = 1 perpendicular at A; then $OB = \sqrt{(1^2 + 1^2)} = \sqrt{2}$; swing an arc centred at O through B down to the line. For $\sqrt{3}$: repeat with the previous hypotenuse - take the segment of length $\sqrt{2}$ as one leg, add a perpendicular leg of 1, get hypotenuse $\sqrt{(2 + 1)} = \sqrt{3}$. Chaining this creates the famous **square root spiral** where each new edge has length $\sqrt{(n+1)}$.

When do I use it? 'Show how to find $\sqrt{5}$ on the number line', 'represent $\sqrt{4.3}$ geometrically', and successive-magnification questions ('zoom twice and tell which two numbers $\sqrt{3}$ sits between at the second decimal'). **Successive magnification.** To locate 2.6 (say $\sqrt{7} \approx 2.6457$): between 2 and 3, then between 2.6 and 2.7, then between 2.64 and 2.65 - each zoom narrows the interval by a factor of ten. **Tiny example.** $\sqrt{5}$: right triangle with legs 2 and 1, hypotenuse $\sqrt{5}$, arc it down to the line between 2.2 and 2.3 (since $2.23^2 \approx 4.98$, $2.24^2 \approx 5.02$).

Figure 3.2: Building √2, √3, √4, √5 on the number line with right triangles and arcs.

3.1.5 Irrationality proofs (the contradiction method)

What is it? The classic proof that $\sqrt{2}$ is irrational, done by assuming the opposite and breaking it. **Skeleton:** Step 1 - suppose $\sqrt{2} = p/q$ in lowest terms (p, q coprime, $q \neq 0$). Step 2 - square: $p^2 = 2q^2$, so 2 divides p^2, hence 2 divides p (because if a prime divides a square, it divides the number itself). Step 3 - write p = 2r; then $4r^2 = 2q^2$, so $q^2 = 2r^2$, hence 2 divides q too. Step 4 - both p and q even contradicts lowest terms. Conclusion: the assumption was false, so $\sqrt{2}$ is irrational. The identical skeleton works for $\sqrt{3}$ (replace 2 by 3) and $\sqrt{5}$.

Extension questions exams adore: (i) $\sqrt{2} + \sqrt{3}$ is irrational - if it were rational, its square $(5 + 2\sqrt{6})$ would be rational, forcing $\sqrt{6}$ rational, contradiction; (ii) rational + irrational = irrational - if $x + \sqrt{2} = r$ were rational then $\sqrt{2} = r - x$ would be rational, contradiction; (iii) nonzero rational × irrational = irrational - same subtraction trick; (iv) the sum of two irrationals may be rational (a + (-a) = 0, or $(1 + \sqrt{2}) + (1 - \sqrt{2}) = 2$) and

the product may be rational ($\sqrt{2} \times \sqrt{2} = 2$) - so NO general rule, must test case by case.

> **Why 'a prime dividing a square divides the number'?**
> If $p \mid n^2$, write $n = p \cdot k + r$ with $0 \le r < p$. Squaring, $n^2 \equiv r^2 \pmod{p}$. For p prime, $n^2 \equiv 0 \pmod{p}$ forces $r^2 \equiv 0 \pmod{p}$, and since p is prime r must be 0. Grade 9 books justify it with prime factorisations: $p \mid n^2$ means p appears in the factorisation of n^2, and every prime in n^2 comes from a prime in n - in pairs. One 'stray' p is impossible.

3.1.6 Surds: simplifying, adding, comparing

What is it? A surd is an irrational root kept in root form, like $\sqrt{2}$, $\sqrt{3}$, $\sqrt{5}$. **Key laws:** $\sqrt{(ab)} = \sqrt{a} \cdot \sqrt{b}$; $\sqrt{(a/b)} = \sqrt{a} / \sqrt{b}$; $\sqrt{(a^2 b)} = a\sqrt{b}$. **Simplify by extracting perfect-square factors:** $\sqrt{50} = \sqrt{(25 \cdot 2)} = 5\sqrt{2}$; $\sqrt{72} = \sqrt{(36 \cdot 2)} = 6\sqrt{2}$; $\sqrt{245} = \sqrt{(49 \cdot 5)} = 7\sqrt{5}$. **Like surds** (same radicand) add like terms: $3\sqrt{2} + 5\sqrt{2} = 8\sqrt{2}$; unlike surds ($3\sqrt{2} + 2\sqrt{3}$) canNOT merge - they just sit there. **Comparing surds:** square both sides (both positive, so order is preserved): which is bigger, $2\sqrt{5}$ or $3\sqrt{2}$? Squares give 20 vs 18, so $2\sqrt{5}$ wins. **Tiny example.** Order $\sqrt{7}$, $2\sqrt{2}$, $\sqrt{5}$: squares are 7, 8, 5, so $2\sqrt{2} > \sqrt{7} > \sqrt{5}$.

3.1.7 Rationalising the denominator

What is it? Rewriting a fraction so no surd remains in the denominator, using the difference-of-squares identity $(a + b)(a - b) = a^2 - b^2$. **Single surd:** multiply top and bottom by that surd: $1/\sqrt{7} = \sqrt{7}/7$. **Binomial denominator ($\sqrt{a} + \sqrt{b}$ or $a + \sqrt{b}$):** multiply by the *conjugate* (flip the middle sign): $1/(2 + \sqrt{3}) = (2 - \sqrt{3})/((2 + \sqrt{3})(2 - \sqrt{3})) = (2 - \sqrt{3})/(4 - 3) = 2 - \sqrt{3}$. The denominator collapses to a clean integer because $(a+b)(a-b) = a^2 - b^2$. **Triple-decker:** for $1/(\sqrt{2} + \sqrt{3} + \sqrt{5})$, group as $(\sqrt{2} + \sqrt{3}) + \sqrt{5}$, rationalise once, then regroup and rationalise again - two rounds of conjugates.

Why bother? Three reasons: standardized answers (everyone writes $2 - \sqrt{3}$, not $1/(2+\sqrt{3})$), easier further computation, and more accurate decimal work (dividing by 1.73... is numerically nastier than multiplying by 0.26...). **Tiny example.** Rationalise $6/(\sqrt{5} - \sqrt{3})$: multiply by $(\sqrt{5} + \sqrt{3})$: $= 6(\sqrt{5} + \sqrt{3})/(5 - 3) = 3(\sqrt{5} + \sqrt{3})$.

Figure 3.3: Density - between any two real numbers, rationals and irrationals are both endless; no 'next' number exists.

3.1.8 Laws of exponents

What is it? The five power rules for a real base $a > 0$ and integer exponents: (1) $a^m \cdot a^n = a^{m+n}$; (2) $a^m / a^n = a^{m-n}$; (3) $(a^m)^n = a^{mn}$; (4) $a^m \cdot b^m = (ab)^m$; (5) $a^0 = 1$, and $a^{-m} = 1/a^m$. Roots join the family: $a^{1/p} = \sqrt[p]{a}$, so $8^{2/3} = (8^{1/3})^2 = 2^2 = 4$. **When do I use them?** Simplifying expressions like $(3^5 \cdot 3^{-2})$, solving $2^x = 128$, or taming products like $(5^{1/2} \cdot 5^{1/3})$. **Tiny example.** $(2^{-3} \times 2^5) = 2^2 = 4$; $(3^0 + 4^0) \times 2^2 = 2 \times 4 = 8$; $64^{-1/3} = 1/4$.

> **Exponent equation strategy**
> See $2^x = 128$? Write both sides with the SAME base: $2^x = 2^7$, so $x = 7$. See $3^x \times 3^{x+1} = 243$? Left side is 3^{2x+1}, so $2x + 1 = 5$, $x = 2$. See $4^x = 8$? Convert: $(2^2)^x = 2^3$, so $2x = 3$, $x = 3/2$. One base to rule them all.

3.1.9 Exam tips for this chapter

1. Classify numbers by writing their DECIMAL EXPANSION in your head first - the expansion type decides the set.

2. For the $2^l \cdot 5^n$ test, reduce the fraction FIRST; the reduced denominator is what counts.
3. Recurring decimal questions: name the variable x, multiply by the correct power of 10, subtract. Show the subtraction line - it carries the marks.
4. Irrationality proofs: memorise the 4-step contradiction skeleton once, reuse it for √2, √3, √5 by changing the prime.
5. Simplify every surd to k√(radicand) form BEFORE adding or comparing.
6. Rationalising: the conjugate is your only tool; write the identity $(a+b)(a-b) = a^2 - b^2$ explicitly in the solution.
7. Exponent problems: force ONE base everywhere; then equate the exponents.
8. Never claim sums/products of irrationals are always irrational - give a counterexample if a true/false question asks.

3.2 Question Types / Concept Map for Chapter 3

Label the question type before you start - in this chapter, recognising 'this is a conjugate problem' or 'this is a same-base problem' IS the solution.

Type	Name	How to recognise it in the question	Core method
T1	Set classification	'Classify into N/Z/Q/I/R', 'true or false about number sets'	Decimal expansion type; nesting $N \subset Z \subset Q \subset R$
T2	Terminating test	'Will the decimal terminate, without division?'	Lowest terms; denominator = $2^m \times 5^n$ only
T3	Recurring → p/q	'Express 0.4777... in p/q form'	x, multiply by 10^k, subtract, solve; reduce
T4	Number line placement	'Represent √5 on the number line', successive magnification	Right-triangle construction; zoom by decimals
T5	Irrationality proofs	'Prove √3 is irrational'	Contradiction: assume p/q lowest terms, prime divides both
T6	Sum/product arguments	'Show √2 + √3 irrational'; true/false on operations	Square it / subtract-the-rational trick; counterexamples
T7	Surd simplification	'Simplify √72 + √32 - √18'	Extract perfect squares; collect like surds
T8	Surds comparison	'Which is larger: 2√5 or 3√2?'	Square both (both positive); compare radicand multiples
T9	Rationalising	'Rationalise 1/(2+√3)', triple surds	Multiply by conjugate; difference of squares; repeat if needed
T10	Exponent laws	'Simplify $(3^5 \times 3^{-2})/3^2$'	Five power rules; combine bases
T11	Exponent equations	'Solve $2^x = 128$', '$5^{x-2} = 125$'	Same base both sides; equate exponents

Type	Name	How to recognise it in the question	Core method
T12	Between-ness and density	'Find two irrationals between 2 and 3'	√ of non-squares in range; a + small irrational tails
T13	Applied / construction	Spiral of roots, geometric representations, real-life decimals	Pythagoras chaining; interpret expansions
T14	Multi-concept HOTS	Nested radicals, x + 1/√x chains, proof + computation mix	Break into labelled mini-steps, verify each junction

3.3 Solved Practice Set (20 Questions)

Har question apne BOX me hai, uske turant baad uska boxed step-by-step solution. Pehle question ko khud solve karne ki koshish karo - jab atak jao tabhi solution dekho. Har step ke saath "Why this step?" diya hai, aur end me Hinglish note jo concept ko dil me basa dega.

QUESTION 1 | Basic | Set classification (T1)

Q1. Classify each number into as many of the sets N, Z, Q, I, R as it belongs to: (i) -7, (ii) 0, (iii) 2/3, (iv) √16, (v) √15, (vi) 0.585858... (the pair 58 repeating).

SOLUTION — step by step

Step 1. Given: six numbers. Needed: set membership for each, using the nesting $N \subset Z \subset Q \subset R$ and $I \cap Q = \emptyset$.

Step 2. -7: negative integer, so Z and Q and R; not N (naturals start at 1); not I.

Step 3. 0: an integer, so Z, Q, R; not N in this book's convention (some books include 0 in N - state your convention).

Step 4. 2/3: a ratio of integers, so Q and R; not N, not Z, not I.

Step 5. √16 = 4: a perfect square under the root, so N, Z, Q, R. Perfect squares are never irrational.

Step 6. √15: 15 is not a perfect square, so √15 is irrational: I and R only.

Step 7. 0.585858... is a recurring decimal, and every recurring decimal is a rational: it equals 58/99. So Q and R.

» **Why this step?** The decimal expansion is the fingerprint: terminating or recurring means rational; endless without a block means irrational.

Answer: (i) Z, Q, R (ii) Z, Q, R (0 ∉ N here) (iii) Q, R (iv) N, Z, Q, R (v) I, R (vi) Q, R (= 58/99)

Hinglish: *Yaad rakho - decimal expansion hi asli ID card hai. Recurring decimal (58-58-58 chalne wala) hamesha rational hota hai, yahan 58/99. √16 jaisi cheez ko dekh ke confuse mat hona - andar perfect square hai toh answer 4 hai, natural number. √15 tabhi irrational hai jab andar non-square ho.*

QUESTION 2 | Basic | Terminating test (T2)

Q2. Without performing long division, decide whether each decimal expansion terminates: (i) 17/8, (ii) 13/3125, (iii) 7/12, (iv) 64/455, (v) 6/15.

SOLUTION — step by step

Step 1. Given: five fractions. Needed: terminating or not, via the $2^m \times 5^n$ test on the REDUCED denominator.

Step 2. (i) 17/8: $8 = 2^3 = 2^3 \times 5^0$. Only 2s, so it terminates (in fact 2.125).

Step 3. (ii) 13/3125: $3125 = 5^5$. Only 5s, so it terminates.

Step 4. (iii) 7/12: $12 = 2^2 \times 3$. The stray 3 can never be absorbed into a power of 10, so it recurs.

Step 5. (iv) 64/455: $455 = 5 \times 7 \times 13$. The primes 7 and 13 force a recurring decimal.

Step 6. (v) 6/15: reduce first! 6/15 = 2/5, and $5 = 5^1$, so it terminates (0.4). Testing $15 = 3 \times 5$ without reducing would give the wrong verdict.

» **Why this step?** $10 = 2 \times 5$, so a fraction terminates exactly when its reduced denominator divides some power of 10 - only primes 2 and 5 can appear.

Answer: (i) terminates (ii) terminates (iii) recurs (iv) recurs (v) terminates (6/15 = 2/5 = 0.4)

Hinglish: *Sabse pehla kaam: fraction ko lowest terms mein lao. Phir denominator ko prime factors mein todo. Sirf 2 aur 5 dikh rahe hain? Decimal ruk jayega. Koi aur prime (3, 7, 13...) dikh gaya? Chalti-firti decimal aayegi. (v) wala question isliye diya hai kyunki students 15 dekh ke turant 'recur' bol dete hain - pehle 3 cancel karo!*

QUESTION 3 | Basic | Pure recurring to p/q (T3)

Q3. Express in p/q form: (i) 0.777... (the 7 repeating), (ii) 0.373737... (the block 37 repeating), (iii) 0.090909... (the block 09 repeating).

SOLUTION — step by step

Step 1. Given: three pure recurring decimals (repeat starts immediately). Needed: exact fractions.

Step 2. (i) Let x = 0.777... . Multiply by 10 (one repeating digit → shift by one place): 10x = 7.777... .

Step 3. Subtract: 10x - x = 7.777... - 0.777... = 7. So 9x = 7 and x = 7/9.

Step 4. (ii) Let x = 0.373737... . Two repeating digits → multiply by 100: 100x = 37.3737... .

Step 5. Subtract: 100x - x = 37. So 99x = 37 and x = 37/99.

Step 6. (iii) Let x = 0.090909... . Two repeating digits → 100x = 9.0909... . Subtract: 99x = 9, x = 9/99 = 1/11.

» **Why this step?** Multiplying by 10^ϵ lines up two identical infinite tails; subtracting cancels them and leaves a finite equation. k repeating digits need a shift of 10^ϵ.

Answer: (i) 7/9 (ii) 37/99 (iii) 1/11

Hinglish: *Rule simple hai: jitne digit repeat ho rahe hain, utne 9 denominator mein. Ek digit repeat = 9, do digit = 99. 0.777... = 7/9, 0.3737... = 37/99. Bas repeating block ke digits upar likh do aur neeche utne hi 9. (iii) mein 9/99 ko 1/11 tak reduce karna mat bhoolna.*

QUESTION 4 | Basic | Mixed recurring to p/q (T3)

Q4. Express in p/q form: (i) 0.4777... (the 4 does not repeat, the 7 repeats), (ii) 0.2333... (the 2 does not repeat, the 3 repeats), (iii) 1.272727... (the block 27 repeats after the integer part).

SOLUTION — step by step

Step 1. Given: mixed recurring decimals. Needed: exact fractions using the shift-and-subtract method.

Step 2. (i) x = 0.4777... . One non-repeating digit then one repeating digit: form 10x = 4.777... and 100x = 47.777... .

Step 3. Subtract: 100x - 10x = 47.777... - 4.777... = 43. So 90x = 43, x = 43/90 (already lowest terms).

Step 4. (ii) x = 0.2333... . 10x = 2.333..., 100x = 23.333... . Subtract: 90x = 21, x = 21/90 = 7/30.

Step 5. (iii) x = 1.272727... . Two repeating digits: 100x = 127.2727... . Subtract x: 99x = 126, x = 126/99 = 14/11.

Step 6. Check (iii): 14/11 = 1.272727... ✓. Check (i): 43 ÷ 90 = 0.4777... ✓.

» **Why this step?** Two shifts are needed so that BOTH copies of the decimal have the same non-repeating head; then the subtraction kills head and tail together.

Answer: (i) 43/90 (ii) 7/30 (iii) 14/11

Hinglish: *Mixed recurring mein trick: pehle non-repeating digits ko 'head' samjho. Do copies banao jisme head same position par ho (10x aur 100x yahan), minus karo - head bhi tail bhi cancel, jo bacha wo equation. Formula version: (poore digits - head)/(utne 9 jitne repeating, utne 0 jitne non-repeating). (i): (47-4)/90 = 43/90. (ii) mein 21/90 ko 7/30 karna zaroori hai.*

QUESTION 5 | Basic | Surd simplification (T7)

Q5. Simplify into k√( ) form: (i) √72, (ii) √245, (iii) √175, (iv) √(147).

SOLUTION — step by step

Step 1. Given: four radicals. Needed: extract every perfect-square factor.

Step 2. (i) 72 = 36 × 2, so √72 = √36 · √2 = 6√2.

Step 3. (ii) 245 = 49 × 5, so √245 = 7√5.

Step 4. (iii) 175 = 25 × 7, so √175 = 5√7.

Step 5. (iv) 147 = 49 × 3, so √147 = 7√3.

» **Why this step?** √(ab) = √a√b lets the perfect-square factor walk out of the root; what remains inside must be square-free.

Answer: (i) 6√2 (ii) 7√5 (iii) 5√7 (iv) 7√3

Hinglish: *Andar sabse bada perfect square dhoondo aur bahar nikalo. 72 = 36×2, 36 ka root 6, toh 6√2. Yaad rakho: 4, 9, 16, 25, 36, 49, 64, 81, 100, 121, 144 - ye aapke dost hain. Jitna bada square nikaloge, utna chhota root; 72 ko 4×18 karke √18 ko phir simplify karna double kaam hai.*

QUESTION 6 | Moderate | Irrationality proof (T5)

Q6. Prove that √3 is irrational.

SOLUTION — step by step

Step 1. Given: a claim to contradict. Assume $\sqrt{3} = p/q$ where p, q are integers, $q \neq 0$, and the fraction is in lowest terms (p and q share no common prime).

Step 2. Squaring both sides: $3 = p^2/q^2$, so $p^2 = 3q^2$.

Step 3. Then 3 divides p^2. Since 3 is prime, 3 divides p itself (a prime dividing a square must divide the number). Write $p = 3r$.

Step 4. Substitute: $(3r)^2 = 3q^2 \Rightarrow 9r^2 = 3q^2 \Rightarrow q^2 = 3r^2$.

Step 5. So 3 divides q^2, hence 3 divides q.

Step 6. Now 3 divides both p and q - contradicting lowest terms. The assumption is false, therefore √3 is irrational. ∎

» **Why this step?** The contradiction must land on the EXACT assumption made (coprimality); that is why the final sentence names lowest terms.

Answer: √3 is irrational (by contradiction; 3 divides both p and q, contradicting lowest terms).

Hinglish: *Proof ka 4-step skeleton yaad karo: (1) assume p/q lowest terms, (2) square karo - prime upar aa jata hai, (3) prime dono (p aur q) ko divide karta nikla, (4) lowest terms se contradiction - done! √2 mein prime 2 tha, yahan 3 hai, √5 mein 5 hoga. Bas prime badlo, skeleton same.*

QUESTION 7 | Moderate | Rationalising a + b√c denominator (T9)

Q7. Rationalise the denominator and simplify: $1/(3 + 2\sqrt{2})$.

SOLUTION — step by step

Step 1. Given: denominator $3 + 2\sqrt{2}$. Its conjugate is $3 - 2\sqrt{2}$ (flip the sign of the surd term).

Step 2. Multiply numerator and denominator by the conjugate:

Step 3. $= (3 - 2\sqrt{2})/((3 + 2\sqrt{2})(3 - 2\sqrt{2}))$.

Step 4. Denominator: $3^2 - (2\sqrt{2})^2 = 9 - 8 = 1$ (difference of squares).

Step 5. So the expression equals $3 - 2\sqrt{2}$. Check numerically: $1/(3 + 2.828...) = 1/5.828... \approx 0.1716$, and $3 - 2.828... = 0.1716$ ✓.

» **Why this step?** The conjugate turns the denominator into $a^2 - b^2$, which has no surds; here it even collapses to 1, giving a free simplification.

Answer: 3 - 2√2

Hinglish: *Conjugate ka matlab: sirf beech ka sign ulta. 3 + 2√2 ka conjugate 3 - 2√2. Multiply karo, denominator (a+b)(a-b) = a²-b² ban jata hai. Yahan 9-8=1 nikla - jackpot! Answer 3-2√2. Aise lucky cases (denominator 1) exam mein bahut aate hain, isliye denominator ko poori tarah simplify karna kabhi mat chhodna.*

QUESTION 8 | Moderate | Comparing surds (T8)

Q8. Arrange in increasing order: √7, 2√2, √5, 3√3/2.

SOLUTION — step by step

Step 1. Given: four positive surds. All are positive, so squaring preserves the order.

Step 2. √7 → 7. $2\sqrt{2} \to (2\sqrt{2})^2 = 4 \times 2 = 8$. √5 → 5. $(3\sqrt{3})/2 \to (9 \times 3)/4 = 27/4 = 6.75$.

Step 3. Order of the squares: $5 < 6.75 < 7 < 8$.

Step 4. So the original order: $\sqrt{5} < 3\sqrt{3}/2 < \sqrt{7} < 2\sqrt{2}$.

Step 5. Check: √5 ≈ 2.236, 3√3/2 ≈ 2.598, √7 ≈ 2.646, 2√2 ≈ 2.828 ✓.

» **Why this step?** For positive quantities $a < b \Leftrightarrow a^2 < b^2$ - comparing squares avoids fiddly decimal roots.

Answer: √5 < 3√3/2 < √7 < 2√2

Hinglish: *Compare karne ka best trick: dono ko square karo. 2√2 ka square $(2)^2 \times 2 = 8$ hota hai - coefficient bhi square hota hai, ye bhoolna sabse common galti hai! $(3\sqrt{3}/2)^2 = 9 \times 3/4 = 6.75$. Ab squares ko order karo aur original order likho. Numerical check se confirm kar lo.*

QUESTION 9 | Moderate | Same-base equations (T11)

Q9. Solve: (i) $3^x = 1/81$, (ii) $5^{(x-2)} = 125$, (iii) $(2/3)^x = 27/8$.

SOLUTION — step by step

Step 1. Given: three exponential equations. Strategy: one base on both sides, then equate exponents.

Step 2. (i) $1/81 = 1/3^4 = 3^{-4}$. So $3^x = 3^{-4} \Rightarrow x = -4$.

Step 3. (ii) $125 = 5^3$. So $5^{(x-2)} = 5^3 \Rightarrow x - 2 = 3 \Rightarrow x = 5$.

Step 4. (iii) $27/8 = 3^3/2^3 = (3/2)^3 = (2/3)^{-3}$. So $x = -3$.

Step 5. Check: $(2/3)^{-3} = (3/2)^3 = 27/8$ ✓.

» **Why this step?** Equality of powers with the same base forces equal exponents; the whole game is rewriting both sides into one base.

Answer: (i) x = -4 (ii) x = 5 (iii) x = -3

Hinglish: *Exponent equation mein dono taraf SAME base lao. 1/81 ko 3^{-4} dekho - negative exponent ka perfect use. $(2/3)^x = 27/8$ mein ulta fraction banana padta hai: $27/8 = (3/2)^3 = (2/3)^{-3}$. Fraction ka negative exponent = fraction flip. Ye teen patterns exam ke favourites hain.*

QUESTION 10 | Moderate | Mixed recurring with integer part (T3)

Q10. Express 2.1363636... (the block 36 repeats) in p/q form.

SOLUTION — step by step

Step 1. Given: x = 2.13636... . One non-repeating digit (3), then the block 36.

Step 2. Multiply by 10: 10x = 21.3636... . Multiply by 1000: 1000x = 2136.3636... .

Step 3. Subtract: 1000x - 10x = 2136.3636... - 21.3636... = 2115 (tails cancel exactly).

Step 4. So 990x = 2115, x = 2115/990.

Step 5. Reduce: HCF(2115, 990) = 45, so x = 47/22.

Step 6. Check: 47 ÷ 22 = 2.13636... ✓.

» **Why this step?** Shifts of 10 and 1000 align the two copies of the repeating tail; 990 = 9×100 with one zero for the non-repeating digit.

Answer: 47/22

Hinglish: *Integer part wale recurring mein panic mat karo - method same hai. 10x aur 1000x banao (ek shift head ke liye, do aur block ke liye), minus karo: 2115/990 aayega. HCF 45 hai, kat ke 47/22. Formula version: (2136 - 21)/990 = 2115/990. Reduce karna compulsory hai!*

QUESTION 11 | Difficult | Irrationality of a surd sum (T6)

Q11. Prove that $\sqrt{2} + \sqrt{3}$ is irrational.

SOLUTION — step by step

Step 1. Suppose $\sqrt{2} + \sqrt{3} = r$ is rational. Square both sides: $2 + 2\sqrt{6} + 3 = r^2$, so $5 + 2\sqrt{6} = r^2$.

Step 2. Solve for the surd: $\sqrt{6} = (r^2 - 5)/2$.

Step 3. r is rational, so $r^2 - 5$ is rational, and $(r^2 - 5)/2$ is rational.

Step 4. So $\sqrt{6}$ would be rational. But 6 is not a perfect square, so $\sqrt{6}$ is irrational (same contradiction proof as $\sqrt{2}$).

Step 5. Contradiction. Hence $\sqrt{2} + \sqrt{3}$ is irrational. ∎

» **Why this step?** Squaring converts a sum of two surds into ONE surd; the known irrationality of $\sqrt{6}$ then finishes the job.

Answer: $\sqrt{2} + \sqrt{3}$ is irrational (rationality would force $\sqrt{6} = (r^2-5)/2$ to be rational).

Hinglish: *Do surds ka sum prove karna hai? Square karo - √2+√3 ka square 5 + 2√6 deta hai. Agar sum rational hota toh √6 = (r²-5)/2 rational hota - jo impossible hai. Yehi pattern √2+√5, √3+√5 ke liye bhi chalega. Ek baar square, phir known irrational se maar do.*

QUESTION 12 | Difficult | Rationalising with clean collapse (T9)

Q12. Simplify completely: $4/(\sqrt{7} - \sqrt{3})$, and express the result in the form $a\sqrt{7} + b\sqrt{3}$ with rational a, b.

SOLUTION — step by step

Step 1. Multiply numerator and denominator by the conjugate $\sqrt{7} + \sqrt{3}$.

Step 2. Denominator: $(\sqrt{7} - \sqrt{3})(\sqrt{7} + \sqrt{3}) = 7 - 3 = 4$.

Step 3. Expression $= 4(\sqrt{7} + \sqrt{3})/4 = \sqrt{7} + \sqrt{3}$.

Step 4. So a = 1 and b = 1.

Step 5. Check numerically: $4/(2.646 - 1.732) = 4/0.914 \approx 4.376$, and $\sqrt{7} + \sqrt{3} \approx 2.646 + 1.732 = 4.376$ ✓.

» **Why this step?** Choosing the conjugate makes the denominator $a^2 - b^2$; here it cancels the numerator perfectly, so no fraction survives.

Answer: $\sqrt{7} + \sqrt{3}$ (a = 1, b = 1)

Hinglish: *Denominator ka conjugate multiply karo: (a-b)(a+b) = a²-b² = 7-3 = 4. Upar wala 4 aur neeche wala 4 cancel - answer √7 + √3, ekdum clean! Ye 'designed' questions mein hamesha ek magic cancellation hota hai; agar aisa nahi ho raha toh calculation check karo.*

QUESTION 13 | Difficult | Conjugate pairs x and 1/x (T14)

Q13. If $x = 2 + \sqrt{3}$, find the value of $x^2 + 1/x^2$ without evaluating x numerically beyond the given form.

SOLUTION — step by step

Step 1. First find 1/x by rationalising: $1/(2 + \sqrt{3}) = (2 - \sqrt{3})/((2 + \sqrt{3})(2 - \sqrt{3})) = (2 - \sqrt{3})/(4 - 3) = 2 - \sqrt{3}$.

Step 2. So x and 1/x are conjugates: $x = 2 + \sqrt{3}$, $1/x = 2 - \sqrt{3}$.

Step 3. Square both: $x^2 = 4 + 4\sqrt{3} + 3 = 7 + 4\sqrt{3}$; $(1/x)^2 = 7 - 4\sqrt{3}$.

Step 4. Add: $x^2 + 1/x^2 = 14$ (the surd terms cancel).

Step 5. Check numerically: $x \approx 3.732$, $x^2 \approx 13.93$, $1/x^2 \approx 0.0718$, sum ≈ 14.0 ✓.

» **Why this step?** Conjugate reciprocals have surd terms in opposite signs; squares keep that symmetry, so the middle terms cancel on adding.

Answer: 14

Hinglish: *Ye conjugate-pair trick poore algebra mein chalega. Pehle 1/x nikaalo - denominator 1 aaya, matlab x aur 1/x perfect conjugates. Square karke jodo: ±4√3 terms aapas mein uda do. Answer 14. Aage ke liye yaad rakho: $x^3 + 1/x^3 = (x + 1/x)^3 - 3(x + 1/x)$.*

QUESTION 14 | Difficult | Factored exponent equation (T11)

Q14. Solve for x: $4^x - 4^{(x-1)} = 24$.

SOLUTION — step by step

Step 1. Factor out the smaller power: $4^{(x-1)}(4^1 - 1) = 24$.

Step 2. So $4^{(x-1)} \times 3 = 24$, giving $4^{(x-1)} = 8$.

Step 3. Convert to one base: $4 = 2^2$ and $8 = 2^3$, so $2^{(2x-2)} = 2^3$.

Step 4. Equate exponents: $2x - 2 = 3$, so $x = 5/2$.

Step 5. Check: $4^{(5/2)} = (\sqrt{4})^5 = 32$; $4^{(3/2)} = (\sqrt{4})^3 = 8$; $32 - 8 = 24$ ✓.

» **Why this step?** Both terms share the base 4, so factoring the smaller power is legal and reduces two powers to one equation.

Answer: x = 5/2

Hinglish: *Do powers ka difference? Common power bahar nikalo - 4^(x-1) common liya, bracket mein 4-1 = 3 bacha. Ab 24/3 = 8. 4 aur 8 dono ko 2 ke base par lao: 2^(2x-2) = 2³. Exponents barabar: x = 5/2. Fraction mein aana bilkul theek hai - check line se confirm karo.*

QUESTION 15 | Difficult | Find a, b after rationalising (T9)

Q15. If $(\sqrt{3} + 1)/(\sqrt{3} - 1) = a + b\sqrt{3}$ where a, b are rational, find a and b.

SOLUTION — step by step

Step 1. Rationalise: multiply by $(\sqrt{3} + 1)/(\sqrt{3} + 1)$.

Step 2. Numerator: $(\sqrt{3} + 1)^2 = 3 + 2\sqrt{3} + 1 = 4 + 2\sqrt{3}$.

Step 3. Denominator: $3 - 1 = 2$.

Step 4. Expression $= (4 + 2\sqrt{3})/2 = 2 + \sqrt{3}$.

Step 5. Compare with $a + b\sqrt{3}$: $a = 2$, $b = 1$.

» **Why this step?** Two expressions of the form $a + b\sqrt{c}$ are equal exactly when the rational parts match and the surd coefficients match (uniqueness of the representation).

Answer: a = 2, b = 1

Hinglish: *Rationalise karo, simplify karo, phir 'compare coefficients' karo. 2 + √3 vs a + b√3: rational part rational se match (a=2), surd coefficient surd coefficient se (b=1). Ye uniqueness isliye chalti hai kyunki agar a≠a' hota toh √3 rational nikal jata - impossible.*

QUESTION 16 | High | Conjugate pair x, y (T14)

Q16. If $x = (\sqrt{3} + 1)/(\sqrt{3} - 1)$ and $y = (\sqrt{3} - 1)/(\sqrt{3} + 1)$, find $x^2 + y^2$ and xy.

SOLUTION — step by step

Step 1. Rationalise x: multiply by $(\sqrt{3} + 1)$: $x = (\sqrt{3} + 1)^2/(3 - 1) = (4 + 2\sqrt{3})/2 = 2 + \sqrt{3}$.

Step 2. Rationalise y: multiply by $(\sqrt{3} - 1)$: $y = (\sqrt{3} - 1)^2/(3 - 1) = (4 - 2\sqrt{3})/2 = 2 - \sqrt{3}$.

Step 3. x and y are conjugates: $xy = (2 + \sqrt{3})(2 - \sqrt{3}) = 4 - 3 = 1$.

Step 4. $x + y = 4$. Then $x^2 + y^2 = (x + y)^2 - 2xy = 16 - 2 = 14$.

Step 5. Check: $x \approx 3.732$, $y \approx 0.268$; $x^2 \approx 13.93$, $y^2 \approx 0.0718$, sum ≈ 14.0 ✓.

» **Why this step?** Reduce both fractions to conjugate form FIRST; then the symmetric identities $x^2 + y^2 = (x+y)^2 - 2xy$ do the rest cheaply.

Answer: $x^2 + y^2 = 14$ and $xy = 1$

Hinglish: *Dono fractions rationalise karo - $x = 2+\sqrt{3}$, $y = 2-\sqrt{3}$ niklega. Ab conjugate pair ke superpowers: $xy = 1$ hamesha, $x+y = 4$. x^2+y^2 ke liye direct square karne ki zaroorat nahi - $(x+y)^2 - 2xy = 16 - 2 = 14$. Symmetric identities se time bachta hai aur galti bhi nahi hoti.*

QUESTION 17 | High | Multi-base exponent relation (T11)

Q17. If $2^x = 3^y = 12^z = t$ ($t \neq 1$), prove that $2/x + 1/y = 1/z$.

SOLUTION — step by step

Step 1. Write each base as a power of the common value t: 2 = t^(1/x), 3 = t^(1/y), 12 = t^(1/z).

Step 2. Use the factorisation $12 = 2^2 \times 3$.

Step 3. So t^(1/z) = (t^(1/x))^2 × t^(1/y) = t^(2/x + 1/y).

Step 4. Same base t ($t \neq 1$) forces equal exponents: $1/z = 2/x + 1/y$. ∎

Step 5. Sanity check with numbers: $x = \log_2 12 \approx 3.585$, $y \approx 2.262$, $z = 1$ gives $2/3.585 + 1/2.262 \approx 0.558 + 0.442 = 1 = 1/1$ ✓.

» **Why this step?** The common value t converts three separate equations into one base; then the prime factorisation of 12 writes the relation.

Answer: $1/z = 2/x + 1/y$ (proved via t-powers and $12 = 2^2 \times 3$).

Hinglish: *Ye 'common value' technique Olympiad ka bread-and-butter hai. Teeno numbers ko t ke power mein likho, phir $12 = 2^2 \times 3$ ka use karo. Exponents add hote- hote relation ban jata hai. Base same aur $t \neq 1$ - tabhi exponent equate karna valid hai, ye condition likhna mat bhoolna.*

QUESTION 18 | High | Telescoping surd sum (T14)

Q18. Evaluate: $1/(\sqrt{2}+1) + 1/(\sqrt{3}+\sqrt{2}) + 1/(2+\sqrt{3})$.

SOLUTION — step by step

Step 1. Key identity: $1/(\sqrt{(n+1)}+\sqrt{n}) = \sqrt{(n+1)} - \sqrt{n}$ (multiply by the conjugate; denominator becomes n+1-n = 1).

Step 2. First term: $1/(\sqrt{2}+1) = \sqrt{2} - 1$.

Step 3. Second: $1/(\sqrt{3}+\sqrt{2}) = \sqrt{3} - \sqrt{2}$.

Step 4. Third: $1/(2+\sqrt{3}) = \sqrt{4} - \sqrt{3} = 2 - \sqrt{3}$.

Step 5. Sum: $(\sqrt{2} - 1) + (\sqrt{3} - \sqrt{2}) + (2 - \sqrt{3}) = 2 - 1 = 1$ (everything in between cancels - telescoping).

» **Why this step?** Each conjugate-rationalised term is a difference of consecutive roots; in the written sum every middle root appears once with plus and once with minus.

Answer: 1

Hinglish: *Ye 'telescoping' pattern dekho: har term rationalise karne par root ka difference banta hai: √2-1, √3-√2, 2-√3. Jodne par beech ke saare roots gayab, sirf first ka minus aur last ka plus bachta hai: 2 - 1 = 1. Agar 1/(√(n+1)+√n) type dikhe, turant conjugate socho.*

QUESTION 19 | HOTS | Infinite nested radical (T14)

Q19. Find the value of $x = \sqrt{(6 + \sqrt{(6 + \sqrt{(6 + \sqrt{(6 + ...)})})})}$ where the pattern continues forever.

SOLUTION — step by step

Step 1. The expression inside is the SAME as the whole: the tail $\sqrt{(6 + \sqrt{(6 + ...)})}$ equals x itself.

Step 2. So x satisfies $x = \sqrt{(6 + x)}$.

Step 3. Square: $x^2 = 6 + x$, i.e. $x^2 - x - 6 = 0$.

Step 4. Factorise: $(x - 3)(x + 2) = 0$, so $x = 3$ or $x = -2$.

Step 5. x is a square root, hence non-negative: $x = 3$.

Step 6. Check: $\sqrt{(6 + 3)} = \sqrt{9} = 3$ ✓ - the value is consistent with itself.

» **Why this step?** Self-similarity lets the infinite tail be replaced by x; the quadratic then follows, and the negative root is discarded on domain grounds.

Answer: x = 3

Hinglish: *Infinite nesting ka trick: andar ka poora structure wahi x hai! x = √(6+x), square karo, quadratic mil gayi. Do roots - 3 aur -2. Root negative nahi ho sakta, toh 3. Ye 'self-similar substitution' technique aage bhi kaam karegi - quadratic banao, domain se filter karo.*

QUESTION 20 | HOTS | Hidden perfect square (T14)

Q20. If $x = 5 + 2\sqrt{6}$, find the value of $\sqrt{x} - 1/\sqrt{x}$.

SOLUTION — step by step

Step 1. Spot the hidden square: $5 + 2\sqrt{6} = 3 + 2 + 2\sqrt{(3\times6/...}$ check: $(\sqrt{3} + \sqrt{2})^2 = 3 + 2\sqrt{6} + 2 = 5 + 2\sqrt{6}$ ✓.

Step 2. So $\sqrt{x} = \sqrt{3} + \sqrt{2}$.

Step 3. Rationalise: $1/\sqrt{x} = 1/(\sqrt{3} + \sqrt{2}) = \sqrt{3} - \sqrt{2}$ (conjugate; denominator $3 - 2 = 1$).

Step 4. $\sqrt{x} - 1/\sqrt{x} = (\sqrt{3} + \sqrt{2}) - (\sqrt{3} - \sqrt{2}) = 2\sqrt{2}$.

Step 5. Check numerically: $x \approx 9.899$, $\sqrt{x} \approx 3.146$, $1/\sqrt{x} \approx 0.318$, difference $\approx 2.828 = 2\sqrt{2}$ ✓.

» **Why this step?** Recognising $a + 2\sqrt{b}$ as $(\sqrt{m} + \sqrt{n})^2$ with $m + n = a$, $mn = b$ turns a messy root into two clean surds.

Answer: $2\sqrt{2}$

Hinglish: *$5 + 2\sqrt{6}$ ko dekho aur socho - kaunse do numbers ka sum 5, product 6? 3 aur 2! Matlab $(\sqrt{3}+\sqrt{2})^2$. Toh $\sqrt{x} = \sqrt{3}+\sqrt{2}$. Uska reciprocal conjugate hi hai $(\sqrt{3}-\sqrt{2})$. Difference: $2\sqrt{2}$. Ye 'hidden perfect square' spotting technique bahut powerful hai - m+n aur mn pattern yaad rakho.*

Chapter 4: Algebraic Identities and Factorisation

An identity is an equality that is true for EVERY value of the letters - not an equation to solve but a tool to rewrite expressions. The uploaded PDF builds the identity toolbox: squares of binomials $(a \pm b)^2$, the difference of squares $a^2 - b^2$, the product $(x + a)(x + b)$, the trinomial square $(a + b + c)^2$, the cubes $(a \pm b)^3$, the sum and difference of cubes $a^3 \pm b^3$, and the crown jewel $a^3 + b^3 + c^3 - 3abc$ with its zero-sum shortcut. It then uses these to factorise polynomials and to evaluate products like 98×102 without a calculator. This practice book runs the toolbox at full power: three-variable cubes, nested substitutions, fourth powers, telescoping evaluations, and proof-style questions that examiners reserve for the last section.

Figure 4.1: The area model of $(a + b)^2$ - a big square cut into a^2, two ab strips and b^2.

4.1 Complete Concept Cheat Sheet (read this before any question)

4.1.1 The square identities and the area model

What is it? $(a + b)^2 = a^2 + 2ab + b^2$ and $(a - b)^2 = a^2 - 2ab + b^2$. **Why true (not just memorised)?** Draw a square of side a + b and cut it: you get a^2 (bottom-left), b^2 (top-right), and TWO ab rectangles - the picture proves the formula. For $(a - b)^2$, start with a^2, remove two strips of width b (areas ab each), then add back b^2 once because it was subtracted twice: $a^2 - 2ab + b^2$. **Companion facts:** $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$ and $(a+b)^2 - (a-b)^2 = 4ab$. The second is the 'multiply quick' engine: $5.5^2 - 4.5^2 = 4 \times 5.5 \times 4.5 = 99$.

When do I use them? Expanding squares, computing squares of numbers near round numbers ($98^2 = (100-2)^2$), building perfect squares ($x^2 + 6x + 9 = (x+3)^2$), and reverse-engineering $a^2 + b^2$ from (a+b) and ab. **Tiny example.** $(3x - 4y)^2 = 9x^2 - 24xy + 16y^2$: square each term, double the product, keep the minus sign.

Figure 4.2: The area model of $(a - b)^2$ - remove two ab strips, add back the overlapping b^2.

> **Sign discipline**
> $(a - b)^2$ has ONE minus, on the middle term only. Writing $a^2 - 2ab - b^2$ is the classic disaster. And $(a+b)^2 \neq a^2 + b^2$ - the cross term 2ab is the whole point.

4.1.2 Difference of squares and (x + a)(x + b)

What is it? $a^2 - b^2 = (a + b)(a - b)$. Verify by expanding: the middle terms a b and -ba cancel exactly. **Use it:** to factorise any two-square difference ($49x^2 - 36y^2 = (7x+6y)(7x-6y)$), to multiply numbers equidistant from a round number ($51 \times 49 = 50^2 - 1^2 = 2499$), and to chain: $x^4 - 81 = (x^2-9)(x^2+9) = (x-3)(x+3)(x^2+9)$. **The (x + a)(x + b) rule:** $= x^2 + (a+b)x + ab$ - coefficient of x is the SUM of the constants, constant term is the PRODUCT. **Tiny example.** $(x + 4)(x + 7) = x^2 + 11x + 28$.

4.1.3 The trinomial square (a + b + c)²

What is it? $(a + b + c)^2 = a^2 + b^2 + c^2 + 2ab + 2bc + 2ca$. Three squares plus twice every product PAIR - three pairs, not two. **Where does it come from?** Group as (a + b) + c and apply the binomial square twice. **Reverse use:** $a^2 + b^2 + c^2 = (a+b+c)^2 - 2(ab+bc+ca)$, which converts between 'sum of squares' and 'sum of pairwise

products' - the most used bridge in three-variable problems. **Consequence:** $a^2 + b^2 + c^2 - ab - bc - ca = \frac{1}{2}[(a-b)^2 + (b-c)^2 + (c-a)^2] \geq 0$, and equals zero only when a = b = c. **Tiny example.** $(2p + 3q - 4r)^2 = 4p^2 + 9q^2 + 16r^2 + 12pq - 24qr - 16pr$: the sign of each cross term is the product of the signs of its parents.

4.1.4 The cube identities

What is it? $(a + b)^3 = a^3 + 3a^2b + 3ab^2 + b^3 = a^3 + b^3 + 3ab(a + b)$; $(a - b)^3 = a^3 - 3a^2b + 3ab^2 - b^3 = a^3 - b^3 - 3ab(a - b)$. Read the signs: all plus for $(a+b)^3$; alternating - + - for $(a-b)^3$ with the LAST term negative. **Numerical use:** $103^3 = (100 + 3)^3 = 1000000 + 90000 + 2700 + 27 = 1092727$; $99^3 = (100 - 1)^3 = 1000000 - 30000 + 300 - 1 = 970299$. **Reverse use:** $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$ - the bridge that computes sums of cubes from the pair-sum and product.

Figure 4.3: $(a + b)^3$ as a cube of side a + b, sliced into 8 blocks: a^3, $3a^2b$, $3ab^2$, b^3.

> **The two cube bridges**
> Given a + b and ab: $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$. Given a - b and ab: $a^3 - b^3 = (a-b)^3 + 3ab(a-b)$. These two lines solve half of all 'find the cube sum' exam questions.

4.1.5 Sum and difference of cubes

What is it? $a^3 + b^3 = (a + b)(a^2 - ab + b^2)$ and $a^3 - b^3 = (a - b)(a^2 + ab + b^2)$. **How to remember the signs:** the binomial keeps the SIGN of the cube; inside the trinomial, the middle term always gets the OPPOSITE sign, and the other two are positive squares. **Verification for a^3+b^3:** expand $(a+b)(a^2-ab+b^2) = a^3 - a^2b + ab^2 + a^2b - ab^2 + b^3 = a^3 + b^3$ - the middle terms cancel pairwise. **Tiny example.** $8a^3 + 27b^3 = (2a)^3 + (3b)^3 = (2a + 3b)(4a^2 - 6ab + 9b^2)$.

4.1.6 The crown identity: $a^3 + b^3 + c^3 - 3abc$

What is it? $a^3 + b^3 + c^3 - 3abc = (a + b + c)(a^2 + b^2 + c^2 - ab - bc - ca)$. **The zero-sum shortcut:** if a + b + c = 0, the right side vanishes, so $a^3 + b^3 + c^3 = 3abc$ exactly. This turns monsters like $12^3 + 1^3 + (-13)^3$ into one multiplication: $3 \times 12 \times 1 \times (-13) = -468$. **The converse (HOTS):** since $a^2+b^2+c^2-ab-bc-ca = \frac{1}{2}[(a-b)^2+(b-c)^2+(c-a)^2]$, the identity forces: if $a^3+b^3+c^3 = 3abc$ then either a + b + c = 0 or a = b = c. **Related identity:** $(a+b+c)^3 = a^3+b^3+c^3 + 3(a+b)(b+c)(c+a)$ - another workhorse.

4.1.7 Factorisation toolkit (the five moves)

Move 1 - Common factor: pull out the HCF first, ALWAYS first. $6x^3 + 9x^2 = 3x^2(2x + 3)$. **Move 2 - Grouping:** rearrange terms so each group shares a factor: $a^2 - b^2 + 2bc - c^2 = a^2 - (b-c)^2 = (a+b-c)(a-b+c)$. **Move 3 - Identity spotting:** perfect squares $(9x^2 + 6x + 1 = (3x+1)^2)$ and cube forms $(x^3 - 8)$. **Move 4 - Middle-term splitting:** for $x^2 + px + q$, find two numbers with sum p and product q: $x^2 + 7x + 12 = (x+3)(x+4)$. For a general quadratic $ax^2 + bx + c$, split b into two numbers whose product is ac. **Move 5 - Substitution:** bracket a repeating chunk, factorise the simpler piece, then back-substitute: $(x^2-2x)^2 - 23(x^2-2x) + 120$ becomes $y^2 - 23y + 120$ with $y = x^2 - 2x$.

The golden sequence: common factor → identity → grouping → middle-term split → substitution. Try in that order and stop as soon as it works. **Always finish:** expand your answer mentally (or on rough paper) and check it rebuilds the original. **Tiny example.** $2x^3 - 8x = 2x(x^2 - 4) = 2x(x+2)(x-2)$ - three moves in one line.

4.1.8 The x + 1/x engine

What is it? A family of chained identities. Given $x + 1/x = s$: $x^2 + 1/x^2 = s^2 - 2$; $x^3 + 1/x^3 = s^3 - 3s$; $x^4 + 1/x^4 = (s^2-2)^2 - 2$; and in general each power follows from the previous two by multiplication and subtraction. Given $x - 1/x = d$: $x^2 + 1/x^2 = d^2 + 2$ and $x^3 - 1/x^3 = d^3 + 3d$. **Why the chains work:** divide $(x + 1/x)^2 = x^2 + 2 + 1/x^2$ by nothing - the middle term is a CONSTANT 2, which is the magic. **Tiny example.** $x + 1/x = 3 \Rightarrow x^2 + 1/x^2 = 7 \Rightarrow x^3 + 1/x^3 = 27 - 9 = 18 \Rightarrow x^4 + 1/x^4 = 49 - 2 = 47$.

4.1.9 Exam tips for this chapter

1. Write the identity NUMBER you are using in the margin - examiners award method marks by identity, not by memory.
2. Common factor first, always: pulling out the HCF shrinks every later step.
3. Numerical products: convert to (round ± small) form before choosing the identity.
4. In three-variable squares, count your cross terms: there must be exactly three pairs, each appearing twice.
5. Sign of a cross term = product of the signs of its two parents. Never 'guess it'.
6. Factorisation is not finished until the last check: expand back, compare term by term.
7. Zero-sum problems: the instant you see $a + b + c = 0$, write $a^3 + b^3 + c^3 = 3abc$ and move on.
8. For x + 1/x chains, keep a running table: s, s^2-2, s^3-3s, ... - never recompute from scratch.

4.2 Question Types / Concept Map for Chapter 4

Fourteen question types cover everything from a two-mark expansion to a five-mark proof. Name the type in your head, then execute.

Type	Name	How to recognise it in the question	Core method
T1	Binomial squares	Expand $(a \pm b)^2$ with letters or numbers	$a^2 \pm 2ab + b^2$
T2	Difference of squares	Two perfect squares subtracted; products like 51 × 49	(a+b)(a-b) both directions
T3	Near-number evaluation	98^2, 103^3, 9.8 × 10.2 without calculator	Write as (100 ± k), apply identities
T4	(x + a)(x + b) products	Two binomials sharing a letter term	$x^2 + (a+b)x + ab$
T5	Trinomial squares	Expand or build $(a + b + c)^2$	Three squares + 2 of each pair
T6	Middle-term splitting	Factorise $x^2 + px + q$ or $ax^2 + bx + c$	Two numbers: sum = p, product = q (or ac)
T7	Cube expansion	Expand $(a \pm b)^3$	$a^3 \pm 3a^2b + 3ab^2 \pm b^3$
T8	Sum/difference of cubes	Factorise $a^3 \pm b^3$	$(a \pm b)(a^2 \mp ab + b^2)$
T9	Crown identity	$a^3 + b^3 + c^3 - 3abc$; zero-sum shortcuts	$(a+b+c)(a^2+b^2+c^2-ab-bc-ca)$
T10	x ± 1/x chains	Given x + 1/x or x - 1/x, find powers	Chain: s^2-2, s^3-3s, ...

Type	Name	How to recognise it in the question	Core method
T11	Symmetric relations	Given a+b, ab; find a^2+b^2, a^3+b^3	$(a\pm b)^2$ and cube bridges
T12	Hard factorisation	Grouping, substitution, a^4+a^2+1 forms	Five moves in golden order
T13	Applied / modelling	Areas, volumes, ages, consecutive integers	Translate to expressions, then T1-T9
T14	Proof / HOTS	'Show that...', converses, chained identities	Label every identity; end with QED line

4.3 Solved Practice Set (20 Questions)

Har question apne BOX me hai, uske turant baad uska boxed step-by-step solution. Pehle question ko khud solve karne ki koshish karo - jab atak jao tabhi solution dekho. Har step ke saath "Why this step?" diya hai, aur end me Hinglish note jo concept ko dil me basa dega.

QUESTION 1 | Basic | Binomial square (T1)

Q1. Expand: (i) $(2x + 3y)^2$, (ii) $(5a - 2b)^2$, (iii) $(x/2 + y/3)^2$.

SOLUTION — step by step

Step 1. Given: three binomial squares. Use $(a \pm b)^2 = a^2 \pm 2ab + b^2$.

Step 2. (i) $a = 2x$, $b = 3y$: $4x^2 + 2(2x)(3y) + 9y^2 = 4x^2 + 12xy + 9y^2$.

Step 3. (ii) $a = 5a$, $b = 2b$: $25a^2 - 2(5a)(2b) + 4b^2 = 25a^2 - 20ab + 4b^2$.

Step 4. (iii) $a = x/2$, $b = y/3$: $x^2/4 + 2(x/2)(y/3) + y^2/9 = x^2/4 + xy/3 + y^2/9$.

» **Why this step?** Square each term, then DOUBLE the product with the binomial's sign - the middle term is where all errors live.

Answer: (i) $4x^2 + 12xy + 9y^2$ (ii) $25a^2 - 20ab + 4b^2$ (iii) $x^2/4 + xy/3 + y^2/9$

Hinglish: *Simple rule: pehla square, plus/minus double product, doosra square. $(2x+3y)^2$ mein $(2x)^2 = 4x^2$, cross term $2\times2x\times3y = 12xy$. Fraction wale case mein fractions ko square karo, cross term fractions ko multiply karta hai - $xy/3$.*

QUESTION 2 | Basic | Difference of squares (T2)

Q2. Factorise: (i) $49x^2 - 36y^2$, (ii) $121 - 25p^2$, (iii) $9(a+b)^2 - 4(a-b)^2$.

SOLUTION — step by step

Step 1. Given: three two-square differences. Use $a^2 - b^2 = (a+b)(a-b)$.

Step 2. (i) $(7x)^2 - (6y)^2 = (7x + 6y)(7x - 6y)$.

Step 3. (ii) $11^2 - (5p)^2 = (11 + 5p)(11 - 5p)$.

Step 4. (iii) $[3(a+b)]^2 - [2(a-b)]^2 = (3a+3b + 2a-2b)(3a+3b - 2a+2b) = (5a + b)(a + 5b)$.

» **Why this step?** Spot each term as a perfect square first; the brackets then collapse by the identity - in (iii) the inner brackets simplify after factoring.

Answer: (i) (7x+6y)(7x-6y) (ii) (11+5p)(11-5p) (iii) (5a+b)(a+5b)

Hinglish: *Dono terms ko square form mein likho: $(7x)^2$ aur $(6y)^2$. Phir (a+b)(a-b). (iii) mein brackets ke andar simplify karo: 3a+3b+2a-2b = 5a+b aur 3a+3b-2a+2b = a+5b. Answer (5a+b)(a+5b) - brackets kholna hi asli kaam tha.*

QUESTION 3 | Basic | Near-number squares (T3)

Q3. Evaluate using identities: (i) 102^2, (ii) 98^2, (iii) 9.7^2.

SOLUTION — step by step

Step 1. Given: three squares near round numbers. Write each as $(100 \pm k)$ or $(10 \pm k)$.

Step 2. (i) $102^2 = (100+2)^2 = 10000 + 400 + 4 = 10404$.

Step 3. (ii) $98^2 = (100-2)^2 = 10000 - 400 + 4 = 9604$.

Step 4. (iii) $9.7^2 = (10 - 0.3)^2 = 100 - 6 + 0.09 = 94.09$.

» **Why this step?** The identity replaces one hard multiplication with three easy pieces; the cross term $2\times100\times k$ does the heavy lifting.

Answer: (i) 10404 (ii) 9604 (iii) 94.09

Hinglish: *Round number ke paas wale squares ka JUGAAD: $(100\pm2)^2$. $98^2 = 10000 - 400 + 4 = 9604$. Dhyan do: 9.7^2 mein cross term $2\times10\times0.3 = 6$, aur $0.3^2 = 0.09$ - decimals ka square chhota hota hai, 0.9 nahi!*

QUESTION 4 | Basic | Square-sum identities (T1/T11)

Q4. If a + b = 10 and a - b = 4, find (i) $a^2 + b^2$ and (ii) ab - all in one line each, using identities.

SOLUTION — step by step

Step 1. Given: the pair-sum and pair-difference. Needed: sum of squares and the product.

Step 2. Add the two squares identities: $(a+b)^2 + (a-b)^2 = 2(a^2 + b^2)$.

Step 3. So $a^2 + b^2 = (10^2 + 4^2)/2 = (100 + 16)/2 = 58$.

Step 4. Subtract: $(a+b)^2 - (a-b)^2 = 4ab$.

Step 5. So $ab = (100 - 16)/4 = 84/4 = 21$.

Step 6. Check: $(a+b)(a-b) = 10\times4 = 40 = a^2 - b^2$; with $a^2+b^2 = 58$ we get $a^2 = 49$, $b^2 = 9$, a = 7, b = 3 ✓ (ab = 21 ✓).

» **Why this step?** The two companion identities convert (a+b, a-b) directly into (a^2+b^2, ab) without ever solving for a and b.

Answer: $a^2 + b^2 = 58$, $ab = 21$

Hinglish: *Ye identity-pair exam ka shortcut bomb hai: $(a+b)^2+(a-b)^2 = 2(a^2+b^2)$ aur $(a+b)^2-(a-b)^2 = 4ab$. Dono values plug karo - 58 aur 21. Solving for a, b optional hai, sirf check ke liye.*

QUESTION 5 | Basic | Difference-of-squares multiplication (T3)

Q5. Evaluate without direct multiplication: (i) 51 × 49, (ii) 103 × 97, (iii) 5.1 × 4.9.

SOLUTION — step by step

Step 1. Given: three products equidistant from a round number. Use $(a+b)(a-b) = a^2 - b^2$.

Step 2. (i) $51 \times 49 = (50+1)(50-1) = 2500 - 1 = 2499$.

Step 3. (ii) $103 \times 97 = (100+3)(100-3) = 10000 - 9 = 9991$.

Step 4. (iii) $5.1 \times 4.9 = (5 + 0.1)(5 - 0.1) = 25 - 0.01 = 24.99$.

» **Why this step?** Rewriting as (round ± same offset) converts multiplication into one subtraction of squares.

Answer: (i) 2499 (ii) 9991 (iii) 24.99

Hinglish: *Dono numbers ka average round number hai? Toh $(a+b)(a-b) = a^2-b^2$. 51×49: average 50, offset 1. 2500-1 = 2499. Ye trick multiplication tables se tez hai.*

QUESTION 6 | Moderate | Cube expansion (T7)

Q6. Expand: $(2x - 3y)^3$ and $(x + 2y)^3$.

SOLUTION — step by step

Step 1. Given: two cube expansions.

Step 2. $(2x - 3y)^3$: a = 2x, b = 3y. $(2x)^3 = 8x^3$; $3(2x)^2(3y) = 36x^2y$; $3(2x)(3y)^2 = 54xy^2$; $(3y)^3 = 27y^3$.

Step 3. With alternating signs: $8x^3 - 36x^2y + 54xy^2 - 27y^3$.

Step 4. $(x + 2y)^3$: $x^3 + 3x^2(2y) + 3x(2y)^2 + (2y)^3 = x^3 + 6x^2y + 12xy^2 + 8y^3$.

» **Why this step?** Write powers of a and b separately first ($8x^3$, $36x^2y$, ...), then attach the sign pattern + - + -.

Answer: $8x^3 - 36x^2y + 54xy^2 - 27y^3$ and $x^3 + 6x^2y + 12xy^2 + 8y^3$

Hinglish: *Cube mein har term ka coefficient pattern 1-3-3-1. Powers pehle nikaalo: $(2x)^2 = 4x^2$, $(3y)^2 = 9y^2$. Coefficients: 8, 3×4×3 = 36, 3×2×9 = 54, 27. Minus wale case mein sirf SIGN pattern badalta hai.*

QUESTION 7 | Moderate | x + 1/x chain (T10)

Q7. If $x + 1/x = 5$, find (i) $x^2 + 1/x^2$ and (ii) $x^4 + 1/x^4$.

SOLUTION — step by step

Step 1. Given: $x + 1/x = 5$. Square both sides: $x^2 + 2 + 1/x^2 = 25$.

Step 2. (i) $x^2 + 1/x^2 = 25 - 2 = 23$.

Step 3. Square again: $(x^2 + 1/x^2)^2 = x^4 + 2 + 1/x^4 = 23^2 = 529$.

Step 4. (ii) $x^4 + 1/x^4 = 529 - 2 = 527$.

» **Why this step?** Every square introduces a constant middle term 2 (because $x \cdot 1/x = 1$); subtract it to keep the chain clean.

Answer: (i) 23 (ii) 527

Hinglish: *Chain method: square karo, 2 minus karo, phir dobara square karo, phir 2 minus. 5 → 23 → 527. Ek hi line mein: $(x+1/x)^2 - 2 = x^2+1/x^2$. Ye chain chalti hi jati hai - x^6 ke liye x^3 wale bridge ka use hota hai.*

QUESTION 8 | Moderate | Trinomial square reverse (T5/T11)

Q8. If $a + b + c = 9$ and $ab + bc + ca = 26$, find $a^2 + b^2 + c^2$.

SOLUTION — step by step

Step 1. Given: the pair-sum of three variables. Use the trinomial square identity:

Step 2. $(a + b + c)^2 = a^2 + b^2 + c^2 + 2(ab + bc + ca)$.

Step 3. $81 = (a^2+b^2+c^2) + 2(26) = (a^2+b^2+c^2) + 52$.

Step 4. So $a^2 + b^2 + c^2 = 81 - 52 = 29$.

» **Why this step?** The identity links the three quantities (sum, pair-sum, sum of squares) - knowing any two gives the third.

Answer: 29

Hinglish: *Trinomial square ka reverse: sum of squares = $(sum)^2$ - 2(pair-sum). 81 - 52 = 29. Ye formula har three-variable question mein kaam aayega - yaad kar lo.*

QUESTION 9 | Moderate | Cube sum bridge (T11)

Q9. If $a + b = 12$ and $ab = 27$, find $a^3 + b^3$ and $a^2 + b^2$.

SOLUTION — step by step

Step 1. Given: pair-sum and product. Cube bridge: $a^3 + b^3 = (a+b)^3 - 3ab(a+b)$.

Step 2. $a^3 + b^3 = 12^3 - 3(27)(12) = 1728 - 972 = 756$.

Step 3. Square identity: $a^2 + b^2 = (a+b)^2 - 2ab = 144 - 54 = 90$.

» **Why this step?** Both bridges express symmetric quantities in terms of (sum, product) - no need to solve for a and b.

Answer: $a^3 + b^3 = 756$, $a^2 + b^2 = 90$

Hinglish: *Symmetric quantities ka rule: sab kuch (sum, product) se nikhlo. Cube bridge: $(a+b)^3 - 3ab(a+b)$. 1728 - 972 = 756. Square: 144 - 54 = 90. Solving a, b se bhi hoga but identity wala path 2 line ka hai.*

QUESTION 10 | Moderate | Crown identity evaluation (T9)

Q10. Using a suitable identity, evaluate $12^3 + 1^3 + (-13)^3$.

SOLUTION — step by step

Step 1. Spot the zero sum: $12 + 1 + (-13) = 0$.

Step 2. Zero-sum shortcut: if $a + b + c = 0$ then $a^3 + b^3 + c^3 = 3abc$.

Step 3. So the value $= 3 \times 12 \times 1 \times (-13) = -468$.

» **Why this step?** The identity $a^3+b^3+c^3-3abc = (a+b+c)(...)$ collapses to 0 on the left whenever $a+b+c = 0$.

Answer: -468

Hinglish: *Numbers dekho: 12, 1, -13 - sum zero! Turant shortcut: cubes ka sum = 3abc = -468. Bina koi cube nikaale. Ye 'zero-sum spotting' exam mein 3 second ka kaam hai - 5 marks bachate hain.*

QUESTION 11 | Difficult | Cubic factorisation by root (T12)

Q11. Factorise completely: $x^3 - 3x^2 - 9x - 5$.

SOLUTION — step by step

Step 1. Hunt for an integer root: try $x = -1$: $(-1)^3 - 3(1) - 9(-1) - 5 = -1 - 3 + 9 - 5 = 0$ ✓. So $(x + 1)$ is a factor.

Step 2. Divide: $x^3 - 3x^2 - 9x - 5 = (x + 1)(x^2 - 4x - 5)$.

Step 3. (Verify by expanding: $x^3 - 4x^2 - 5x + x^2 - 4x - 5 = x^3 - 3x^2 - 9x - 5$ ✓.)

Step 4. Factorise the quadratic: $x^2 - 4x - 5$: sum -4, product -5 → -5 and +1: $(x - 5)(x + 1)$.

Step 5. Final: $(x + 1)^2(x - 5)$.

» **Why this step?** An integer root r of a polynomial always yields the factor (x - r); testing small integers (0, ±1, ±2) is the fastest opening move.

Answer: $(x + 1)^2(x - 5)$

Hinglish: *Cubic factorisation ka opening: chhote integers try karo. x = -1 par zero mila, toh (x+1) factor. Long division ya coefficient comparison se quadratic mila: x^2-4x-5, wo phir middle-term split. Final $(x+1)^2(x-5)$.*

QUESTION 12 | Difficult | Substitution factorisation (T12)

Q12. Factorise completely: $(x^2 - 2x)^2 - 23(x^2 - 2x) + 120$.

SOLUTION — step by step

Step 1. Substitute $y = x^2 - 2x$. The expression becomes $y^2 - 23y + 120$.

Step 2. Split the middle: two numbers with sum -23, product 120: -8 and -15.

Step 3. $y^2 - 23y + 120 = (y - 8)(y - 15)$.

Step 4. Back-substitute: $(x^2 - 2x - 8)(x^2 - 2x - 15)$.

Step 5. Factor each: $x^2 - 2x - 8 = (x-4)(x+2)$; $x^2 - 2x - 15 = (x-5)(x+3)$.

Step 6. Final: $(x - 4)(x + 2)(x - 5)(x + 3)$.

» **Why this step?** The repeated chunk $x^2 - 2x$ is the signal for substitution; every later step is then standard quadratic work.

Answer: $(x - 4)(x + 2)(x - 5)(x + 3)$

Hinglish: *(x^2-2x) do baar repeat ho raha hai - isko y maan lo. Quadratic ho gayi easy. Factor karke WAPAS substitute karna zaroori hai, aur phir dono quadratics ko bhi factor karo - 'completely' tabhi hota hai. Char linear factors final.*

QUESTION 13 | Difficult | Cube bridge with scaled variables (T11)

Q13. If $2x + 3y = 13$ and $xy = 6$, find $8x^3 + 27y^3$.

SOLUTION — step by step

Step 1. Recognise $8x^3 + 27y^3 = (2x)^3 + (3y)^3$.

Step 2. Cube bridge: $A^3 + B^3 = (A+B)^3 - 3AB(A+B)$ with $A = 2x$, $B = 3y$.

Step 3. $(A + B)^3 = 13^3 = 2197$.

Step 4. $AB = (2x)(3y) = 6xy = 36$.

Step 5. $(2x)^3 + (3y)^3 = 2197 - 3(36)(13) = 2197 - 1404 = 793$.

» **Why this step?** Scaling turns the question into the standard (sum, product) bridge - AB is where xy enters.

Answer: 793

Hinglish: *$8x^3 = (2x)^3$, $27y^3 = (3y)^3$ - scaling pehchano. Bridge lagao: $(2x+3y)^3 - 3(2x)(3y)(2x+3y)$. AB = 6xy = 36. 2197 - 1404 = 793. Cubes nikaale bina answer!*

QUESTION 14 | Difficult | Cube pair simplify (T7)

Q14. Simplify completely: $(a + b)^3 + (a - b)^3$.

SOLUTION — step by step

Step 1. Expand $(a+b)^3$: $a^3 + 3a^2b + 3ab^2 + b^3$.

Step 2. Expand $(a-b)^3$: $a^3 - 3a^2b + 3ab^2 - b^3$.

Step 3. Add: the $\pm 3a^2b$ and $\pm b^3$ terms cancel; survivors are $2a^3 + 6ab^2$.

Step 4. Factor the result: $2a(a^2 + 3b^2)$.

» **Why this step?** Adding opposite-sign expansions kills the odd terms - the same mechanism as $(a+b)^2 + (a-b)^2 = 2(a^2+b^2)$.

Answer: $2a^3 + 6ab^2 = 2a(a^2 + 3b^2)$

Hinglish: *Sum mein odd terms ($\pm 3a^2b$, $\pm b^3$) cancel ho jate hain, even terms double. Result $2a^3 + 6ab^2$, factor: $2a(a^2+3b^2)$. Ye square-identity ka cube bhai hai: $(a+b)^3+(a-b)^3 = 2a(a^2+3b^2)$.*

QUESTION 15 | Difficult | Crown identity as factorisation (T9)

Q15. Factorise: $a^3 - 8b^3 - 64c^3 - 24abc$.

SOLUTION — step by step

Step 1. See the shape: $a^3 + (-2b)^3 + (-4c)^3 - 3 \cdot a \cdot (-2b) \cdot (-4c)$.

Step 2. Check the cross term: $3 \times a \times (-2b) \times (-4c) = 24abc$, and the expression has $-24abc$ ✓.

Step 3. Apply the crown identity with x = a, y = -2b, z = -4c:

Step 4. $x^3+y^3+z^3-3xyz = (x+y+z)(x^2+y^2+z^2-xy-yz-zx)$.

Step 5. x + y + z = a - 2b - 4c.

Step 6. $x^2+y^2+z^2 = a^2+4b^2+16c^2$; xy = -2ab, yz = 8bc, zx = -4ca; so -xy-yz-zx = +2ab - 8bc + 4ca.

Step 7. Final: $(a - 2b - 4c)(a^2 + 4b^2 + 16c^2 + 2ab - 8bc + 4ca)$.

» **Why this step?** The crown identity factorises ANY three cubes whose product term appears as -3xyz - negative letters are welcome.

Answer: $(a - 2b - 4c)(a^2 + 4b^2 + 16c^2 + 2ab - 8bc + 4ca)$

Hinglish: *Teen cubes dikh rahe hain: a^3, $(-2b)^3$, $(-4c)^3$ aur cross term $-24abc = -3xyz$. Crown identity seedha lag jata hai. Negative letters ke signs cross terms mein multiply hote hain: $yz = (-2b)(-4c) = +8bc$, isliye $-yz = -8bc$. Sign discipline hi yahan ka skill hai.*

QUESTION 16 | High | Cube of x + 1/x (T10)

Q16. Expand $(x + 1/x)^3$ and hence find $x^3 + 1/x^3$ when $x + 1/x = 3$.

SOLUTION — step by step

Step 1. Use $(a+b)^3 = a^3 + b^3 + 3ab(a+b)$ with a = x, b = 1/x, ab = 1:

Step 2. $(x + 1/x)^3 = x^3 + 1/x^3 + 3(x + 1/x)$.

Step 3. Given $x + 1/x = 3$: cube both sides directly: $27 = x^3 + 1/x^3 + 3 \times 3$.

Step 4. $x^3 + 1/x^3 = 27 - 9 = 18$.

» **Why this step?** ab = 1 compresses the whole expansion to (cubes) + 3(sum) - the cleanest bridge in the chapter.

Answer: $x^3 + 1/x^3 = 18$

Hinglish: *$(x+1/x)^3 = x^3+1/x^3+3(x+1/x)$. Given value 3: $27 = x^3+1/x^3+9$. Answer 18. Yaad rakho: cube bridge s → s^3 - 3s hota hai. s = 3: 27-9 = 18.*

QUESTION 17 | High | Crown identity with product (T9)

Q17. If $x + y + z = 15$, $x^2 + y^2 + z^2 = 83$ and $xyz = 30$, find $x^3 + y^3 + z^3$.

SOLUTION — step by step

Step 1. Pair-sum: $xy + yz + zx = [(x+y+z)^2 - (x^2+y^2+z^2)]/2 = (225 - 83)/2 = 71$.

Step 2. Crown identity: $x^3+y^3+z^3 - 3xyz = (x+y+z)(x^2+y^2+z^2 - xy-yz-zx) = 15(83 - 71) = 180$.

Step 3. So $x^3+y^3+z^3 = 180 + 3xyz = 180 + 90 = 270$.

» **Why this step?** Two bridges in sequence: first recover the pair-sum, then the crown identity, then add back 3xyz.

Answer: 270

Hinglish: *Pehle pair-sum: (225-83)/2 = 71. Crown se cubes-minus-product: 15×12 = 180. Phir 3xyz = 90 wapas jodo: 270. Teen givens, teen steps - symmetric quantity ka poora game.*

QUESTION 18 | High | Cubic factorisation, three roots (T12)

Q18. Factorise completely: $x^3 - 6x^2 + 11x - 6$.

SOLUTION — step by step

Step 1. Test integer roots. x = 1: 1 - 6 + 11 - 6 = 0 ✓, so (x - 1) is a factor.

Step 2. Divide: $x^3 - 6x^2 + 11x - 6 = (x - 1)(x^2 - 5x + 6)$.

Step 3. Split the middle: $x^2 - 5x + 6 = (x - 2)(x - 3)$.

Step 4. Final: $(x - 1)(x - 2)(x - 3)$.

Step 5. Check: roots 1, 2, 3 sum to 6 = -(-6) ✓, pairwise products 11 ✓, product 6 ✓ (Vieta consistency).

» **Why this step?** A cubic with three integer roots yields to root-hunting + division + quadratic split - three standard moves.

Answer: (x - 1)(x - 2)(x - 3)

Hinglish: *x = 1 pehle try karo (sum of coefficients zero hai tab). Factor mila, divide karo, quadratic split: (x-2)(x-3). Final teen linear factors. Vieta check: roots ka sum 6, pair-sum 11, product 6 - sab original coefficients se match.*

QUESTION 19 | HOTS | Sum-cube identity (T14)

Q19. Prove that $(a + b + c)^3 = a^3 + b^3 + c^3 + 3(a + b)(b + c)(c + a)$.

SOLUTION — step by step

Step 1. Start from the right side's known relative: $(a+b+c)^3 = a^3+b^3+c^3 + 3(a+b+c)(ab+bc+ca) - 3abc$ (standard expansion).

Step 2. Expand $3(a+b)(b+c)(c+a)$: as computed in Q49, it equals $3(ab+bc+ca)(a+b+c) - 3abc$.

Step 3. Indeed: $(a+b)(b+c)(c+a) = (a+b+c)(ab+bc+ca) - abc$ (expand and collect).

Step 4. Then $3(a+b)(b+c)(c+a) = 3(a+b+c)(ab+bc+ca) - 3abc$.

Step 5. Adding $a^3+b^3+c^3$ gives exactly $(a+b+c)^3$. ∎

Step 6. Direct verification with numbers: a = 1, b = 2, c = 3: LHS = 216. RHS = 36 + 3(3)(5)(4) = 36 + 180 = 216 ✓.

» **Why this step?** The two forms differ only in packaging: one counts pair-sums with the total sum, the other uses three pairwise sums.

Answer: Proved (both forms differ by the identity $(a+b)(b+c)(c+a) = (a+b+c)(ab+bc+ca) - abc$).

Hinglish: *Do forms hain - ek $(a+b+c)(ab+bc+ca)$ wali, ek $(a+b)(b+c)(c+a)$ wali. Dono equal hain kyunki $(a+b)(b+c)(c+a) = (a+b+c)(ab+bc+ca) - abc$. Numeric check 1,2,3 se 216 dono taraf - proof pakka.*

QUESTION 20 | HOTS | Sum of squares is zero (T5/T14)

Q20. If $x^2 + y^2 + 2x - 4y + 5 = 0$, find $x^3 + y^3$.

SOLUTION — step by step

Step 1. Rearrange: $(x^2 + 2x) + (y^2 - 4y) + 5 = 0$.

Step 2. Complete both squares: $(x + 1)^2 - 1 + (y - 2)^2 - 4 + 5 = 0$,

Step 3. so $(x + 1)^2 + (y - 2)^2 = 0$.

Step 4. A sum of squares is zero only when both squares are zero: x = -1, y = 2.

Step 5. $x^3 + y^3 = -1 + 8 = 7$.

» **Why this step?** Completing squares converts one equation in two unknowns into two forced equations - because squares cannot be negative.

Answer: $x = -1, y = 2$, so $x^3 + y^3 = 7$

Hinglish: *Squares complete karo: $(x+1)^2 + (y-2)^2 = 0$. Dono non-negative - dono zero hone padenge. $x = -1$, $y = 2$. Cubes: $-1 + 8 = 7$. Ye 'sum of squares = 0' trick real numbers ka superpower hai.*

Chapter 6: Perimeter and Area

Measurement questions are the most reliable marks in any paper - and the most reliable marks LOST to careless formulas. The uploaded PDF assembles the full toolkit: a review of perimeters and areas of triangles, squares, rectangles and parallelograms; the meaning of π; circumference and area of circles; arc length and sector area; Heron's formula for triangles with known sides; its special cases for equilateral and isosceles triangles; area of a quadrilateral split by a diagonal; Brahmagupta's formula for cyclic quadrilaterals; and rings, paths and composite regions. This book hardens every tool: reverse questions (area back to radius), sector-perimeter traps, grazing goats, tracks with semicircular ends, and Hippocrates-style shaded regions that look impossible and collapse in one line.

Figure 6.1: Heron's machinery - from three sides s(a+b+c)/2 to a clean area, no heights needed.

6.1 Complete Concept Cheat Sheet (read this before any question)

6.1.1 Perimeter and area: the review set

What is it? Perimeter is the length of the boundary; area is the size of the enclosed surface. **Core formulas:** rectangle $P = 2(l+b)$, $A = l \times b$; square $P = 4s$, $A = s^2$, diagonal $= s\sqrt{2}$; triangle $A = \frac{1}{2} \times \text{base} \times \text{height}$; parallelogram $A = \text{base} \times \text{height}$; rhombus $A = \frac{1}{2} \times d_1 \times d_2$; trapezium $A = \frac{1}{2}(a+b)h$. **The perimeter-area trap:** equal perimeters do NOT mean equal areas - among all shapes with a fixed perimeter, the circle encloses the most area. **When do I use which?** Perimeter answers 'how much boundary/fencing/frame', area answers 'how much surface/paint/carpet/grass'.

Units discipline: 1 m = 100 cm but 1 m^2 = 10000 cm^2; 1 hectare = 10000 m^2. Converting lengths squares the conversion factor - forget this once and your answer is off by a factor of 100.

6.1.2 π, circumference and area of a circle

What is π? The ratio circumference/diameter - the same constant for EVERY circle, approximately 3.14159... irrational. Exam convention: $\pi = 22/7$ when the radius is a multiple of 7, otherwise 3.14. **Formulas:** $C = 2\pi r = \pi d$; $A = \pi r^2$. **Reverse uses (the exam favourites):** given C, $r = C/(2\pi)$ then $A = C^2/(4\pi)$; given A, $r = \sqrt{(A/\pi)}$ then $C = 2\pi r$. **Scaling laws:** double the radius → double the circumference but QUADRUPLE the area; radii in ratio k → areas in ratio k^2. **Tiny example.** r = 14 cm: $C = 2 \times (22/7) \times 14 = 88$ cm; $A = (22/7) \times 196 = 616$ cm^2.

6.1.3 Arc length and sector area

What is it? A sector is the pizza slice between two radii and an arc of angle θ. The full circle is 360°, so a sector owns the fraction θ/360 of everything. **Arc length:** $l = (\theta/360) \times 2\pi r$. **Sector area:** $A = (\theta/360) \times \pi r^2$. **The half-angle gem:** $A = \frac{1}{2} \times l \times r$ (treat the arc as a curved 'base' and the radius as the 'height'). **Sector PERIMETER (the trap):** P = arc + TWO radii = l + 2r. Writing l + r loses marks every single year. **Tiny example.** r = 21, θ = 120°: $l = (120/360)(2 \cdot (22/7) \cdot 21) = 44$ cm; $A = (1/3)\pi(441) = 462$ cm^2; P = 44 + 42 = 86 cm.

Figure 6.2: Sector anatomy - arc l, two radii, angle θ; the slice owns θ/360 of the circle.

6.1.4 Heron's formula

What is it? Area of a triangle from its three sides alone, no height needed: A = √(s(s-a)(s-b)(s-c)) where s = (a+b+c)/2 is the semi-perimeter. **Why it matters:** most real triangles (a field, a plot) give you sides, not heights - Heron converts three tape-measure readings into an area. **Worked rhythm:** (1) add sides, halve for s; (2) compute the three differences s-a, s-b, s-c; (3) multiply all four numbers; (4) take the square root - and SIMPLIFY the surd if it is not a perfect square. **Tiny example.** Sides 5, 12, 13: s = 15; product 15·3·2 = 900; A = 30. **Validity check:** each side must be less than the sum of the other two (triangle inequality), otherwise the quantity under the root goes negative - the 'triangle' does not exist.

> **Verification habit**
> If the three sides satisfy $a^2 + b^2 = c^2$, the triangle is right-angled and the area must equal $\frac{1}{2}ab$. Run this cross-check whenever the sides look Pythagorean - 5-12-13, 8-15-17, 7-24-25, 9-40-41 and their scaled cousins.

6.1.5 Equilateral and isosceles shortcuts

Equilateral triangle (side a): height = $(\sqrt{3}/2)a$, area = $(\sqrt{3}/4)a^2$, perimeter = 3a. The area formula saves one Heron round-trip. **Isosceles triangle:** drop the height to the base; it bisects the base. Given equal sides l and base b: height = $\sqrt{(l^2 - b^2/4)}$, area = $\frac{1}{2} \cdot b \cdot$ height, and Heron must agree. **Reverse equilateral:** area given as $k\sqrt{3}$, side a satisfies $(\sqrt{3}/4)a^2 = k\sqrt{3} \Rightarrow a = 2\sqrt{k}$. **Tiny example.** Equilateral side 12: A = $(\sqrt{3}/4)(144) = 36\sqrt{3} \approx 62.35$.

6.1.6 Quadrilateral by diagonal; Brahmagupta

Diagonal split: any quadrilateral ABCD splits along diagonal AC into triangles ABC and ACD. Find each area (heights or Heron) and ADD. This is the master method for four-sided fields with unequal sides. **Brahmagupta's formula (cyclic quadrilaterals):** if all four vertices lie on a circle and sides are a, b, c, d with s = (a+b+c+d)/2, then A = √((s-a)(s-b)(s-c)(s-d)) - Heron with the (s - ...) pattern of four sides and no explicit height. It is valid ONLY for cyclic quadrilaterals; applying it to a general quadrilateral is a silent error. **Tiny example.** Sides 13, 10, 13, 10 (cyclic): s = 23; product 10·13·10·13 = 16900; A = 130.

Figure 6.3: A quadrilateral split by diagonal BD - two triangles, two Herons, one sum.

6.1.7 Rings, paths and composite regions

Ring (annulus): $A = \pi R^2 - \pi r^2 = \pi(R^2 - r^2) = \pi(R+r)(R-r)$. Given width w = R - r and area, use $(R+r)(R-r) = A/\pi$ to recover both radii. **Inside border path:** rectangular field L × B with a path of width w inside: path area = L·B - (L-2w)(B-2w). **Outside circular path:** $\pi((r+w)^2 - r^2) = \pi w(2r + w)$. **Composite figures:** decompose into rectangles, triangles, sectors and rings; add or subtract; keep every π exact until the last line. **Chord touching the inner circle (HOTS classic):** a chord of the outer circle tangent to the inner one has half-chord^2 = $R^2 - r^2$, so ring area = π × (half-chord)^2 - no radii needed!

Figure 6.4: Ring geometry - a chord of the outer circle touching the inner circle fixes $R^2 - r^2$.

6.1.8 Exam tips for this chapter

1. Choose $\pi = 22/7$ when a radius or diameter is a multiple of 7; otherwise use 3.14 - and say which you used.
2. Sector perimeter = arc + 2r. Circle perimeter = $2\pi r$. Never mix them.
3. Reduce every surd area (e.g. $\sqrt{1215} = 9\sqrt{15}$) - unsimplified surds lose the final mark.
4. Check for right triangles (Pythagoras) before Heron; $\frac{1}{2} \times \text{leg} \times \text{leg}$ is faster and safer.
5. Write units at every line: cm for length, cm^2 for area, m^2 for fields, hectares for land.
6. In reverse questions (area → radius → circumference) keep π symbolic until the numbers force it in.
7. Ratio of radii k ⇒ ratio of areas k^2 - read ratios as ratios, never as differences.
8. Cost questions: compute the geometric quantity FIRST, then multiply by rate; never mix the two steps.

6.2 Question Types / Concept Map for Chapter 6

Type	Name	How to recognise it in the question	Core method
T1	Basic perimeter/area	Rectangle, square, triangle, parallelogram, rhombus, trapezium	Direct formulas with correct units
T2	Heron direct	Three sides given, area asked	$s = (a+b+c)/2$; $\sqrt{(s(s-a)(s-b)(s-c))}$
T3	Equilateral / isosceles	'all sides equal', perimeter + base given	$(\sqrt{3}/4)a^2$; height by Pythagoras
T4	Quadrilateral by diagonal	Four sides plus a diagonal	Two Herons, add
T5	Brahmagupta	'cyclic quadrilateral' with four sides	$\sqrt{((s-a)(s-b)(s-c)(s-d))}$
T6	Circle basics + reverse	C ↔ r ↔ A chains	$C = 2\pi r$, $A = \pi r^2$, reversed
T7	Arc and sector	Angle θ at centre	$(\theta/360)$ fractions; $A = \frac{1}{2}lr$; $P = l + 2r$
T8	Segments and combinations	'segment', chord with angle	sector − triangle
T9	Rings / annulus	Two concentric circles	$\pi(R^2-r^2)$; chord trick
T10	Paths and borders	'path inside/outside', track, frame	Outer minus inner
T11	Composite figures	Mixed shapes, shaded regions	Decompose; add/subtract
T12	Ratio and scaling	Radii in ratio, same perimeter comparisons	Areas scale as k^2
T13	Cost applications	Rates for fencing, ploughing, tiling	Geometry first, money second
T14	HOTS shaded/proofs	Lunes, hexagons, grazing animals, AP sides	Label regions; prove then compute

6.3 Solved Practice Set (20 Questions)

Har question apne BOX me hai, uske turant baad uska boxed step-by-step solution. Pehle question ko khud solve karne ki koshish karo - jab atak jao tabhi solution dekho. Har step ke saath "Why this step?" diya hai, aur

end me Hinglish note jo concept ko dil me basa dega.

QUESTION 1 | Basic | Triangle base-height (T1)

Q1. A triangle has base 14 cm and height 9 cm. (i) Find its area. (ii) A second triangle has the same area but height 7 cm; find its base.

SOLUTION — step by step

Step 1. Given: b = 14, h = 9. Area = $\frac{1}{2} \times 14 \times 9 = 63$ cm^2.

Step 2. Second triangle: $\frac{1}{2} \times b \times 7 = 63 \Rightarrow 7b = 126 \Rightarrow b = 18$ cm.

» **Why this step?** Area = ½(base)(height) works because a triangle is half the parallelogram built on the same base.

Answer: (i) 63 cm² (ii) 18 cm

Hinglish: *Area = aadha × base × height. Pehla: 63 cm². Doosre mein same area, height 7 - base 18. Reverse questions mein equation banao, shortcut mat dhoondo.*

QUESTION 2 | Basic | Circumference and area (T6)

Q2. For a circle of radius 14 cm, find (i) the circumference and (ii) the area (use $\pi = 22/7$).

SOLUTION — step by step

Step 1. Given: r = 14 (a multiple of 7 → use $\pi = 22/7$).

Step 2. Circumference $C = 2\pi r = 2 \times (22/7) \times 14 = 88$ cm.

Step 3. Area $A = \pi r^2 = (22/7) \times 14^2 = (22/7) \times 196 = 616$ cm^2.

» **Why this step?** The 7 in the denominator cancels with the radius multiple - that is exactly when 22/7 gives exact answers.

Answer: C = 88 cm, A = 616 cm²

Hinglish: *Radius 7 ka multiple hai, isliye 22/7 perfect. C = 88 cm, A = 616 cm². Dono answers clean aane chahiye - decimal aaya toh kahin galti hai.*

QUESTION 3 | Basic | Equilateral shortcut (T3)

Q3. Find the perimeter and the area of an equilateral triangle of side 12 cm (leave $\sqrt{3}$ in your answer, then approximate with $\sqrt{3} \approx 1.73$).

SOLUTION — step by step

Step 1. Perimeter = 3 × 12 = 36 cm.

Step 2. Area = $(\sqrt{3}/4)a^2 = (\sqrt{3}/4) \times 144 = 36\sqrt{3}$ cm^2.

Step 3. Numerically: 36 × 1.73 = 62.28 cm^2 (with 1.732: 62.35).

» **Why this step?** $(\sqrt{3}/4)a^2$ is Heron's result pre-simplified for equal sides - one line instead of four.

Answer: P = 36 cm, A = $36\sqrt{3} \approx 62.3$ cm²

Hinglish: *Equilateral ka ready-made formula: $(\sqrt{3}/4)a^2$. 144/4 = 36, toh $36\sqrt{3} \approx 62.3$ cm². Heron se bhi milega but formula fast hai.*

QUESTION 4 | Basic | Heron direct (T2)

Q4. Find the area of a triangle with sides 5 cm, 12 cm and 13 cm.

SOLUTION — step by step

Step 1. s = (5 + 12 + 13)/2 = 15 cm.

Step 2. Differences: s - a = 10, s - b = 3, s - c = 2.

Step 3. A = $\sqrt{(15 \times 10 \times 3 \times 2)} = \sqrt{900} = 30$ cm².

Step 4. Cross-check: $5^2 + 12^2 = 169 = 13^2$, so it is right-angled; $\frac{1}{2} \times 5 \times 12 = 30$ ✓.

» **Why this step?** Spotting the right angle gives an instant verification - Heron must agree with $\frac{1}{2}ab$.

Answer: 30 cm²

Hinglish: *Heron ka 4-step rhythm: s nikaalo, differences nikaalo, multiply, root. 15×10×3×2 = 900, root = 30. Bonus: 5-12-13 Pythagorean triple hai - ½×5×12 = 30 se confirm.*

QUESTION 5 | Basic | Arc and sector (T7)

Q5. For a circle of radius 7 cm, find (i) the length of an arc subtending 90° at the centre and (ii) the area of the corresponding sector.

SOLUTION — step by step

Step 1. Fraction of the circle: 90/360 = 1/4.

Step 2. Arc = (1/4) × 2πr = (1/4) × 44 = 11 cm.

Step 3. Sector area = (1/4) × πr^2 = (1/4) × 154 = 38.5 cm².

» **Why this step?** Everything scales by θ/360 - compute the full-circle value once, then take the fraction.

Answer: arc = 11 cm, sector = 38.5 cm²

Hinglish: *90° matlab quarter circle. Arc = 44/4 = 11 cm, area = 154/4 = 38.5 cm². Fraction pehle likho, calculation aadhi ho jati hai.*

QUESTION 6 | Moderate | Heron with larger numbers (T2)

Q6. Find the area of a triangle with sides 40 cm, 30 cm and 14 cm.

SOLUTION — step by step

Step 1. s = (40 + 30 + 14)/2 = 42 cm.

Step 2. Differences: 2, 12, 28.

Step 3. Product: 42 × 2 × 12 × 28 = 28224.

Step 4. A = √28224 = 168 cm^2.

» **Why this step?** Large products factor stepwise: 42×2 = 84, 12×28 = 336, 84×336 = 28224; keep factors paired to make the root visible.

Answer: 168 cm^2

Hinglish: *s = 42, differences 2, 12, 28. Product 28224. Root 168. Bade numbers mein partial products likhte jao - ek hi line mein sab multiply karna errors bulata hai.*

QUESTION 7 | Moderate | Heron classic (T2)

Q7. A triangular park has sides 122 m, 120 m and 22 m. Find its area.

SOLUTION — step by step

Step 1. s = (122 + 120 + 22)/2 = 132 m.

Step 2. Differences: 132-122 = 10, 132-120 = 12, 132-22 = 110.

Step 3. Product: 132 × 10 × 12 × 110 = 1,742,400.

Step 4. A = √1742400 = 1320 m^2.

» **Why this step?** Pair the factors smartly: 132×110 = 14520 and 10×12 = 120; 14520×120 = 1742400. Grouping reveals the square: $(1320)^2$.

Answer: 1320 m^2

Hinglish: *s = 132, differences 10, 12, 110. Product 1742400, root 1320 m^2. Note the beauty: root = s × 10 = 1320 - aise patterns pehchano.*

QUESTION 8 | Moderate | Quadrilateral by diagonal (T4)

Q8. In quadrilateral ABCD, AB = 24 cm, BC = 10 cm, CD = 25 cm, DA = 17 cm and diagonal AC = 26 cm. Find the area of ABCD.

SOLUTION — step by step

Step 1. Triangle ABC: sides 24, 10, 26. Spot $24^2 + 10^2 = 576 + 100 = 676 = 26^2$: right angle at B.

Step 2. Area(ABC) = $\frac{1}{2} \times 24 \times 10 = 120$ cm^2.

Step 3. Triangle ACD: sides 26, 25, 17. s = 34; differences 8, 9, 17.

Step 4. Product $34 \times 8 \times 9 \times 17 = 41616$; Area(ACD) = $\sqrt{41616} = 204$ cm^2.

Step 5. Total area = 120 + 204 = 324 cm^2.

» **Why this step?** Splitting along the diagonal reduces one quadrilateral to two triangle problems; Pythagoras spotting saves a full Heron on the first one.

Answer: 324 cm^2

Hinglish: *Diagonal ke saath do triangles. ABC: 24-10-26 right triangle hai! Area ½×24×10 = 120, Heron ki zaroorat nahi. ACD: Heron se 204. Total 324 cm². Pehle Pythagoras check karo - time bachta hai.*

QUESTION 9 | Moderate | Brahmagupta (T5)

Q9. A cyclic quadrilateral has sides 13 cm, 10 cm, 13 cm and 10 cm (in order). Find its area using Brahmagupta's formula.

SOLUTION — step by step

Step 1. Cyclic quadrilateral ⇒ Brahmagupta applies: A = $\sqrt{(s-a)(s-b)(s-c)(s-d)}$.

Step 2. s = (13 + 10 + 13 + 10)/2 = 23 cm.

Step 3. Differences: 10, 13, 10, 13.

Step 4. Product: $10 \times 13 \times 10 \times 13 = 16900$. A = $\sqrt{16900} = 130$ cm^2.

» **Why this step?** Heron's pattern extends to four sides; the symmetric sides make every difference reappear - a designed clean answer.

Answer: 130 cm^2

Hinglish: *Cyclic quadrilateral mein Heron ka bhai Brahmagupta: same (s - side) pattern, char baar. s = 23, differences 10,13,10,13. Product 16900, root 130 cm².*

QUESTION 10 | Moderate | Wheel revolutions (T13)

Q10. A wheel of diameter 70 cm makes a journey of 1.1 km. How many complete revolutions does it make ($\pi = 22/7$)?

SOLUTION — step by step

Step 1. Circumference = $\pi d = (22/7) \times 70 = 220$ cm.

Step 2. Convert the journey: 1.1 km = 1100 m = 110000 cm.

Step 3. Revolutions = 110000/220 = 500.

» **Why this step?** One revolution moves the wheel exactly one circumference; total distance divided by circumference gives the count.

Answer: 500 revolutions

Hinglish: *Ek chakkar = circumference = 220 cm. Distance 110000 cm. 110000/220 = 500 revolutions. Units same karo pehle - km vs cm sabse common galti.*

QUESTION 11 | Moderate | Sector area and arc (T7)

Q11. A sector of a circle of radius 21 cm has central angle 120°. Find (i) the arc length, (ii) the sector area, (iii) the sector's perimeter.

SOLUTION — step by step

Step 1. Fraction: 120/360 = 1/3.

Step 2. Full circle: $C = 2\pi(21) = 132$ cm; $A = \pi(441) = 1386$ cm^2.

Step 3. Arc = 132/3 = 44 cm.

Step 4. Sector area = 1386/3 = 462 cm^2.

Step 5. Perimeter = arc + 2 radii = 44 + 42 = 86 cm.

» **Why this step?** The sector owns one third of the circumference AND one third of the area; its perimeter needs the two straight edges too.

Answer: arc 44 cm; area 462 cm^2; perimeter 86 cm

Hinglish: *120° = one-third. Arc 44, area 462. Perimeter mein DO radii: 44 + 42 = 86 cm. Arc + ek radius likhna classic mistake hai.*

QUESTION 12 | Difficult | Quadrilateral via diagonal (T4)

Q12. In quadrilateral ABCD, AB = 7 cm, BC = 24 cm, CD = 26 cm, DA = 17 cm and AC = 25 cm. Find the area of ABCD.

SOLUTION — step by step

Step 1. Triangle ABC: sides 7, 24, 25. Check $7^2 + 24^2 = 49 + 576 = 625 = 25^2$: right angle at B.

Step 2. Area(ABC) = $\frac{1}{2} \times 7 \times 24 = 84$ cm^2.

Step 3. Triangle ACD: sides 25, 26, 17. s = 34; differences 9, 8, 17.

Step 4. Product: $34 \times 9 \times 8 \times 17 = 41616$; Area(ACD) = 204 cm^2.

Step 5. Total = 84 + 204 = 288 cm^2.

» **Why this step?** One diagonal creates two independent triangle problems; a hidden right triangle in one of them halves the work.

Answer: 288 cm^2

Hinglish: *ABC: 7-24-25 triple! ½×7×24 = 84. ACD: Heron se 204. Total 288 cm². Diagonal questions mein pehle Pythagoras triple hunt karo.*

QUESTION 13 | Difficult | Segment area (T8)

Q13. Find the area of the minor segment of a circle of radius 14 cm cut off by a chord subtending 90° at the centre ($\pi = 22/7$).

SOLUTION — step by step

Step 1. Segment = sector - triangle.

Step 2. Sector (90°) = $(1/4)\pi(196) = 154$ cm^2.

Step 3. The chord and two radii form a right isosceles triangle with legs 14: area = $\frac{1}{2} \times 14 \times 14 = 98$ cm^2.

Step 4. Segment = 154 - 98 = 56 cm^2.

» **Why this step?** Every segment problem is 'pizza minus triangle'; the triangle's area uses the two radii as legs when the angle is 90°.

Answer: 56 cm^2

Hinglish: *Segment = sector - triangle. Sector 154, triangle ½×14×14 = 98. Answer 56 cm². 90° wale case mein triangle right-angled hota hai - radii legs.*

QUESTION 14 | Difficult | Inside rectangular path (T10)

Q14. A rectangular garden is 30 m by 20 m. A path 2 m wide runs inside it along all four sides. Find the area of the path.

SOLUTION — step by step

Step 1. Outer area = 30 × 20 = 600 m^2.

Step 2. The inner rectangle loses 2 m on EACH side: inner length = 30 - 4 = 26 m; inner width = 20 - 4 = 16 m.

Step 3. Inner area = 26 × 16 = 416 m^2.

Step 4. Path area = 600 - 416 = 184 m^2.

» **Why this step?** An inside border subtracts the width TWICE per dimension - once for each of the two parallel sides.

Answer: 184 m^2

Hinglish: *Path andar hai: dono dimensions se 2w = 4 minus. Inner 26×16 = 416. Path = 600-416 = 184 m^2. '2w' rule yaad rakho.*

QUESTION 15 | Difficult | Outside circular path (T10)

Q15. A circular park of radius 17 m is surrounded by a path 4 m wide running outside it. Find the area of the path ($\pi = 22/7$).

SOLUTION — step by step

Step 1. Outer radius R = 17 + 4 = 21 m.

Step 2. Path area = $\pi(R^2 - r^2)$ = (22/7)(441 - 289) = (22/7)(152).

Step 3. = 3344/7 ≈ 477.7 m^2.

» **Why this step?** An outside path adds to the radius once; the ring formula $\pi w(2r + w)$ is the fast version: (22/7)(4)(38).

Answer: $152\pi \approx 477.7$ m^2

Hinglish: *Bahar wala path: R = 21. $\pi(152) \approx 477.7$ m^2. Shortcut: $\pi w(2r+w) = (22/7)\times4\times38$ - same answer, direct.*

QUESTION 16 | High | Pythagorean ratio (T2/T3)

Q16. The sides of a triangle are in the ratio 3 : 4 : 5 and its perimeter is 72 cm. Find the area by TWO methods and confirm they agree.

SOLUTION — step by step

Step 1. Sides: 18k... let 3k + 4k + 5k = 72 ⇒ k = 6: sides 18, 24, 30 cm.

Step 2. Method 1 (right triangle): $18^2 + 24^2 = 324 + 576 = 900 = 30^2$, so right angle between the legs 18 and 24.

Step 3. Area = ½ × 18 × 24 = 216 cm^2.

Step 4. Method 2 (Heron): s = 36; differences 18, 12, 6; product 36×18×12×6 = 46656; √46656 = 216 ✓.

» **Why this step?** A 3-4-5 ratio triangle is always right-angled (the ratio is scale-invariant), so both methods must match.

Answer: 216 cm^2 (both methods)

Hinglish: *3:4:5 hamesha right triangle hota hai. Sides 18, 24, 30. ½×18×24 = 216. Heron bhi 216 deta hai - dono match. Ratio dekh ke triple pehchano.*

QUESTION 17 | High | Trapezium via Heron (T4)

Q17. A field is in the shape of a trapezium whose parallel sides are 77 m and 60 m, and the non-parallel sides are 25 m and 26 m. Find its area.

SOLUTION — step by step

Step 1. Draw the height through the shorter parallel side's endpoint; it creates a helper triangle whose base is 77 - 60 = 17 m (after transferring the full 60 to the other side).

Step 2. Helper triangle sides: 25, 26, 17. s = 34; differences 9, 8, 17.

Step 3. Area of helper triangle = √(34×9×8×17) = √41616 = 204 m^2.

Step 4. Height: 204 = ½ × 17 × h ⇒ h = 24 m.

Step 5. Trapezium area = ½(77 + 60)(24) = ½ × 137 × 24 = 1644 m^2.

» **Why this step?** A general trapezium hides a triangle built from the slanted sides and the difference of the parallel sides; Heron extracts the height.

Answer: 1644 m^2

Hinglish: *Trapezium ka height chhupa hai helper triangle mein: base = 77-60 = 17, sides 25, 26. Heron se area 204. h = 2×204/17 = 24. Phir ½(a+b)h = 1644 m^2. Ye 4-step method ratta maar lo.*

QUESTION 18 | High | Running track (T11)

Q18. A running track is 90 m long on each straight and has two semicircular ends of diameter 56 m. Find (i) the perimeter of the track and (ii) the area enclosed ($\pi = 22/7$).

SOLUTION — step by step

Step 1. Perimeter = two straights + two semicircular arcs = 90 + 90 + one full circle of diameter 56.

Step 2. Circle circumference = $\pi d = (22/7)(56) = 176$ m.

Step 3. Perimeter = 180 + 176 = 356 m.

Step 4. Area = rectangle 90×56 + full circle $\pi(28^2) = 5040 + (22/7)(784) = 5040 + 2464 = 7504$ m^2.

» **Why this step?** Two semicircles of equal diameter combine into one full circle - for both length and area.

Answer: (i) 356 m (ii) 7504 m^2

Hinglish: *Do semicircles = ek full circle. Perimeter 180 + 176 = 356 m. Area 5040 + 2464 = 7504 m². Track problems mein halves ko jod kar full banao.*

QUESTION 19 | HOTS | Grazing goat (T13/T14)

Q19. A goat is tied to one corner of a rectangular field 40 m by 30 m with a rope 21 m long. Find the area it can graze, assuming it cannot leave the field ($\pi = 22/7$).

SOLUTION — step by step

Step 1. At a corner of a rectangle, only 90° of the circle's sweep lies inside the field: a quarter circle.

Step 2. Grazing area = $(1/4)\pi r^2 = (1/4) \times (22/7) \times 441 = 346.5$ m^2.

» **Why this step?** The two fences of the corner cut the reachable disk to a quarter; the rope length is the radius.

Answer: 346.5 m^2

Hinglish: *Corner par sirf 90° ka sector available - quarter circle. ¼ × π × 441 = 346.5 m². Rope = radius, corner = center.*

QUESTION 20 | HOTS | Hippocrates lunes (T14)

Q20. On the legs of a right triangle with legs 12 cm and 16 cm, semicircles are drawn OUTWARD. On the hypotenuse, the semicircle is drawn inward (toward the triangle). Show that the total area of the two lunes (the crescent-shaped regions between the outward semicircles and the hypotenuse semicircle) equals the area of the triangle.

SOLUTION — step by step

Step 1. Right triangle: hypotenuse = $\sqrt{(144 + 256)} = 20$ cm; triangle area = $\frac{1}{2} \times 12 \times 16 = 96$ cm^2.

Step 2. Semicircle on leg 12: area $\frac{1}{2}\pi(6^2) = 18\pi$; on leg 16: $\frac{1}{2}\pi(8^2) = 32\pi$.

Step 3. Semicircle on hypotenuse 20: $\frac{1}{2}\pi(10^2) = 50\pi$.

Step 4. Key identity: $18\pi + 32\pi = 50\pi$ - the two leg semicircles exactly fill the hypotenuse semicircle (because $6^2 + 8^2 = 10^2$).

Step 5. The two lunes together = (leg semicircles) + (triangle) - (hypotenuse semicircle) = $18\pi + 32\pi + 96 - 50\pi = 96$ cm^2. ∎

» Why this step? Pythagoras on the RADII makes the curved areas cancel, leaving the flat triangle - Hippocrates' classic squaring of lunes.

Answer: The lunes total exactly 96 cm^2, equal to the triangle's area.

Hinglish: *Lunes ka magic: leg semicircles ka sum = hypotenuse semicircle ($18\pi+32\pi = 50\pi$). Lunes = 96 cm^2 = triangle ka area exactly. Curved areas cancel ho gaye - 2500 saal purana result!*

Chapter 7: Probability - Measuring Chance

Probability turns 'maybe' into a number between 0 and 1 - and every dice, card and coin question in every exam is built on a handful of ideas done carefully. The uploaded PDF introduces random experiments and trials, outcomes and sample spaces, the classical formula P(E) = favourable outcomes / total outcomes, the complement rule P(not E) = 1 - P(E), sure and impossible events, and empirical probability from observed data. It works through coins, dice, cards and everyday experiments. This practice book sharpens all of it: two-dice tables, three-coin tree diagrams, without-replacement draws, 'at least one' reasoning, geometric probability on lengths and areas, and proof-style questions that show why probabilities must always add to 1.

Figure 7.1: A tree diagram - every path is one outcome; multiply along branches, add across paths.

7.1 Complete Concept Cheat Sheet (read this before any question)

7.1.1 Experiment, outcome, sample space, event

What is it? A *random experiment* is any action with an uncertain result (tossing a coin, drawing a card). Each possible result is an *outcome*. The set of ALL outcomes is the *sample space* S, and any subset of S is an *event*. **Key counts:** one coin: S = {H, T}, |S| = 2. One die: |S| = 6. Two coins: 4 outcomes. Two dice: 36 ordered outcomes. Three coins: 8 outcomes. A standard deck: 52 cards. **The golden habit:** before computing ANY probability, WRITE the sample space or its size. Half of all errors are wrong totals, not wrong fractions.

When do I count 'ordered' outcomes? When the experiment distinguishes them: two dice give 36 outcomes because (2, 5) and (5, 2) are different throws; drawing two cards one after another gives 52 × 51 ordered draws. If you prefer unordered counting, you must be consistent in numerator AND denominator.

7.1.2 Classical probability and its three laws

What is it? For equally likely outcomes, P(E) = n(E)/n(S) - favourable outcomes over total outcomes. **Three laws:** (1) $0 \le P(E) \le 1$ always; (2) P(sure event) = 1 and P(impossible event) = 0; (3) P(E) + P(not E) = 1, so P(not E) = 1 - P(E). The complement rule converts 'at least one' and 'none' questions into their easier partners. **Tiny example.** P(rolling a prime on a die) = {2, 3, 5}/6 = 1/2; P(not prime) = 1 - 1/2 = 1/2.

Reading the answer: probability may be written as a fraction in lowest terms, a decimal, or a percentage - fractions are the exam default. A probability of 7/4 or -0.2 is a signal that the COUNTS are wrong, not that the formula is wrong.

7.1.3 Cards: the anatomy of a deck

What is it? 52 cards: 4 suits (hearts ♥, diamonds ♦ - red; spades ♠, clubs ♣ - black) × 13 ranks each (A, 2, 3, ..., 10, J, Q, K). **Standard sub-counts:** aces: 4; face cards (J, Q, K): 12; kings: 4; red queens: 2; hearts: 13; numbered cards 2-10: 36; cards between 3 and 8 (inclusive): 24; honour cards (A, K, Q, J, 10): 20. **Overlaps matter:** 'a king OR a heart' is not 4 + 13 = 17 outcomes because the king of hearts was counted twice - it is 16. The inclusion-exclusion habit starts here.

7.1.4 Dice: singles, pairs and the 36-table

What is it? One die: six outcomes. Two dice: 36 ordered pairs - and the smart move is a 6×6 grid in rough work. **Star facts from the table:** sum 7 has 6 pairs (most likely); sums 2 and 12 have 1 pair each (least likely); P(sum = 7) = 6/36 = 1/6; P(sum > 9) = 6/36 = 1/6 (pairs for 10, 11, 12); P(doubles) = 6/36 = 1/6; P(product even) = 27/36 = 3/4; P(same parity) = 18/36 = 1/2. **Products:** a product is odd only when BOTH dice are odd: 9/36 = 1/4. **Tiny example.** P(sum ≥ 10) = 6/36 = 1/6 (pairs: 4-6, 5-5, 6-4, 5-6, 6-5, 6-6).

7.1.5 Coins and tree diagrams

What is it? Each coin doubles the sample space: 1 coin → 2, 2 coins → 4, 3 coins → 8, n coins → 2^n. A tree diagram draws every path: two branches (H/T) per coin, and each complete path is one outcome. **Star facts (three coins):** P(exactly 2 heads) = 3/8; P(at least 1 head) = 7/8 (only TTT fails); P(at least 2 heads) = 4/8 = 1/2; P(all same) = 2/8 = 1/4. **Why trees work:** multiplying along a branch counts outcomes systematically - it is the visual version of 2 × 2 × 2.

Figure 7.2: The probability scale - impossible 0, sure 1, everything else between.

7.1.6 Draws without replacement

What is it? When drawn objects are NOT returned, the second draw's counts shrink: a bag with 5 red and 3 blue balls, two drawn one after another - P(both red) = (5/8)(4/7) = 20/56 = 5/14. The denominator fell from 8 to 7 because one ball left. **Two-draw combinations via pairs:** P(one red one blue in any order) = 2 × (5/8)(3/8) - the factor 2 handles the two orders (RB, BR). **Alternative counting:** total ways to choose 2 balls from 8 = C(8,2) = 28; both red: C(5,2) = 10; P = 10/28 = 5/14 - same answer, cleaner for 'any order' questions.

7.1.7 Empirical probability

What is it? When real data (not symmetry) defines the chances: P(E) = (number of times E occurred)/(total number of trials). It is an estimate from observation - the more trials, the closer it typically drifts to the true probability. **Example shape:** 200 screws inspected, 12 defective → P(defective) = 12/200 = 0.06, so P(good) = 0.94. **Warning:** empirical probabilities need not be 'nice' fractions, and they describe the PAST data - classical probability describes idealised symmetric setups.

7.1.8 Geometric probability

What is it? When outcomes form a continuum - a point chosen on a segment, a dart hitting a board - probability equals the ratio of measures (lengths, areas): P = favourable length (or area) / total length (or area). **Tiny example.** A point chosen on a 10 cm segment; P(it lies in the first 3 cm) = 3/10. A dart hitting a 20 cm square board with a 10 cm circle target: P(hit circle) = $\pi(5^2)/400 = 25\pi/400 = \pi/16$. **Guard:** both numerator and denominator must use the SAME kind of measure.

7.1.9 Exam tips for this chapter

1. Always write n(S) first; a wrong total poisons the whole question.
2. Two dice: draw the 6×6 grid once and reuse it - sums, products, parity all read off it.
3. Complement first for 'at least one': P(at least one H) = 1 - P(no H).
4. Without replacement: denominators shrink (8 → 7); numerators shrink too if a favourable item left.
5. Cards: memorise 4, 12, 13, 26 and the overlap warning.
6. Reduce every fraction to lowest terms - un-reduced answers lose the presentation mark.

7. Check the range: if $P > 1$ or $P < 0$, recount - the arithmetic, not the theory, is wrong.
8. Empirical questions: read the table column-wise; the total row is n(S).

7.2 Question Types / Concept Map for Chapter 7

Type	Name	How to recognise it in the question	Core method
T1	Sample space building	'List the outcomes', 'find n(S)'	Coins/dice/cards/objects; 2^n, 36, 52
T2	Single die events	One die, one event	Count favourable ÷ 6
T3	Two-dice events	'two dice thrown', sums/products	36-table; systematic pairs
T4	Card events	Deck language: suit, face, ace	Standard counts; overlaps
T5	Complement rule	'at least one', 'not', 'none'	1 - P(opposite)
T6	Bag/ball single draw	Coloured objects, one draw	Favourable colour ÷ total
T7	Two draws without replacement	'second ball is not replaced'	Shrinking denominators; ×2 for orders
T8	Tree diagrams	Multi-stage: 2-3 coins, draw twice	Multiply along branches
T9	Empirical probability	Frequency tables, surveys, trials	occurrences ÷ trials; complement from table
T10	Geometric probability	spinners, dartboards, points on segments	favourable measure ÷ total measure
T11	Event relations	sure/impossible, P(E) + P(not E), mutually exclusive	Range laws; complement
T12	Fairness and decisions	'is the game fair?', expected ties	Compare P(win) values
T13	Missing probability algebra	P(one outcome) unknown, sum = 1	Set up equation; solve
T14	HOTS multi-stage	Three dice, replacement mixtures, proofs	Decompose; tree or table; verify

7.3 Solved Practice Set (20 Questions)

Har question apne BOX me hai, uske turant baad uska boxed step-by-step solution. Pehle question ko khud solve karne ki koshish karo - jab atak jao tabhi solution dekho. Har step ke saath "Why this step?" diya hai, aur end me Hinglish note jo concept ko dil me basa dega.

QUESTION 1 | Basic | Single die (T2)

Q1. A die is rolled once. Find P(i) an even number, (ii) a number greater than 4, (iii) a number that is even AND greater than 4.

SOLUTION — step by step

Step 1. n(S) = 6.

Step 2. (i) Even: {2, 4, 6}: P = 3/6 = 1/2.

Step 3. (ii) Greater than 4: {5, 6}: P = 2/6 = 1/3.

Step 4. (iii) Both: {6} only: P = 1/6.

» **Why this step?** 'AND' shrinks the event: only outcomes satisfying both conditions count.

Answer: (i) 1/2 (ii) 1/3 (iii) 1/6

Hinglish: *Even teen numbers, greater-than-4 do numbers, dono mein sirf 6. 1/2, 1/3, 1/6. AND ka matlab intersection.*

QUESTION 2 | Basic | Two coins (T1/T8)

Q2. Two coins are tossed together. Find P(i) exactly one head, (ii) at least one head, (iii) no head.

SOLUTION — step by step

Step 1. S = {HH, HT, TH, TT}: n(S) = 4.

Step 2. (i) Exactly one head: {HT, TH}: P = 2/4 = 1/2.

Step 3. (ii) At least one head: {HH, HT, TH}: P = 3/4.

Step 4. (iii) No head: {TT}: P = 1/4. Note (ii) and (iii) are complements: 3/4 + 1/4 = 1 ✓.

» **Why this step?** 'At least' questions pair with 'none' through the complement - checking they sum to 1 validates both.

Answer: (i) 1/2 (ii) 3/4 (iii) 1/4

Hinglish: *Chaar outcomes. Exactly one head = HT, TH = 1/2. At least one = 3/4 (complement of TT). At least wale questions mein 'none' nikaalo, 1 minus karo.*

QUESTION 3 | Basic | Single draw from a bag (T6)

Q3. A bag contains 4 red, 5 blue and 3 green balls. One ball is drawn at random. Find P(i) red, (ii) not green, (iii) red or blue.

SOLUTION — step by step

Step 1. Total = 4 + 5 + 3 = 12.

Step 2. (i) P(red) = 4/12 = 1/3.

Step 3. (ii) P(not green) = 1 - 3/12 = 9/12 = 3/4.

Step 4. (iii) P(red or blue) = (4+5)/12 = 9/12 = 3/4 (which matches (ii), since 'not green' = 'red or blue').

» **Why this step?** Disjoint categories add; 'not green' and 'red or blue' describe the same set here - a built-in consistency check.

Answer: (i) 1/3 (ii) 3/4 (iii) 3/4

Hinglish: *Total 12. Red 1/3. Not green 3/4. Red-or-blue bhi 3/4 - same cheez hai dono! Consistency check hamesha karo.*

QUESTION 4 | Basic | Card basics (T4)

Q4. One card is drawn from a well-shuffled deck of 52. Find P(i) an ace, (ii) a red card, (iii) a face card, (iv) the queen of hearts.

SOLUTION — step by step

Step 1. (i) Aces: 4 → P = 4/52 = 1/13.

Step 2. (ii) Red cards: 26 → P = 26/52 = 1/2.

Step 3. (iii) Face cards (J, Q, K): 3 × 4 = 12 → P = 12/52 = 3/13.

Step 4. (iv) One specific card: P = 1/52.

» **Why this step?** Four deck counts (4, 12, 13, 26) answer a third of all card questions instantly.

Answer: (i) 1/13 (ii) 1/2 (iii) 3/13 (iv) 1/52

Hinglish: *Yaad rakho: aces 4, reds 26, face cards 12, hearts 13. 1/13, 1/2, 3/13, 1/52. Ye counts ratta - deck questions 90% inhi par.*

QUESTION 5 | Basic | Sum to 1 with missing probability (T13)

Q5. A random experiment has outcomes A, B, C with P(A) = 0.3 and P(B) = 0.25. Find P(C) if A, B, C are the only outcomes.

SOLUTION — step by step

Step 1. The probabilities of all outcomes sum to exactly 1.

Step 2. P(C) = 1 - 0.3 - 0.25 = 0.45.

» **Why this step?** Total probability = 1 is the accounting equation of probability theory.

Answer: 0.45

Hinglish: *Sab probabilities ka sum 1 hota hai. 1 - 0.55 = 0.45. Ye 'missing probability' pattern bahut common hai.*

QUESTION 6 | Moderate | Two-dice conditions (T3)

Q6. Two dice are rolled. Find P(i) the sum is a prime number, (ii) the product is odd, (iii) both dice show the same number.

SOLUTION — step by step

Step 1. (i) Prime sums: 2 (1 pair), 3 (2), 5 (4), 7 (6), 11 (2): total 15 pairs → P = 15/36 = 5/12.

Step 2. (ii) Product odd ⇔ both dice odd: 3 × 3 = 9 pairs → P = 9/36 = 1/4.

Step 3. (iii) Doubles: (1,1), ..., (6,6): 6 pairs → P = 6/36 = 1/6.

» **Why this step?** Products split by parity cleanly: odd needs odd×odd; even is the complement 27/36.

Answer: (i) 5/12 (ii) 1/4 (iii) 1/6

Hinglish: *Prime sums ka count: 1+2+4+6+2 = 15 → 5/12. Product odd = dono odd = 1/4. Doubles 1/6. 36-grid se teeno direct padho.*

QUESTION 7 | Moderate | Cards with overlap (T4)

Q7. One card is drawn from a deck of 52. Find P(i) a king or a heart, (ii) a red face card, (iii) neither a king nor a queen.

SOLUTION — step by step

Step 1. (i) Kings: 4; hearts: 13; king of hearts counted twice → 4 + 13 - 1 = 16 → P = 16/52 = 4/13.

Step 2. (ii) Red face cards: J, Q, K of hearts and diamonds: 6 → P = 6/52 = 3/26.

Step 3. (iii) Kings and queens: 8 cards; neither: 52 - 8 = 44 → P = 44/52 = 11/13.

» **Why this step?** Overlap correction (add both counts, subtract the intersection) is the whole skill in (i).

Answer: (i) 4/13 (ii) 3/26 (iii) 11/13

Hinglish: *King-or-heart mein 4+13 = 17 galat - king of hearts do baar gina! 16 sahi → 4/13. Red face 6. Neither king nor queen 11/13.*

QUESTION 8 | Moderate | Empirical probability (T9)

Q8. In a survey of 400 students, 96 walk to school, 160 take a bus, 84 cycle and the rest come by car. A student is chosen at random. Find P(i) walks, (ii) does not cycle, (iii) comes by car.

SOLUTION — step by step

Step 1. Car users: 400 - 96 - 160 - 84 = 60.

Step 2. (i) P(walks) = 96/400 = 0.24.

Step 3. (ii) P(not cycle) = 1 - 84/400 = 316/400 = 0.79.

Step 4. (iii) P(car) = 60/400 = 0.15.

» **Why this step?** Empirical probability = occurrences ÷ trials; the missing category comes from the total.

Answer: (i) 0.24 (ii) 0.79 (iii) 0.15

Hinglish: *Car wale: 60. 0.24, 0.79, 0.15. Sum check: 0.24+0.40+0.21+0.15 = 1 ✓. Empirical mein decimals normal hain.*

QUESTION 9 | Moderate | Three coins (T8)

Q9. Three coins are tossed together. Find P(i) exactly two heads, (ii) at least two heads, (iii) all three same.

SOLUTION — step by step

Step 1. S has $2^3 = 8$ outcomes.

Step 2. (i) Exactly two heads: HHT, HTH, THH → 3/8.

Step 3. (ii) At least two heads: those 3 plus HHH → 4/8 = 1/2.

Step 4. (iii) All same: HHH, TTT → 2/8 = 1/4.

» **Why this step?** 'Exactly' and 'at least' differ by the all-heads outcome - read the wording like a lawyer.

Answer: (i) 3/8 (ii) 1/2 (iii) 1/4

Hinglish: *8 outcomes. Exactly 2: teen paths = 3/8. At least 2: +HHH = 1/2. All same 1/4. Tree diagram banao - 8 paths dikh jayengi.*

QUESTION 10 | Moderate | Bag: two draws, same colour (T7)

Q10. A bag has 5 red and 3 blue balls. Two balls are drawn one after another WITHOUT replacement. Find P(both red).

SOLUTION — step by step

Step 1. First red: 5/8. After removing one red, second red: 4/7.

Step 2. P(both red) = (5/8) × (4/7) = 20/56 = 5/14.

Step 3. Combination check: C(5,2)/C(8,2) = 10/28 = 5/14 ✓.

» **Why this step?** Without replacement, the second draw's denominator drops to 7 - the shrinking universe is the whole idea.

Answer: 5/14

Hinglish: *Pehla red 5/8, doosra red 4/7 (ek red ja chuka!). Multiply: 5/14. Denominator 8 se 7 hua - yahi 'without replacement' hai.*

QUESTION 11 | Difficult | Two dice: difference events (T3)

Q11. Two dice are rolled. Find P(the absolute difference of the two numbers is 2).

SOLUTION — step by step

Step 1. Pairs with |a - b| = 2: (1,3), (2,4), (3,1), (3,5), (4,2), (4,6), (5,3), (5, ... list systematically:

Step 2. Difference 2 rows: a - b = 2: (3,1), (4,2), (5,3), (6,4): 4 pairs. b - a = 2: (1,3), (2,4), (3,5), (4,6): 4 pairs.

Step 3. Total 8 pairs → P = 8/36 = 2/9.

» **Why this step?** Both diagonals above and below the main diagonal count; symmetry doubles one direction's count.

Answer: 2/9

Hinglish: *Difference 2: do diagonals - 4 + 4 = 8 pairs. 2/9. Difference d ke liye formula: 2(6-d)/36 pairs for d = 1..5.*

QUESTION 12 | Difficult | Cards: two aces (T7)

Q12. Two cards are drawn from a deck without replacement. Find P(both are aces).

SOLUTION — step by step

Step 1. First ace: 4/52 = 1/13. Second ace: 3/51 = 1/17.

Step 2. P = (1/13)(1/17) = 1/221.

Step 3. Combination check: C(4,2)/C(52,2) = 6/1326 = 1/221 ✓.

» **Why this step?** Two-stage multiplication with shrinking counts; 51 = the deck after one card leaves.

Answer: 1/221

Hinglish: *1/13 × 1/17 = 1/221. Doosra ace 51 cards mein se 3 - shrinking deck.*

QUESTION 13 | Difficult | Geometric: point on segment (T10)

Q13. A point P is chosen at random on a segment AB of length 12 cm. Find the probability that AP > 4 cm.

SOLUTION — step by step

Step 1. Total measure: length 12 cm.

Step 2. Favourable: positions with AP > 4 cm form a sub-segment of length 12 - 4 = 8 cm.

Step 3. P = 8/12 = 2/3.

» **Why this step?** Geometric probability replaces counting with measuring lengths of intervals.

Answer: 2/3

Hinglish: *AP > 4 matlab P, 4 cm se aage kahin bhi - 8 cm ka region. 8/12 = 2/3. Length ratio hi probability hai.*

QUESTION 14 | Difficult | Geometric: dartboard (T10)

Q14. A dart hits a square board of side 20 cm at a random point. A circle of diameter 10 cm is painted at the centre. Find P(the dart lands inside the circle).

SOLUTION — step by step

Step 1. Board area = 400 cm^2. Circle radius = 5 cm; area = 25π cm^2.

Step 2. P = 25π/400 = π/16.

Step 3. With π = 22/7: 22/112 = 11/56 ≈ 0.196.

» **Why this step?** Both measures are areas; the π cancels if kept symbolic.

Answer: π/16 ≈ 0.196

Hinglish: *Circle/board = 25π/400 = π/16. Dono same measure (area) - yahi rule hai. π symbolic rakho jab tak ho sake.*

QUESTION 15 | Difficult | Mixed bag with replacement vs not (T7)

Q15. A bag has 3 red and 2 green balls. Two balls are drawn. Compare P(both red) when (i) the first ball IS replaced, (ii) it is NOT replaced.

SOLUTION — step by step

Step 1. (i) With replacement: P = (3/5)(3/5) = 9/25.

Step 2. (ii) Without: P = (3/5)(2/4) = 6/20 = 3/10.

Step 3. Comparison: 9/25 = 0.36 vs 3/10 = 0.30 - replacement keeps the odds constant, so both-red is more likely.

» **Why this step?** Replacement freezes the probabilities; without it, drawing a red makes the next red rarer.

Answer: (i) 9/25 ≈ 0.36 (ii) 3/10 = 0.30; replacement gives the larger probability.

Hinglish: *Replace karo: 9/25. Nahi: 3/10. Replace > no-replace for same colour. Denominator shrink hi difference hai.*

QUESTION 16 | High | Two dice: one even one odd (T3)

Q16. Two dice are rolled. Find P(one die shows an even number and the other odd).

SOLUTION — step by step

Step 1. Favourable: (even, odd) or (odd, even): 3×3 + 3×3 = 18 pairs.

Step 2. P = 18/36 = 1/2.

Step 3. Complement check: same parity (both even or both odd): 9 + 9 = 18 → 1/2 ✓.

» **Why this step?** Mixed parity and same parity split the 36 outcomes exactly in half.

Answer: 1/2

Hinglish: *Even-odd mix: 18 pairs = 1/2. Same parity bhi 18 - poora balance. Symmetry se check karo.*

QUESTION 17 | High | At least one with bag draws (T5/T7)

Q17. A bag has 4 red and 4 blue balls. Three balls are drawn without replacement. Find P(at least one red).

SOLUTION — step by step

Step 1. Complement: no red = all three blue.

Step 2. P(all blue) = (4/7)(3/6)(2/5) = 24/210 = 4/35.

Step 3. P(at least one red) = 1 - 4/35 = 31/35.

» **Why this step?** 'At least one' with three draws is far easier through the complement - one cascade instead of three cases.

Answer: 31/35

Hinglish: *At least one red = 1 - (sab blue). Sab blue: 4/7×3/6×2/5 = 4/35. Answer 31/35. Complement hi sahi raasta hai.*

QUESTION 18 | High | Tree diagram three-stage (T8)

Q18. A coin is tossed and then a die is rolled. Using a tree diagram, find P(head and an even number) and P(tail and a number greater than 4).

SOLUTION — step by step

Step 1. Tree: 2 branches (H/T), each splitting into 6 die branches: 12 paths.

Step 2. Head + even: H connects to {2, 4, 6}: 3 paths → P = 3/12 = 1/4.

Step 3. Tail + >4: T connects to {5, 6}: 2 paths → P = 2/12 = 1/6.

» **Why this step?** Each path has probability (1/2)(1/6) = 1/12; counting paths is counting twelfths.

Answer: 1/4 and 1/6

Hinglish: *12 paths. H+even: 3 → 1/4. T+big: 2 → 1/6. Tree = paths ginna.*

QUESTION 19 | HOTS | Sum probability distribution proof (T14)

Q19. Two dice are rolled. Tabulate the probability of each possible sum (2 through 12) and verify that the probabilities add to 1.

SOLUTION — step by step

Step 1. Counts: sum 2:1, 3:2, 4:3, 5:4, 6:5, 7:6, 8:5, 9:4, 10:3, 11:2, 12:1.

Step 2. Each probability = count/36.

Step 3. Total counts: 1+2+3+4+5+6+5+4+3+2+1 = 36.

Step 4. Sum of probabilities = 36/36 = 1 ✓ - the distribution is complete.

» **Why this step?** The pyramid count (1,2,3,4,5,6,5,4,3,2,1) is the two-dice signature; completeness is guaranteed by covering every pair exactly once.

Answer: P(s = k) = count/36 with counts 1,2,3,4,5,6,5,4,3,2,1; total 1.

Hinglish: *Pyramid pattern: 1,2,3,4,5,6,5,4,3,2,1. Total 36 = 1. Ye table ratta maar lo - har two-dice question isi se nikalta hai.*

QUESTION 20 | HOTS | Draw until first success (T14)

Q20. A bag has 3 red and 3 blue balls. Balls are drawn one at a time WITHOUT replacement until the first red appears. Find P(exactly two draws are needed).

SOLUTION — step by step

Step 1. Exactly two draws means: first is blue AND second is red.

Step 2. P = (3/6)(3/5) = 9/30 = 3/10.

» **Why this step?** Stopping questions translate directly into 'first k-1 draws fail, draw k succeeds'.

Answer: 3/10

Hinglish: *Exactly 2 draws = pehla blue, doosra red. (1/2)(3/5) = 3/10. 'Until' questions ka pattern.*